Gregory Budzban and Philip Feinsilver

Algebraic Methods in Automata Theory

Discussion Draft for Colleague Review – 2026-10-22

October 30, 2026

Springer

Scope and assumptions

General theory

The general theory concerns finite transformation semigroups, their kernels, partitions, ranges, idempotents, representations and associated operators. It includes the partial-circulant family (PtC). Simplicity, the absence of loops, two-out regularity, irreducibility and aperiodicity are not blanket assumptions on this general theory. Probabilistic results state their own hypotheses on the stochastic matrix and generator probabilities; existence, positivity, uniqueness and convergence of stationary laws must not be inferred without the relevant hypotheses.

Assumptions for road-coloring questions

In questions concerning the road coloring problem, the underlying graph G is a finite simple directed graph: there are no loops and no parallel edges with the same ordered endpoints. Opposite arrows $x \to y$ and $y \to x$ are permitted. Every vertex has exactly two distinct outgoing neighbours. The graph is strongly connected and aperiodic. Write $\mathscr{A}$ for its adjacency matrix and set

$$\mathscr{A} \in \{0,1\}^{n\times n}, \qquad \mathscr{A}_{xx} = 0, \qquad \mathscr{A}\mathbf{1} = 2\mathbf{1}, \qquad A = \tfrac{1}{2}\mathscr{A} = \tfrac{1}{2}(R+B).$$

Thus A is stochastic, traceless, irreducible and aperiodic. For a coloring,

$$R_{xx} = B_{xx} = 0, \qquad R_{xy}B_{xy} = 0, \qquad xR \neq xB.$$

The stationary row $\boldsymbol{\pi}$ is strictly positive and normalized by $\boldsymbol{\pi}\mathbf{1} = 1$; $\boldsymbol{\pi}A = \boldsymbol{\pi}$ and $A^m \to \mathbf{1}\boldsymbol{\pi}$. Within this setting A denotes the normalized transition matrix, not the adjacency matrix. Recoloring changes R, B but not $\mathscr{A}$ or A.

The auxiliary target-pair graph has one edge $\{xR, xB\}$ for each source. Under these road-coloring assumptions it has no loops. Distinct sources may give the same auxiliary edge without violating simplicity of G.

The road-coloring assumptions are not imposed on PtC or on the general semigroup theory. Each theorem retains its own explicitly stated hypotheses.

Note to readers

This is a provisional discussion draft, version 2026-10-22, circulated for comments on mathematical content, organization, hypotheses, and exposition. It assembles the current working material in a dependency-ordered form; it is not a publication-ready manuscript and has not been compilation-tested.

The main development concerns algebraic and probabilistic methods for transformation semigroups. The road coloring theorem provides background; a complete exposition of Trahtman's proof is not included. The partial circulant family serves as a running laboratory and a synthesis chapter.

Readers should distinguish inherited exposition, recently added material, and explicitly marked research questions. Chapter 5 uses a single expectation-operator convention. Later inherited blocks retain their locally announced historical conventions. Some historical prose and cross-references still need revision. The Lie-constant converse and the general description of opposite-letter kernel coordinates are not asserted as proved results.

The thirteen chapters have differing levels of development. Chapters on permanental Markov chains and graph lifts presently provide a core treatment, not a complete transfer of their source papers. Appendices B and C are scope notes rather than finished appendices. Appendix D preserves alternative proofs and historical strategies for critical review.

Comments are especially welcome on prerequisites, redundant proofs, missing hypotheses, operator conventions, the balance between theory and examples, and the scope of the specialty topics. Please identify comments by chapter, section title, and theorem label where possible; numbering may change.

Contents

Index of Notation xiii

Part I Foundations

1 Graphs, Transformations, and Synchronization 3
1.1 The Generalized Road Coloring Problem 3

2 Finite Semigroups and Kernel Geometry 7
2.1 Introduction to Semigroup Theory 7
2.2 Completely Simple Semigroups 9
2.3 The Rees-Sushkewitch Theorem 11
2.4 The Semigroup $\mathscr{F}_X$ 15
2.5 Finite Semigroups and the Road Coloring Problem 18
2.6 The Kernel of a Coloring Semigroup 20

3 Probability on Finite Semigroups 23
3.1 Random Walks on Semigroups 23
3.2 Semigroup Algebras 27
3.3 Convergence of Convolution Powers 29
3.4 Trees, unicycles, and the stationary distribution 36

Part II Kernel geometry and coloring

4 Tensor Squares and Equipartition 41
4.1 Setup and statement 41
4.2 The collision matrix 42
4.3 Positive semidefiniteness and stationarity 43
4.4 Minimum-rank ranges and equal fiber masses 44
4.5 Consequences and scope 45

5 Kernel Expectations and Their Spectra 47
5.1 Basic observables and the kernel e-projections 47
5.1.1 The two basic expectation operators 47
5.1.2 Column projections: hold the range fixed 48
5.1.3 Row projections: hold the partition fixed 48
5.1.4 The average of the distinguished idempotents 49
5.2 Basic Terminology and Notations 49
5.3 Expectation operators and their fixed-point equations 50
5.4 Basic data and properties 50
5.5 Norms 51
5.6 Equipartitioning 52
5.7 Algebra generated by the expectation operators 53

6 Delta, Canonical Colorings, and Recoloring 55
6.1 Questions about a fixed graph and its colorings 55
6.2 The matrix Δ 55
6.3 Existence of canonical colorings from the transition matrix . . 57
6.3.1 Constructing and counting the CC colorings 58
6.3.2 A four-state illustration of the target-pair test 58
6.3.3 A test involving only matrix operations on A 60
6.3.4 Singularity is necessary but not sufficient 60
6.3.5 Existence versus kernel rank 61
6.4 The matrix Δ annihilates only rank one kernels 61
6.5 Special rank one matrices in $\mathrm{Vec}(S)$ 62
6.6 The Rank Condition 63
6.7 Coloring and recoloring 64
6.8 Main Result 65

7 Rank Tests, Cycles, and Potential Operators 69
7.1 Ranking 69
7.1.1 Ranking with Δ 70
7.1.2 Ranking with $\mathbf{A}+\varepsilon\Delta$ 70
7.2 Divisibility 71
7.3 Applications 72
7.3.1 Prime Cycle 72
7.3.2 Friedman 72
7.4 A potential operator and changes of letter probabilities 72
7.4.1 A necessary correction concerning the complex spectrum 74
7.4.2 Endpoint cycle measures and a wave operator 74
7.4.3 An explicit partial-circulant potential 74
7.4.4 Adjugate and notation qualifications 75

Part III Higher representations and constructions

8 Tensor, Symmetric, and Zeon Hierarchies 79
8.1 From particle configurations to the zeon quotient 79
8.1.1 The invariant collision filtration 79
8.1.2 Averages and limiting projections 80
8.2 Zeon algebra and structure of the kernel 81
8.2.1 Introduction 81
8.2.2 Basic Identities 82
8.2.3 Trace Identities 83
8.2.4 Kernel structure 86
8.3 Levels 89
8.4 Distributions 90
8.4.1 Interpretation 90
8.5 Absorbing states 91
8.6 Inclusion operators and normalized compatible fields 91

9 Permanental Markov Chains 93
9.1 Permanents as killed independent-particle transitions 93
9.2 A positivity hypothesis in the converse 93
9.3 Independent versus common-letter motion 94

10 Coupled Semigroups and Opposite Letters 95
10.1 Opposite-letter tensor coupling and its kernel 95
10.1.1 The original kernel alone is insufficient 96
10.1.2 The running examples 96
10.1.3 Same-letter and opposite-letter chains 96
10.1.4 A partial-circulant consequence 97

11 Graph Lifts and the Double Construction 99
11.1 The edge-state lift 99
11.2 Kernel rank and kernel elements under the lift 99
11.3 A systematic source of CC examples 100

Part IV Synthesis

12 Lie Methods and Lie Constants 103
12.1 Completely Simple Semigroups, Lie Algebras, and the Road Coloring Problem 103
12.1.1 Background 103
12.1.2 Semigroups and kernels 104
12.2 Solvability and right groups 106
12.3 The constant ideal 109
12.4 An exact computational procedure 109
12.5 The kernel-partition obstruction 110
12.6 A short explicit extraction 111
12.7 Computations and the remaining problem 111

13 Partial Circulants: A Running Laboratory 113
13.1 Definitions and statement 113
13.2 Congruence classes and elementary collapses 114
13.3 Proof of the theorem 115
13.4 Further structure 116
13.5 The ladder chain and its stationary distribution 117
13.6 The spectrum of the collision operator 117
13.7 Why permanental and common-letter hierarchies agree here . . 118
13.8 The rank condition: a hypothesis test 118
13.9 A worked example and exercises 119
13.10 Research directions, not asserted theorems 119

A Weak Convergence and Compact Semigroups 121
A.1 Weak Convergence of Probability Measures 121
A.2 Some Initial Results of Convergence 123
A.3 Convolution Powers 129

B The Symmetric-Function Excursus 137

C Exact Computation and the Example Bank 139
C.1 Running all recolorings of the four-state graph 139

D Alternative Proofs and Historical Arguments 143
D.1 Introduction 143
D.2 Semigroup Proofs of Selected Results 144
D.3 A technique to complete the solution to the generalized road coloring conjecture 152
D.4 Conclusion 153

References 155

Subject Index 157

Index of Notation

This grouped list records the working convention and gives the page of a definition or principal discussion. It is a notation guide, not a list of every occurrence. Local hypotheses remain those of the referenced text.

Historical conventions.

Some inherited passages use N_0 for the present N, N for $J-N$, M_0 for the present M, and M for $M-\tau\omega$. These are historical aliases, not additional operators. The ongoing notation revision removes those aliases chapter by chapter. In the partial-circulant collision-spectrum section, the locally used $N_0 = J-N$ is a different alias and should be read only with its local definition.

Graphs and transformations

$\mathscr{A}$ Adjacency matrix; distinguish from the stochastic normalization and the locally reused family symbol in the partial-circulant chapter. p. 3.

R, B Letter transformations, represented by binary stochastic matrices. p. 3.

A Normalized transition matrix; in the two-letter equal-weight setting, $(R+B)/2$. p. 49.

Δ Letter-difference matrix $(R-B)/2$. p. 49.

u All-ones row, of the dimension required by the representation level. p. 49.

J All-ones matrix $u^{\mathsf{T}}u$. p. 49.

Semigroups, kernel partitions, and ranges

$\mathscr{S}$ Transformation semigroup generated by the letters. p. 49.
$\mathscr{K}$ Minimum-rank ideal (semigroup kernel); not the nullspace of a matrix. p. 49.
r Kernel rank, the minimum transformation rank; not generally the rank of the averaged collision matrix. p. 47.
e_{ij} Distinguished idempotent with kernel partition i and range j. p. 47.
$\mathscr{P}_i, \mathscr{R}_j$ Kernel partition and kernel range; the latter is a transversal of each kernel partition. p. 47.
$\rho(k)$ Range-indicator row of a kernel transformation; shape $1 \times n$. p. 49.
ρ_j Diagonal indicator matrix of range $\mathscr{R}_j$; shape $n \times n$. p. 48.

Probability and stationary data

λ Limiting kernel probability measure used in kernel expectations; use the appropriate Cesaro convention. p. 47.
α_i, β_j Partition and range marginals of the kernel measure. p. 47.
π Normalized stationary row of A. p. 49.
Ω Stationary projection $u^{\mathsf{T}}\pi$ in the irreducible setting. p. 49.
ω Diagonal matrix $\operatorname{diag}(\pi)$. p. 49.

Kernel expectations and projections

N Collision expectation $\mathbb{E}_\lambda[KK^{\mathsf{T}}]$; symmetric $n \times n$ matrix. p. 47.
M Weighted range expectation $\mathbb{E}_\lambda[K^{\mathsf{T}}JK]$; symmetric $n \times n$ matrix. p. 47.
$J - N$ Pair-survival matrix, complementary to the collision expectation. p. 47.
τ Trace $\operatorname{tr} M$ in the kernel-operator convention; the potential-operator section also uses this letter as an interpolation parameter. p. 50.
$M - \tau\omega$ Centered range expectation; distinguish from M. p. 50.
$\widetilde{M}$ Pair-range observable $\mathbb{E}[K^{\mathsf{T}}(J-I)K]$; distinct from the centered operator $M - \tau\omega$. p. 86.
C_i Partition coincidence matrix $e_{ij}e_{ij}^{\mathsf{T}}$; independent of the range index j. p. 47.
P_j Range-fixed projection $N \times n$, acting on row vectors. p. 48.
Q_i Partition-fixed projection; $rC_i\omega$ under the stated equipartition hypotheses. p. 48.
$\bar{e}$ Product-marginal average of distinguished idempotents; not generally an idempotent. p. 49.

Higher representations and inclusion maps

$f^{\otimes k}$ Common-transformation action on the kth tensor power. p. 79.
$\mathrm{Sym}^k(V)$ Symmetric power: unordered configurations with multiplicity. p. 79.
F_ℓ Collision filtration by at most ℓ occupied vertices, at fixed degree. p. 79.
$f^{\vee k}$ Degree-k zeon action: a k-subset maps to its image or to zero after a collision. p. 79.
A_k Common-letter average at degree k; distinguish it from an independent-particle power of A. p. 79.
Ω_k Higher-level limiting projection; interpreted through the stated Abel or Cesaro limit. p. 90.
u_ℓ Probability column that an ℓ-subset is split by a kernel partition. p. 91.
η_ℓ Normalized range-subset probability row; distinguish from inherited higher-level row normalizations. p. 91.
$E^{\ell,k}$ Inclusion matrix from ℓ-subsets to k-subsets. p. 91.
$D_{\ell,k}$ Normalized inclusion matrix $\binom{\ell}{k}^{-1} E^{\ell,k}$. p. 91.

Rank tests and potential operators

$\Sigma(v,T)$ Cycle matrix with rows $vT,\dots,vT^d$, where $d=\mathrm{rank}\,T$. p. 69.
Z Fundamental inverse $(I-A+\Omega)^{-1}$. p. 72.
V Potential operator ΔZ; elsewhere V may denote a vector space or vertex set. p. 72.
A_τ Biased transition matrix $A+au\Delta$; here au is an interpolation parameter, not $\mathrm{tr}\,M$. p. 73.

Partial-circulant local notation

M_i Partial-circulant generator: $1M_i=i$ and $xM_i=x-1$ for $x>1$; not the expectation operator M. p. 113.
m,d,C Maximum generator index, gcd of generator indices, and recurrent core $\{1,\dots,m\}$. p. 113.
P_a,C_a Residue block modulo d and its intersection with the core (printed as P_r,C_r with a local residue index r); not the range projection P_j. p. 113.

Part I
Foundations

Chapter 1
Graphs, Transformations, and Synchronization

This edition separates the existence of a synchronizing coloring from the behavior of a fixed coloring. The classical aperiodic constant-outdegree road coloring statement is a theorem; see [21]. Historical terminology in inherited source blocks is not a claim of current open status. The precise generalized formulation is retained as a separate formulation requiring its own status review. PtC and the four-, six-, and nine-state examples will recur throughout the text.

Scope

The front matter distinguishes the general semigroup theory from the additional assumptions imposed in road-coloring questions. The latter are not blanket assumptions on the book.

1.1 The Generalized Road Coloring Problem

The definitions below allow a general constant outdegree d so that the historical formulations can be read precisely. In our road-coloring applications we specialize to the simple, loopless, irreducible and aperiodic two-out setting stated at the opening. The generalized periodic formulation at the end is recorded separately and does not relax those application hypotheses.

Let $\mathscr{A}$ be the adjacency matrix of a strongly connected digraph $G = (V, \mathscr{E})$. We will assume that the row sums of $\mathscr{A}$ are constant or, equivalently, that the outdegree of each vertex of G is constant.

Definition 1.1. A coloring of G is a decomposition of $\mathscr{A}$ into d binary (that is, 0–1) stochastic matrices $R_1, R_2, ..., R_d$ such that $\mathscr{A} = R_1 + ... + R_d$.

Intuitively, one can think of a matrix R_i as assigning the ith of d colors to the edge (j,k) if $(R_i)_{j,k} = 1$. Each $n \times n$ binary stochastic matrix R can also be thought of as a function on the vertices $\{1,2,...,n\}$ where $R_{j,k} = 1$ iff $jR = k$. (We will write functions to the right of domain elements to duplicate the order of matrix multiplication.)

Let $S = \langle R_1, R_2, ..., R_d \rangle$ be the semigroup generated by $R_1, R_2, ..., R_d$ under multiplication. Clearly, S is a finite semigroup. We will refer to S as a coloring semigroup, and will write $E(S)$ for the idempotents in S.

Definition 1.2 (Target-pair graph).

For a two-out coloring (R,B), use $A = (R+B)/2$. On $V = \{1,\dots,n\}$, put

$$t_x = \{xR, xB\}, \qquad \mathscr{T}(A) = (t_1,\dots,t_n), \qquad H_A = (V, \{t_x : x \in V\}).$$

Here H_A is undirected: each source contributes an edge joining its two destinations, not an edge from the source to a destination. The source-indexed list $\mathscr{T}(A)$ retains repeated auxiliary edges; H_A suppresses their multiplicities when discussing connectivity or bipartiteness. For single-digit vertex labels, write ij for the unordered pair $\{i, j\}$, with $i < j$, to display the list compactly.

Remark 1.1. The target-pair graph is invariant under every interchange of the two outgoing colors at a source; it depends on the uncolored two-out graph, not its coloring. Under the simple road-coloring hypotheses it is loopless, although different sources may yield the same auxiliary edge. In the more general two-generator setting, a coincident pair $xR = xB$ contributes a loop. The connected components, their bipartiteness, and hence the ranks of A and Δ are fixed throughout a recoloring run. The two bipartition sides, when they exist, describe CC target types; they are not generally kernel blocks. The component partition is the graph-intrinsic canonical partition. These distinctions will be used in the rank and CC discussions.

Definition 1.3. Let $S = \langle R_1, R_2, \dots, R_d \rangle$ be a coloring semigroup for the digraph G. We will say that S has a synchronizing instruction, if S contains a matrix of rank one.

Notice that a rank one binary stochastic matrix has a column k of all ones. We can think of this matrix as a function mapping every vertex to vertex k, which intuitively captures the idea of synchronization.

The kernel K of a finite transformation semigroup consists of its minimum-rank transformations. Its rank is one precisely when the coloring is synchronizing. The structure of completely simple kernels, their Rees coordinates, partitions, ranges, local groups and right groups will be developed systematically in the semigroup chapter; we do not need their full structure merely to state the coloring questions.

The final concept required to precisely state the road coloring problem is the notion of the period of a graph.

Definition 1.4. Let $\mathscr{P} = \{P_1, P_2, \ldots, P_t\}$ be a partition of the vertices V such that if $(i, j) \in \mathscr{E}$ and $i \in P_k$, then $j \in P_{k+1}$ ($i \in P_t$ implies $j \in P_1$). Then $\mathscr{P}$ will be called a periodic partition of order $t \geq 2$.

Definition 1.5. A digraph $G = (V, \mathscr{E})$ is periodic of period $t \geq 2$ iff

$$t = \max_{\mathscr{P}}[\text{card}(\mathscr{P})]$$

where this maximum is taken over all periodic partitions of G. If G has no periodic partitions, then G is aperiodic. In this case, we will refer to $\mathscr{P} = \{V\}$ as the periodic partition.

The above definition can be shown to be equivalent to the more familiar one that uses the period of a vertex, where $\text{per}(v) = \gcd\{|c_1|, \ldots, |c_n|\}$ where $c_1, \ldots, c_n$ are the simple cycles containing v.

The road coloring problem can now be stated in the following way.

Classical road coloring statement. Let $G = (V, \mathscr{E})$ be a d-out, strongly connected digraph with adjacency matrix $\mathscr{A}$. Let $\left\{\left\{R_{i_1}, R_{i_2}, \ldots, R_{i_d}\right\} : \mathscr{A} = R_{i_1} + R_{i_2} + \ldots + R_{i_d}\right\}$ be the set of all colorings of G and let $S_i = \left\langle R_{i_1}, R_{i_2}, \ldots, R_{i_d}\right\rangle$ be the associated coloring semigroups. If G is aperiodic, then for some i, S_i contains a synchronizing instruction.

In order to generalize the road coloring conjecture to all strongly connected graphs, one needs to observe that if the completely simple kernel of a semigroup of functions is a right group, then each element in the kernel induces the same partition on the set of vertices. Here, of course, the partition $\mathscr{P}$ induced by a function k is the collection of subsets of V of the form $k^{-1}(v)$, for $v \in V$. In the case of the constant functions, the partition induced by the kernel is simply $\{V\}$.

Generalized Road Coloring Conjecture. Let $G = (V, \mathscr{E})$ be a strongly connected d-out digraph. Then G is periodic of period $t \geq 1$ with maximal periodic partition $\mathscr{P} = \{P_1, \ldots, P_t\}$ iff G has a minimal (with respect to the rank of its kernel) coloring semigroup $S = \langle R_1, \ldots, R_d\rangle$ whose kernel K is a right group with partition $\mathscr{P}$.

Clearly if $\mathscr{P} = \{P_1, \ldots, P_t\}$ is the partition induced by the kernel K, then $\text{rank}(K) = t$. The semigroup discussion explains why having a minimal coloring semigroup whose kernel is a right group with rank t is a sufficient condition for the graph to be periodic of order t.

Chapter 2
Finite Semigroups and Kernel Geometry

2.1 Introduction to Semigroup Theory

For our initial purposes, the definition of a semigroup will be very straightforward. No topological notions will be included.

Definition 2.1. A set of elements S equipped with a binary operation $*$ will be called a semigroup if

(a) for all $s,t \in S$, $s*t \in S$ (closure);
(b) for all $s,t,u \in S$, $(s*t)*u = s*(t*u)$ (associativity);

(we will usually drop the $*$ notation and write $s*t = st$).

We will be utilizing certain canonical examples.

Example 2.1. Let X be a set and let $\mathscr{F}_X$ be the set of all functions on the set X with the operation composition. We will write composition in the following way

$$f,g \in \mathscr{F}_X.$$

Then,

$$(x)f \circ g = ((x)\mathscr{F})g.$$

Note this is opposite the standard direction for $\mathscr{F} \circ g$ and is done due to the matrix representation of $\mathscr{F}_X$, when X is finite. (See Example 2.2.)

Note. $X = \{1,2,3\}$

$$\begin{array}{ccccccc} & f & & g & & & \\ 1 & \to & 1 & \to & 1 & & \\ 2 & \searrow & 2 & \to & 2 & & \\ 3 & \to & 3 & \nearrow & 3 & & \end{array} \qquad \begin{array}{ccc} & f \circ g & \\ 1 & \to & 1 \\ 2 & \to & 2 \\ 3 & \nearrow & 3 \end{array}$$

An extremely useful concept in all of algebra is that of a representation. Due to the usefulness of matrices, it is often preferable to find an algebraically equivalent matrix semigroup to work with.

Definition 2.2. Let S be a semigroup. Then a semigroup of matrices M is called a representation of S, if there exists a homomorphism $T : S \to M$. If T is an isomorphism, then the representation M is called faithful.

Example 2.2. Let $X = \{1,2,\ldots,n\}$ and let $S = \mathscr{F}_X$.

For $f \in \mathscr{F}_X$ define

$$[T(f)]_{ij} = \begin{cases} 1 & (i)f = j \\ 0 & \text{otherwise.} \end{cases}$$

Let $M = \{T(f) \,|\, f \in \mathscr{F}_X\}$.

Homework. Show that $T : S \to M$ is a semigroup isomorphism. That is, show that M is a faithful representation of S.

Let $X = \{1,2\}$.

$\operatorname{card}(\mathscr{F}_X) = 4$

$\mathscr{F}_1 \begin{matrix} 1 \to 1 \\ 2 \to 2 \end{matrix}$ $\quad \mathscr{F}_2 \begin{matrix} 1 \searrow 1 \\ 2 \nearrow 2 \end{matrix}$ $\quad \mathscr{F}_3 \begin{matrix} 1 \to 1 \\ 2 \nearrow 2 \end{matrix}$ $\quad \mathscr{F}_4 \begin{matrix} 1 \searrow 1 \\ 2 \to 2 \end{matrix}$

$$T(\mathscr{F}_1) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad T(\mathscr{F}_2) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad T(\mathscr{F}_3) = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \quad T(\mathscr{F}_4) = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

Note. The set of matrices M in the above example are also binary (i.e., 0,1) stochastic matrices. Their row sums are trivially 1.

Some notation. If $s \in S$, and A, B are subsets of S then

(1) $sA = \{sa \,|\, a \in A\}$
(2) $AB = \{ab \,|\, a \in A,\ a \in B\}$

Definition 2.3.(1) $T \subset S$ is a subsemigroup of S if $T \cdot T \subset T$. If T is a group, we will call T a subsgroup of S.

(2) A subset L of S is called a left ideal, if $SL \subset L$. Right ideals are similarly defined ($RS \subset R$). Note that L, R are necessarily subsemigroups of S.
(3) A subset I of S is a (two-sided) ideal if it is both a right ideal and a left ideal.
(4) S is said to be left (right) simple if it contains no proper left (right) ideals. If S contains no proper ideal, then S is said to be simple. (S is trivially an ideal.)

Homework. If I_1, I_2 are ideals, show that $I_1 \cap I_2$ is an ideal. (In particular, it is not empty.)

Homework. Show that S is left simple iff for any $a, b \in S$ there exists $s \in S$ such that $sa = b$. (Hint: Note that for any $a \in S$, Sa is a left ideal of S.)

(5) A left ideal is called principal if it is of the form $\{a\} \cup Sa$ for some $a \in S$.

(6) An element $u \in S$ is idempotent if $a^2 = a$. If an idempotent e is such that $es = s$ ($se = s$) then e is called a left (right) identity. If e is both a left and right identity, then it is called the identity of S. It is easy to see that an identity must be unique.
(7) An element z is called a left (right) zero if $zs = z$ ($sz = z$) for all $s \in S$. z is a zero, if it is both a left and right zero.

Homework. Show that a semigroup S is a group iff it is both left simple and right simple.

Example 2.3. Let $S = \left\{ \begin{bmatrix} a & b \\ b & a \end{bmatrix} \mid a,b \in GF(q),\ a \neq b,\ q = p^n \right\}$. Then S is an abelian semigroup. (Show this.)

$$S = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \right\}$$

Notice

$$\det \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = 3 \equiv 0 \bmod 3$$

$$\det \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = 0 \bmod 3$$

In fact, both matrices are rank 1, mod 3. Notice also, that $G = \left\{ \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \right\}$ forms an ideal in S, has no ideals within it (G is a minimal ideal) and in fact, G is a group.

Example 2.4. This example will be of paramount importance!!

Let X, Y be sets and G a group with $\phi : Y \times X \to G$ any function. Let $S = X \times G \times Y = \{(x,g,y)\}$ define

$$(x_1,g_1,y_1)(x_2,g_2,y_2) = (x_1, g_1\phi(y_1,x_2)g_2, y_2).$$

Then S becomes a semigroup under this product.

We will begin our next lecture by discussing this example at length.

2.2 Completely Simple Semigroups

Let us again consider a semigroup $S = X \times G \times Y$ where for $(x_1,g_1,y_1),(x_2,g_2,y_2) \in S$

$$(x_1,g_1,y_1)(x_2,g_2,y_2) = (x_1, g_1\phi(y_1x_2)g_2, y_2)$$

where $\phi : Y \times X \to G$ is arbitrary. (We will usually write $\phi(y,x) = (yx)$. Notice the following:

(1) Each set of the form $\{x_1\} \times G \times \{y_1\}$ is isomorphic to G under the isomorphism

$$g \mapsto (x_1, g(y_1x_1)^{-1}, y_1)$$
$$\text{or} \quad g \mapsto (x_1, (y_1x_1)^{-1}g, y_1).$$

(Show this.)

(2) Note that the local idempotent is of the form $(x,(yx)^{-1},y)$. In fact, suppose $(x,h,y)(x,h,y) = (x,h,y)$

$$\text{then} \quad h(yx)h = h$$
$$-\text{or} \quad h(yx) = e$$
$$h = (yx)^{-1}.$$

(3) One can picture $X \times G \times Y$ as a union of isomorphic copies of G with a different "center" in each box ("center" is the local identity).

	y_1	y_2	y_3	y_4	y_5
x_1					
x_2					
x_3					
x_4					

Homework.

(4) Sets of the form $\{x\} \times G \times Y$ form minimal right ideals, while minimal left ideals are $X \times G \times \{y\}$.

Homework. Show that if L is a left ideal of S. Then $L = X \times G \times Y$, where $Y_1 \subset Y$. (A similar result holds for right ideals.)

(5) Since the local groups of the form $\{x\} \times G \times \{Y\}$ are disjoint from each other (that is, $\{x_1\} \times G \times \{y_1\} \cap \{x_2\} \times G \times \{y_2\} = \emptyset$ iff $x_1 \neq x_2$ or $y_1 \neq y_2$), then these form a partition of S.

$$S = \bigcup_{\substack{x_i \in X \\ y_j \in Y}} \{x_i\} \times G \times \{y_j\} \equiv \bigcup_{i,j} G_{ij}$$

Also note that $G_{ij} \cdot G_{mn} = G_{in}$. That is, the partition $\{G_{ij}\}$ is preserved under the semigroup operation. If we call ρ the equivalence relation that generates $\{G_{ij}\}$ then

Homework. S/ρ is a quotient semigroup isomorphic to $x \times \{e\} \times Y$.

Proposition 2.1. Let $S = X \times G \times Y$. Then S is simple.

Proof. Suppose $I \subsetneq S$ was a nontrivial ideal of S. Then for all $a \in I$

$$Sa \subset I$$
$$\Rightarrow SaS \subset IS \subset I$$

Let $a = (x,g,y)$ be arbitrary. Then

$$(X \times G \times Y)(x,g,y)(X \times G \times Y)$$
$$= X \times G(Yx)g(yX)G \times Y$$

but $GhG = G$ for any $h \in G$. Thus $SaS = X \times G \times Y$ and $I = S$ which implies that S is simple. □

Definition 2.4. An idempotent e is called primitive iff e is minimal w.r.t. the partial order $e \leq f \Leftrightarrow ef = fe = e$.

Definition 2.5. A semigroup S is called completely simple iff S is simple and contains a primitive idempotent.

Example 2.5.

$$S = \left\{ \begin{bmatrix} a & 0 \\ b & 1 \end{bmatrix} \middle\| a,b > 0 \right\}$$

Observe that if

$$\begin{bmatrix} a_1 & 0 \\ b_1 & 1 \end{bmatrix} \begin{bmatrix} a_1 & 0 \\ b_1 & 1 \end{bmatrix} = \begin{bmatrix} a_1 & 0 \\ b_1 & 1 \end{bmatrix},$$

then $a_1^2 = a_1$ and $a_1 b_1 + b_1 = b_1$. But $a_1^2 = a_1$ implies that $a_1 = 1$. Thus $2b_1 = b_1$, which forces $b_1 = 0$. Thus S contains no idempotent.

Homework. Show that the semigroup S in Example 2.5 is simple.

Homework. Show that every finite semigroup contains an idempotent.

2.3 The Rees-Sushkewitch Theorem

Lemma 2.1. Let S be a simple semigroup.

If $e \in S$ is a primitive idempotent, then Se, eS are minimal left and right ideals respectively and the set eSe is a group. In addition, $eSe = eS \cap Se$.

Proof. Consider eS. Clearly it is a right ideal. Suppose $I \subset eS$ is a right ideal. Let $a \in I$.

– Notice that for $a \in eS$, e acts like left identity.

$$a \in eS \Rightarrow a = es_1 \Rightarrow ea = ea = e^2 s_1 = es_1 = a.$$

Now $SaS = S$ since S is simple. Thus $xay = e$ for some $x,y \in S$. Notice $(exe)(ea)(ye) = e$. Let $exe = x'$, $ye = y'$, and $z' = ay'x'$. Then

$$\begin{aligned} z'z' &= ay'x'ay'x' \\ &= ay'(exe)(ea)(ye)x' \\ &= ay'ex' \\ &= ay'x' = z'. \end{aligned}$$

Thus z' is an idempotent. Also

$$\begin{aligned} z'e = (ay'x')e = ay'x' = z' \\ ez' = eay'x' = ay'x' = z'. \end{aligned}$$

But e was assumed to be primitive, thus $e = z'$. But $e = ay'x' \in aS \subset IS \subset I$, thus

$$\begin{aligned} eS \subset IS \subset I \\ \Rightarrow eS = I \end{aligned}$$

and eS is minimal. □

Homework. Show that Se is a minimal left ideal.

Now let $G = eSe$, and $g \in G$. Then

$$\begin{aligned} Gg = G(ese) &= (eSe)(ese) \\ &= e(Sese) \\ &= eSe = G \quad \text{by minimality} \\ gG = (ese)G &= (eseS)e = eSe = G. \end{aligned}$$

Thus G is a group, since it is left simple and right simple.

Homework. Show that $eSe = eS \cap Se$.

In fact, we have the following.

Theorem 2.1. Let S be a finite semigroup, with R a minimal right ideal and L a minimal left ideal. Then RL is a group.

Proof. Let $RL = G$, $g \in G$. Then

$$\begin{aligned} gG &= rlRL \\ &= RL \end{aligned}$$

since $(rl)R \subset R$ and is a right ideal. Similarly,

$$\begin{aligned} Gg &= RLrl \qquad rl \in L \\ &= RL. \end{aligned}$$

Thus RL is both left and right simple and thus is a group. □

Lemma 2.2. Let L be a minimal left ideal in a semigroup S. Then Lc is a minimal left ideal for all $c \in S$.

Proof. Suppose not. Let $A \subsetneq Lc$ be a left ideal. Let $B = \{x \in L : xc \in A\} \subset L$. By definition, $Bc \subset A \subsetneq Lc$. Notice that if $a \in A$, $a = lc$ for some $l \in L \Rightarrow l \in B \Rightarrow lc = a \in Bc \Rightarrow Bc = A$.

Now let $b \in B$, $s \in S$. Then,

$$(sb)c \subset sA \subset A \qquad (A \text{ is a left ideal})$$

Also,

$$sb \subset sL \subset L.$$

Thus

$$\begin{array}{lll} & (sb)c \in A & \text{where } sb \in L \\ \Rightarrow & (sb) \in B & \text{for all } s \in S \\ \Rightarrow & B & \text{is a left ideal} \\ \Rightarrow & B = L & \text{by minimality} \\ \Rightarrow & Bc = Lc = A. & \otimes \end{array}$$

Thus

$$Lc \text{ is minimal.} \qquad \square$$

Homework. Prove the analogous result for right ideals

Lemma 2.3. Let S be completely simple. Then $S = \bigcup_i L_i = \bigcup_j R_j$ where $\{L_i\}$, $\{R_j\}$ are the set of all minimal left and right ideals, respectively.

Proof. Notice that the union of two left ideals is a left ideal. (Homework)

Let $A = \bigcup_i L_i$ where each L_i is a minimal left ideal. We will show A is a right ideal as well. Now

$$\begin{array}{ll} & a \in L_i \qquad \text{for some } i \\ \text{and} & as \in L_i s \subset A, \end{array}$$

since $L_i s$ is a minimal left ideal by Lemma 2. Thus

$$as \in A \quad \text{for all} \quad s \in S.$$

Thus A is a right ideal, therefore an ideal. But S is simple $\Rightarrow A = S$.

Notation. For any subset $T \subset S$, $E(T)$ will be the set of idempotents of T. Consider $E(L)$ where L is a minimal left ideal and let $e \in E(L)$. Then since $e \in L$,

$$Se \subset SL \subset L$$

and Se is a left ideal $\Rightarrow Se = L$. (In fact, every minimal left ideal is of the form Ss for some s.)

Now let $l \in L$. Then $l = se$ for some s and $le = (se)e = se = l$. Thus each idempotent of L acts like a right identity of L. Take $e_1, e_2 \in E(L)$. Suppose

$$\begin{aligned} & e_1 e_2 = e_2 e_1 \\ \Rightarrow\ & e_1 = e_2. \end{aligned}$$

Thus all idempotents are primitive. (Since $S = \bigcup L_i$, every idempotent is in some L.)

Note. $E(L)$ is a left zero semigroup. Similarly, $E(R)$ is a right zero semigroup.

Rees-Sushkewitch Theorem. A semigroup is completely simple iff there exist sets X, Y, and a group G such that $S \cong X \times G \times Y$, where $X \times G \times Y$ is a Rees product semigroup.

Proof. Let S be completely simple and $e \in E(s)$. Let $X = E(Se)$, $Y = E(eS)$, $G = eSe$. Recall we have shown that G is a group. Then from the previous discussion $X = E(Se)$. $Y = E(eS)$ are left zero and right zero semigroups respectively.

Define $\psi : X \times G \times Y \to S \qquad (x,g,y) \mapsto xgy$. Suppose that $\psi(x_1,g_1,y_1) = \psi(x_2,g_2,y_2)$.

$$\begin{aligned} \text{Then} \qquad & x_1 g_1 y_1 = x_2 g_2 y_2 \\ \text{and} \qquad & e x_1 g_1 y_1 e = e x_2 g_2 y_2 e \\ y_i e = e,\ e x_i = e \qquad & e g_1 e = e g_2 e \\ \Rightarrow \qquad & g_1 = g_2. \end{aligned}$$

Also

$$\begin{aligned} & x_1 g y_1 = x_2 g y_2 \\ \Rightarrow\ & x_1 g y_1 e g^{-1} = x_2 g y_2 e g^{-1} \\ \Rightarrow\ & x_1 e = x_2 e \\ \Rightarrow\ & x_1 = x_2. \end{aligned}$$

Similarly, $y_1 = y_2$. Thus, ψ is one-to-one.

Now let $s \in S$. Associate with s the following element in $X \times G \times Y$

$$\begin{aligned} \alpha : & s \to X \times G \times Y \\ & s \mapsto \big[E(sSe), ese, E(eSs)\big] \\ & \quad = (e_1, ese, e_2) \end{aligned}$$

(well defined since $E(sSe)$, $E(eSs)$ are unique). Notice that $[(s)\alpha]\psi$ is the identity since

$$\begin{aligned} & e_1(ese)e_2 = e_1 s e_2 = s \\ \text{since} \quad & e_1 \in E(sS) \qquad \text{and acts like a left identity} \\ & e_2 \in E(Ss) \qquad \text{and acts like a left identity.} \end{aligned}$$

Thus, ψ is one-to-one and onto. $\square$

Now, let S be any finite semigroup and let $I = \{I_1, I_2, \ldots, I_n\}$ be the set of all ideals of S. Let $K = \bigcap_{j=1}^{n} I_j$. Then K is the unique minimal ideal of S. By construction K is a simple semigroup. To show K is completely simple we must show it contains a primitive idempotent. Now K is also a semigroup, and in particular is a left ideal, and a right ideal. Since there are only a finite number of left ideals possible, if we order them by inclusion there must exist minimal elements. Similarly, minimal right ideals exist.

Homework. Show that $RL = R \cap L$. Let e be the identity of RL. Suppose $f = f^2$ is such that $ef = fe = f \Rightarrow \mathscr{F} \in R \cap L \Rightarrow e = f$ by uniqueness.

Combining the lemma with the previous discussion we have:

Theorem 2.2. Let S be a finite semigroup. Then S has a minimal ideal K and K is a completely simple semigroup. K is often referred to as the kernel of S.

2.4 The Semigroup $\mathscr{F}_X$

As before, let $X = \{x_1, x_2, \ldots, x_n\}$ be any finite set, and let $\mathscr{F}_X = \{\mathscr{F} \mid f : x \to x\}$ with

$$x(f \circ g) = \big((x)f\big)g.$$

Now, for $f \in \mathscr{F}_X$, $R(f) = \{y \in X \mid f(x) = y \text{ for some } x \in X\}$. $R(f)$ is the range of f. $\pi(f) = \{(y)f^{-1} \mid y \in R(f)\}$ or $x_1 \equiv x_2$ iff $f(x_1) = f(x_2) = y$ where $(y)f^{-1} = \{x \in X \mid (x)f = y\}$. Then $\pi(f)$ is a partition of X and the cardinality of $\pi(f)$, $|\pi(f)| = |R(f)|$ is called the rank of f.

Now suppose $x_1 \equiv x_2$ under $\pi(f)$. Then

$$\begin{aligned} & (x_1)f = (x_2)f \\ \Rightarrow \quad & (x_1 f)g = (x_2 f)g \\ \Rightarrow \quad & x_1 \equiv x_2 \text{ under } \pi(f \circ g) \\ \Rightarrow \quad & \pi(f) \subset \pi(f \circ g). \end{aligned}$$

Thus we have the following.

Proposition 2.2. Let π_0 be an arbitrary partition of X. Let $R(\pi_0) = \{f \in \mathscr{F}_X \mid \pi_0 \subset \pi(f)\}$. Then $R(\pi_0)$ is a right ideal of $\mathscr{F}_X$.

Now, perhaps the most important elements in a semigroup are its idempotents.

Proposition 2.3. Let $f \in \mathscr{F}_X$. Then f is idempotent iff f is the identity map on $R(f)$.

Proof. Suppose $f^2 = f$ and let $y \in R(f)$. Then for some $x \in X$, $(x)f = y$.

Now,

$$\begin{aligned} & f^2 = f \\ \Rightarrow \quad & (x)f^2 = (x)f \\ \Rightarrow \quad & (xf)f = (x)f \\ \Rightarrow \quad & (y)f = y. \end{aligned}$$

Thus f is the identity on $R(f)$.

Now suppose f is the identity on its range and let $x \in X$. Then

$$(x)f \in R(f)$$

and

$$\big((x)f\big)f = (x)f$$

since f is the identity on $R(f)$

$$\Rightarrow \quad f^2 = f.$$

Thus if $f \in K'$,

$$\begin{aligned} \text{rank}(f \circ g) &= \text{rank}(f) \\ \text{rank}(g \circ f) &= \text{rank}(f) \end{aligned}$$

by minimality. Thus K' is an ideal of S. Now it suffices to show that $K' < K$.

Let $f \in K'$, $h \in K$. Then $fhf \in K' \cap K$. Notice $\text{range}(fhf) = \text{range}(f)$, by minimality, since $\text{range}(fhf) \subset \text{range}(f)$.

Now K is a completely simple semigroup. Let e be the local identity of the group containing fhf. Then $\text{range}(fhf) = \text{range}(fhfe) \subset \text{range}(e)$. Also, there is a local inverse element $g \ni gfhf = e$. Thus $\text{range}(e) = \text{range}(gfhf) \subset \text{range}(fhf)$. Thus $\text{range}(e) = \text{range}(f)$. But for all $x \in X$, $(x)f = (xf)e$ since e is the identity on its range.

$$\Rightarrow \quad f = fe \in K$$

since $e \in K$

$$\Rightarrow \quad K' \subset K$$

and we are done. □

Now let K be the kernel of some subsemigroup S of $\mathscr{T}_X$. Then since K is a completely simple set of functions it is a union of isomorphic groups. Can we describe those groups? Of course.

$$\begin{aligned} \bar{R} &= \{R_j \mid R(f) = R_j \text{ for some } f \in K\} \\ \prod &= \{\pi_i \mid \pi(f) = \pi \text{ for some } f \in K\} \end{aligned}$$

Now for $f \in K$, $f \in H$ for some subgroup H. In fact, $H \subset H(\pi(f), R(f)) \cong S_r$, $r = |\pi(f)| = |R(f)|$ where S_r is the symmetric group on r elements. Now if $\pi_i \in \prod$, $R_j \in \bar{R}$ there exists f with $\pi(f) = \pi_i$, g with $R(g) = R_j$. Then consider

$f \circ g$. Clearly

$$\pi(f \circ g) = \pi(f) = \pi_i \qquad R(f \circ g) = R(g) = R_j$$
$$\Rightarrow \quad f \circ g \in H(\pi_i, R_j).$$

Thus for every pair (π_i, R_j) there exists some function belonging to it from K. Thus every pair (π_i, R_j) is compatible.

Homework.

(a) Show that for each pair $(\pi_i, R_j) \exists f \in K \ni f \in H(\pi_i, R_j)$.
(b) Show that each pair (π_i, R_j) is compatible.

Thus

$$K = \bigcup_{\substack{\pi_i \in \prod \\ R_j \in R}} K(\pi_i, R_j) \subset \bigcup_{\substack{\pi_i \in \prod \\ R_j \in R}} H(\pi_i, R_j)$$

where $K(\pi_i, R_j)$ is some subgroup contained in $H(\pi_i, R_j) \cong S_r$, $r = |\pi_i| = |R_j|$.

Now suppose π, R are given with $|\pi| = |R|$. When is it possible to construct an idempotent, $e(\pi, R)$ with the given partition and range. It is possible precisely when

$$|P \cap R| = 1 \text{ for each } P \in \pi.$$

Thus we are led to the following definition. □

Definition 2.6. A partition π and range R will be called compatible iff $|P \cap R| = 1$ for all $P \in \pi$.

Notation. Let π, R be such that $|\pi| = |R|$. Denote $H(\pi, R) = \{f \in \mathscr{F}_X \mid \pi(f) = \pi,\ R(f) = R\}$.

Homework.

Proposition 2.4. $H(\pi, R)$ is a group iff π and R are compatible.

Proposition 2.5. Let (π_1, R_1), (π_2, R_2) be compatible. If (π_2, R_1) are compatible then $\mathrm{rank}(f) = \mathrm{rank}(f \circ g) = \mathrm{rank}(g)$ and $f \circ g \in H(\pi_1, R_2)$.

Proof. Clearly, since π_2, R_1 are compatible then $\mathrm{rank}(f) = \mathrm{rank}(g)$. Now $R(f \circ g) \subset R(g)$. Thus if we show that $R(f \circ g) = R(g)$ we are done. We must show that g/R_1 is one-to-one. Let $R_1 = \{j_1, j_2, \ldots\}$, $R_2 = \{k_1, k_2, \ldots, k_r\}$, and $\pi_2 = \{(k_1)g^{-1}, (k_2)g^{-1}, \ldots, (k_r)g^{-1}\}$. Now since π_2, R_1 are compatible, for each $|R_1 \cap (k_j)g^{-1}| = 1$ let $\{j_i\} = R_1 \cap (k_j)g^{-1}$ but then j_i is the unique element such that $(j_i)g = k_j$ and g/R_1 is one-to-one. □

Now we have the very important,

Theorem 2.3. Let S be a subsemigroup of $\mathscr{F}_X$. Then the kernel of S is precisely the set of elements of S with minimal rank.

Proof. Let $r = \min\{\text{rank}(f) \mid f \in S\}$.

Let $K' = \{f \in S \mid \text{rank}(f) = r\}$. We wish to show that K' is the minimal ideal of S.

Now for any $f, g \in \mathscr{F}_X$ it is true that

$$\begin{aligned} |(X)f \circ g| &\leq |(X)f| \\ &\leq |(X)g| \end{aligned} \qquad \square$$

Homework. Let $X = \{1,2,3,4\}$.

$$\pi_1 = \{(1,2),(3,4)\}$$
$$\pi_2 = \{(1,3),(2,4)\} \qquad S = \bigcup_{\substack{i=1,2 \\ j=1,2}} H(\pi_i, R_j)$$

$$R_1 = \{1,4\}$$
$$R_2 = \{2,3\}$$

Determine all possible Rees product structure for S.

2.5 Finite Semigroups and the Road Coloring Problem

A finite semigroup is a finite set, S, which is closed under a binary associative operation. For any subset T of S, we will write $E(T)$ to refer to the set of idempotents in T. Let $G = (V, \mathscr{E})$ be a strongly connected digraph with b vertices having adjacency matrix A. We will assume that the outdegree of each vertex of G or, equivalently, the sum of each row of A is some constant d.

Definition 2.7. A coloring of G is a decomposition of A into d binary (that is, $0-1$) stochastic matrices $R_1, R_2, \ldots, R_d$ such that $A = R_1 + \cdots + R_d$.

Intuitively, one can think of a matrix R_i as assigning the ith of d colors to the edge (j,k) if $(R_i)_{j,k} = 1$. Each $b \times b$ binary stochastic matrix R can also be thought of as a function on the vertices $\{1,2,\ldots,n\}$ where $R_{j,k} = 1$ iff $jR = k$. As a notational convention, writing functions on the right allows us to be indifferent to whether we consider $jR = k$ as a function mapping j to k or R as a binary stochastic matrix with j a row vector having a one in the jth position and zeros elsewhere.

Let $S = \langle R_1, R_2, \ldots, R_d \rangle$ be the semigroup generated by $R_1, R_2, \ldots, R_d$ under matrix multiplication (or composition). Clearly, S is a finite semigroup. We will refer to S as a coloring semigroup. The following theorem is extremely important in all that follows.

Theorem 2.4 ([7]). Let S be a finite semigroup. Then S contains a minimal ideal $\mathscr{K}$ called the kernel which is a disjoint union of isomorphic group. In fact, $\mathscr{K}$ is isomorphic to $\mathscr{X} \times \mathscr{G} \times \mathscr{Y}$ where if $e \in E(S)$, then $e\mathscr{K}e$ is a group and

$$\mathscr{X} = E(\mathscr{K}e) \qquad \mathscr{G} = e\mathscr{K}e \qquad \mathscr{Y} = E(e\mathscr{K})$$

and if (x_1, g_1, y_1), $(x_2, g_2, y_2) \in \mathscr{X} \times \mathscr{G} \times \mathscr{Y}$ then

$$(x_1, g_1, y_1)(x_2, g_2, y_2) = (x_1, g_1(y_1x_2)g_2, y_2)$$

The product structure $\mathscr{X} \times \mathscr{G} \times \mathscr{Y}$ is sometimes called a Rees product and any semigroup that has a Rees product is called completely simple. Thus the kernel of a finite semigroup is always completely simple.

The next theorem is extremely useful for characterizing the elements of the kernel. It is an elementary implication of a more general result by Clark [6].

Theorem 2.5. Let S be a finite semigroup of matrices. Then the kernel of S is the set of matrices with minimal rank.

Example 2.6. Let us reconsider the road coloring problem using the following example. Suppose the adjacency matrix of a graph is given by

$$A = \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

Let

$$R_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \qquad B_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad B_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Then $A = R_1 + B_1 = R_2 + B_2$.

Let $S_1 = \langle R_1, B_1 \rangle$ and $S_2 = \langle R_2, B_2 \rangle$. Notice that both R_2 and B_2 are permutation matrices, thus S_2 is a group and contains only matrices of rank three. However,

$$(B_1)^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore, the kernel of S_1 is a set of rank one matrices. In particular, $(B_1)^2$ is a synchronizing instruction. Following the blue road twice will take each person to building three. Thus we can make the following observations from this example:

(1) Rank one matrices (equivalently, constant functions) have a natural interpretation as synchronizing instructions
(2) Each graph will, in general, have many colorings and not every coloring will produce a synchronizing instruction.

The existence of a synchronizing instruction has ramifications for the "period" of a given graph. We first define the period of a vertex.

Definition 2.8. For each $v \in V$, define the period of v as

$$\text{per}(v) = \gcd\{|W| : W \text{ is a walk from } v \text{ to } v\}.$$

If the $\text{per}(v) = 1$, then v is called aperiodic. It is a well known fact in Markov chain theory that the period of a vertex is a so-called class property. That is, each vertex in a given strongly connected component (irreducible component, in Markov chain terminology) has the same period. Our assumption of strong connectedness for the graph allows us to speak of the period of the graph. The following theorem was originally proven in [1].

Theorem 2.6. If G has a synchronizing instruction, then G is aperiodic.

Proof. Let $S = \langle R_1, R_2, \ldots, R_d \rangle$ be a coloring semigroup for G that contains a synchronizing instruction $W = M_1 M_2 \ldots M_k$ where each M_i is one of the colors R_j. Then W is a rank one binary stochastic matrix with one column, say w, which is nonzero. Then for each vertex v in V, $vW = w$. In particular, W represents a walk of length k from w to itself. Suppose $u = wR_1$. Now $wR_1W = uW = w$, thus R_1W is a walk of length $k+1$ from w to itself. Since the period of w is $\gcd(k, k+1)$, G is aperiodic. □

Thus aperiodicity is a necessary condition for the existence of a synchronizing instruction. The road coloring problem seeks to determine if aperiodicity is also a sufficient condition.

Road Coloring Conjecture Let $G = (V, \mathscr{E})$ be a d-out, strongly connected digraph with adjacency matrix A. Let $\left\{\left\{R_{i_1}, R_{i_2}, \ldots, R_{i_d}\right\} : A = R_{i_1} + \cdots + R_{i_d}\right\}$ be the set of all colorings of G and let $S_i = \left\langle R_{i_1}, R_{i_2}, \ldots, R_{i_d}\right\rangle$ be the associated coloring semigroups. If G is aperiodic, then for some i, S_i contains a synchronizing instruction.

2.6 The Kernel of a Coloring Semigroup

Suppose $S = \langle R_1, R_2, \ldots, R_d \rangle$ is a coloring semigroup for some d-out digraph G. Even if G is not strongly connected, S will be a finite semigroup with kernel $\mathscr{K}$. As we stated in the last section, $\mathscr{K} = \mathscr{X} \times \mathscr{G} \times \mathscr{Y}$. Let M, N be elements

of $\mathscr{K}$. Then with respect to the Rees product structure $M = (M_1, M_2, M_3)$ and $N = (N_1, N_2, N_3)$. We will consider the structure of $\mathscr{K}M$, $N\mathscr{K}$, and $N\mathscr{K}M$:

(1) $\mathscr{K}M = \mathscr{X} \times \mathscr{G} \times \{M_3\}$ is a minimal left ideal in $\mathscr{K}$ whose elements all have the same range, or nonzero columns as M. This type of semigroup is also referred to as a left group.

(2) $N\mathscr{K} = \{N_1\} \times \mathscr{G} \times \mathscr{Y}$ is a minimal right ideal in $\mathscr{K}$ whose elements all have the same partition of the vertices as N. An element P_j in the partition can be assigned to each nonzero column of N by

$$P_j = \left\{ i : N_{i,j} = 1 \right\}$$

A semigroup like $N\mathscr{K}$ is usually called a right group.

(3) $N\mathscr{K}M$ is the intersection of $N\mathscr{K}$ and $\mathscr{K}M$ and is a maximal group in $\mathscr{K}$ (an H-class in the language of semigroups). In this instance, it is best thought of as a set of one-to-one functions specified by the partition of N and the range of M. The idempotent of $N\mathscr{K}M$ is the function which is the identity when restricted to the range of M.

The next result establishes the connection between the kernel and the property of strong connectedness.

Theorem 2.7. Let $S = \langle R_1, R_2, \ldots, R_d \rangle$ be any coloring semigroup for $G = (V, E)$. Let S have kernel $\mathscr{K}$ and let M be any element of $\mathscr{K}$. Then G is strongly connected iff for all w in V, $wM\mathscr{K} = V$.

Proof. Notice that $wM\mathscr{K} = V$ implies strong connectedness immediately, since this implies there is a path between any two vertices. Now, given w_1 in V, let M be any element in $\mathscr{K}$ and suppose $w_1 M = w_2$. Let w_3 be any vertex in V. There is a T in S such that $w_2 T = w_3$, since G is strongly connected. Thus $w_1 MT = w_3$. But MT is in $\mathscr{K}$ and has the same partition as M, since $\mathrm{rank}(M) \leq \mathrm{rank}(T)$. Thus MT is in $M\mathscr{K}$. □

Next we seek to demonstrate that it is the structure of the maximal groups in the kernel that determines whether a coloring semigroup contains a synchronizing instruction. This will follow easily from the ensuing result.

Theorem 2.8. Let u, v be arbitrary elements of V. If S is any coloring semigroup with kernel $\mathscr{K}$, then for each maximal group $\mathscr{G}$ in $\mathscr{K}$, $u\mathscr{G} = v\mathscr{G}$.

Proof. By the previous result, within each minimal right ideal $M\mathscr{K}$, there exists a T_1 such that $uT_1 = v$. Now T_1 belongs to the maximal group $T_1\mathscr{K}T_1 = \mathscr{G}_1 \subset M\mathscr{K}$.

For this group

$$v\mathscr{G}_1 = uT_1\mathscr{G}_1 = u\mathscr{G}_1.$$

If $\mathscr{G}_2$ is any other maximal group in $M\mathscr{K}$ and $T_2 \in \mathscr{G}_2$, then $\mathscr{G}_1 T_2 = \mathscr{G}_2$ and

$$v\mathscr{G}_2 = v\mathscr{G}_1 T_2 = u\mathscr{G}_1 T_2 = u\mathscr{G}_2.$$

But $M\mathscr{K}$ was an arbitrary minimal right ideal. Thus for every maximal group $\mathscr{G}$ in $\mathscr{K}$,

$$u\mathscr{G} = v\mathscr{G}. \qquad \square$$

Theorem 2.9. A coloring semigroup S with kernel $\mathscr{K}$ contains a synchronizing instruction iff for every idempotent E in $\mathscr{K}$, $E\mathscr{K}E = \{E\}$.

Proof. Clearly, if S contains a synchronizing instruction E, then $E = E^2$ is an idempotent. Since E is a rank one binary stochastic matrix, $E\mathscr{K}E = \{E\}$.

Now suppose for every idempotent E in $\mathscr{K}$, $E\mathscr{K}E = \{E\}$. Since E is an idempotent function, E is the identity when restricted to its range. Suppose E is not a constant function. Then there exist u, v in the range of E with $u \neq v$. But $uE = u \neq v = vE$. Since $\{E\}$ is a maximal group in $\mathscr{K}$, this contradicts the previous result. $\square$

The maximal groups in the kernel carry a great deal of information. In fact, we believe that the structure of the maximal groups in the kernels of certain coloring semigroups will provide the periodic structure of the graph. Making this idea precise will require the following definition of periodicity that can be shown to be equivalent to the previous one.

Definition 2.9. A digraph $G = (V, \mathscr{E})$ is periodic of period $t \geq 2$ iff there exists a partition of the vertices $\{P_1, P_2, \ldots, P_t\}$ such that if $(i, j) \in E$ and $i \in P_k$ then $j \in P_{k+1}$ ($i \in P_t$ implies $j \in P_1$). If no such partition exists, then G is aperiodic. In this case, $\pi = \{V\}$ is the "periodic" partition.

The cyclic nature of a periodic graph suggests the following generalization of the road coloring problem.

Conjecture 2.1. Let $G = (V, \mathscr{E})$ be a strongly connected d-out digraph. Then G is periodic of period $t \geq 1$ with periodic partition $\pi = \{P_1, \ldots, P_t\}$ iff G has a minimal (with respect to the rank of its kernel) coloring semigroup $S = \langle R_1, \ldots, R_d \rangle$ whose kernel $\mathscr{K}$ is a right group with partition π and whose maximal groups are all cyclic of order t.

Computer simulations provide evidence for the conjecture. In addition, a closely-related result in the context of Markov chains has been proven in a very interesting paper by Högnäs [11]. It should be mentioned that in Högnäs' result, (many) more than d colors are often required. Also note that in the aperiodic case, the set of constant functions would be the kernel. The constant functions are a right group with partition $\pi = \{V\}$ and whose maximal groups are all order one, thus trivially cyclic.

Chapter 3
Probability on Finite Semigroups

3.1 Random Walks on Semigroups

Now we will begin to do some probability.

Let $x_1, x_2, x_3, \ldots$ be a sequence of independent random variables taking values in a semigroup S.

(Recall that this means that there exists a measure space (Ω, P, B) where B consists of a sigma-algebra of P-measurable subsets of Ω (the measurable sets) and X_i is a measurable function from Ω into S.

Now each X_i induces a measure on S via P called the distribution of X_i. That is, for each X_i define a measure μ_i on S such that for any subset $T \subset S$

$$\mu_i(T) = P\{w \in \Omega \mid (w)X_i \in T\}.$$

Now given (X_k), define a random walk on S in the following way. Let

$$\begin{aligned} w_1 &= x_1 \\ w_2 &= x_1 \cdot x_2 \\ w_3 &= x_1 \cdot x_2 \cdot x_3 \\ &\vdots \\ w_n &= x_1 \cdot x_2 \cdot x_3 \cdots x_n \end{aligned}$$

Now each w_i, itself has a distribution, say ν_i, where $\nu_i(s) = P(w_i = s) = P(x_1 \cdot x_2 \cdots x_i = S)$. $P(w_i = s)$ always means $P\{w \mid w(w_i) = S\}$. Let's compute ν_2.

$$\nu_2(S) = P(w_2 = S) = P(x_1 \cdot x_2 = S)$$

Now, what is $P(x_1 \cdot x_2 = S)$

$$\begin{aligned}\nu_2(s) = P(x_1 \cdot x_2 = s) &= \sum_{t\in S} P(x_1 \cdot x_i = s \text{ and } x_2 = t) \\ &= \sum_{t\in S} P(x_1 \cdot x_i = s \mid x_2 = t)P(x_2 = t) \\ &= \sum_{t\in S} P(x_1 \cdot t = s \mid x_2 = t)P(x_2 = t) \\ &= \sum_{t\in S} P(x_1 \cdot t = s)P(x_2 = t) \\ &= \sum_{t\in S} \mu_1(st^{-1})\mu_2(t)\end{aligned}$$

where $st^{-1} = \{x \in S \mid xt = s\}$. Notice that if S is a group, st^{-1} is a single element.

We will call $\sum \mu_1(st^{-1})\mu_2(t)$ the convolution of μ_1, μ_2 and denote it as $\mu_1 * \mu_2$.

Homework.

1) Show that $\mu_1 * \mu_2(s) = \sum_{t\in S} \mu_2(t^{-1}s)\mu_1(t)$.
2) $\mu_1 * \mu_2(s) = \sum_{S_1S_2=s} \mu_1(s_1)\mu_2(s_2)$.
3) Show that $\mu_1 * \mu_2 \in \text{Prob}(S)$.
4) Show that $\mu_1 * (\mu_2 * \mu_3) = (\mu_1 * \mu_2) * \mu_3$.

Notice that 3,4 together show that $(\text{Prob}(S), *)$ is a semigroup. In fact, $(\text{Prob}(S), *)$ is a topological semigroup.

Definition 3.1. Let $(S, \cdot)$ be a semigroup with S a topological space as well. Then S is a topological semigroup if the mapping

$$\alpha : S \times S \to S \qquad \alpha(s,t) = s \cdot t$$

is continuous. Here $S \times S$ is given the product topology.

Example 3.1. Let $(R, +)$ be endowed with the standard topology $[d(x,y) = |x-y|]$. Then $(R, +)$ is a topological group. Since there is a countable basis for the topology we can use sequence to detect limit points. Thus $(x,y) \mapsto x+y$ being continuous is equivalent to

$$x_n \to x, \quad y_n \to y$$

then

$$x_n + y_n \to x + y$$
$$|x_n + y_n - (x+y)| \le |x_n - x| + y_n - y|.$$

Example 3.2. $S = [0,1]$ under multiplication, again under the standard (subspace) topology.

$$(x,y) \mapsto x \cdot y$$

$$\begin{aligned}|x_n y_n - xy| &= |(x_n - x)(y_n - y) + x(y_n - y) + y(x_n - x)| \\ &\leq |(x_n - x)(y_n - y)| + |x||y_n - y| + |y||x_n - x|\end{aligned}$$

$$(x_n - x)(y_n - y) = x_n y_n - x y_n - y x_n + xy + xy_n + y x_n - 2xy$$

Now let S be a two point semigroup and let $\mu \in \mathrm{Prob}(S)$. Let $S = \{s_1, s_2\}$.

$$\mu(s_1) + \mu(s_2) = 1$$

	s_1	s_2
s_1	s_1	s_2
s_2	s_1	s_2

	s_1	s_2
s_1	s_1	s_2
s_2	s_2	s_1

$$\mu_1 * \mu_2(s_1) = \sum_t \mu_1(s, t^{-1})\mu_2(t) = \mu_1\{s\}\mu_2(s_1) = \mu_2(s_1)$$

$$s_1 t^{-1} = \{y \mid yt = s_1\} = \begin{cases} \{s_1, s_2\} & t = s_1 \\ \emptyset & t = s_2 \end{cases}$$

$$s_2 t^{-1} = \begin{cases} (s_1, s_2) & t = s_2 \\ \emptyset. \end{cases}$$

Thus

$$\mu_1 * \mu_2 = \mu_2.$$

$$\begin{aligned} \mu_1(s_1) &= p_{11} & \mu_2(s_1) &= p_{21} \\ \mu_1(s_2) &= p_{12} & \mu_2(s_2) &= p_{22} \end{aligned}$$

$$\begin{aligned} \mu_1 * \mu_2(s_1) &= \mu_1(s_1)\mu_2(s_1) + \mu_1(s_2)\mu_2(s_2) \\ &= p_{11}p_{21} + p_{12}p_{22} \\ \mu_1 * \mu_2(s_2) &= \mu_1(s_1)\mu_2(s_2) + \mu_1(s_2)\mu_2(s_1) \\ &= p_{11}p_{22} + p_{12}p_{21} \end{aligned}$$

$$\begin{aligned} \mathrm{Ent}(\mu) &= -\sum_{i=1}^{n} p_i \ln p_i \\ \mathrm{Ent}(\mu_1) &= -(p_{11}\ln p_{11} + p_{12}\ln p_{12}) \\ \mathrm{Ent}(\mu_2) &= -(p_{21}\ln p_{21} + p_{22}\ln p_{22}) \\ \mathrm{Ent}(\mu_1 * \mu_2) &= (p_{11}p_{21} + p_{12}p_{22})\ln(p_{11}p_{21} + p_{12}p_{22}) \\ &\quad + (p_{11}p_{22} + p_{12}p_{21})\ln(p_{11}p_{22} + p_{12}p_{21}) \end{aligned}$$

Example 3.3. $\mu_1 = \mu_2 = \left(\frac{1}{4}, \frac{3}{4}\right)$

$$\mu_1 * \mu_2 = \left(\left(\frac{1}{4}\right)^2 + \left(\frac{3}{4}\right)^2,\ 2\left(\frac{1}{4}, \frac{3}{4}\right) \right)$$
$$= \left(\frac{10}{10}, \frac{6}{10}\right) = \left(\frac{5}{8}, \frac{3}{8}\right)$$

Graph of a Semigroup Digraph induced by right translation

$$\text{If } s_1 s_j = s_k \text{ then } (s_i, s_k) \text{ has edge labeled } s_j$$

for two element group

right zero semigroup

Then if vertices s_1, s_2 have "energy" $\mu_1(s_1), \mu_1(s_2)$ and μ_2 represents the proportion of the energy transferred over the edges, then $\mu_1 * \mu_2(s_i)$ is the energy that ends up at vertex (s_i).

3.2 Semigroup Algebras

Let $S = \{s_1, s_2, \ldots, s_n\}$ be a finite semigroup. Let $A(S) = \{a_1 s_1 + a_2 s_2 + \cdots + a_n s_n \mid a_i \in \mathbb{F}\}$ where $\mathbb{F}$ is some field of scalars (usually $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$). Then define the following operations. Let

$$v_1 = a_1 s_1 + \cdots + a_n s_n$$
$$v_2 = b_1 s_1 + \cdots + b_n s_n.$$

Then

(1) $v_1 + v_2 = (a_1 + b_1)s_1 + \cdots + (a_n + b_n)s_n$.
(2) for $c \in \mathbb{F}$, $cv_1 = (ca_1)s_1 + (ca_2)s_2 \cdots (ca_n)s_n$.
(3) $v_1 \cdot v_2 = (a_1 s_1 \cdots a_n s_n)(b_1 s_1 + \cdots + b_n s_n)$

$$= \sum_{i,j=1}^{n} a_i b_j (s_i \cdot s_j)$$

where $s_i \cdot s_j$ occurs in the semigroup. Notice

$$v_1 \cdot v_2 = \sum_{i,j=1}^{n} a_i b_j (s_i \cdot s_j)$$
$$= \sum_{k=1}^{n} c_k s_k$$

where

$$c_k = \sum_{\substack{i,j \\ s_i \cdot s_j = s_k}} a_i b_j.$$

Example 3.4. Let $G = \{e, g_1, g_2\}$ with $g_1^2 = g_2$, $g_1^3 = e$ be the three element cyclic group. Then $A(G) = \{a_0 e + a_1 g_1 + a_2 g_2\}$.

$$v_1 = a_0 e + a_1 g_1 + a_2 g_2$$
$$v_2 = a_0 e + b_1 g_1 + b_2 g_2$$

$$\begin{aligned} v_1 \cdot v_2 = {} & a_0 b_0 e^2 + a_0 b_1 e g_1 + a_0 b_2 e g_2 \\ & + a_1 b_0 g_1 e + a_1 b_1 g_1^2 + a_1 b_2 g_1 g_2 \\ & + a_2 b_0 g_2 e + a_2 b_1 g_2 g_1 + a_2 b_2 g_2^2 \\ = {} & (a_0 b_0 + a_1 b_2 + a_2 b_1)e + \ldots \end{aligned}$$

Now let $P \subset A(S)$ be defined by

$$P = \left\{ a_1 s_1 + \cdots + a_n s_n \mid \sum_{i=1}^{n} a_i = 1 \right\}.$$

Notice if $v_1 = a_1 s_1 + \cdots + a_n s_n,\ v_2 = b_1 s_1 + \cdots + b_n s_n$ are elements of P then since

$$v_1 \cdot v_2 = \sum_{i,j=1}^{n} a_i b_j (s_i \cdot s_j)$$

then $v_1 \cdot v_2 \in P$ because

$$\sum_{i,j=1}^{n} a_i b_j = \left(\sum_{i=1}^{n} a_i \right) \left(\sum_{j=1}^{n} b_j \right) = 1 \cdot 1 = 1.$$

Now let $\psi : \mathrm{Prob}(S) \to A(S)$

$$\mu \mapsto \mu(s_1)s_1 + \mu(s_2)s_2 + \cdots + \mu(s_n)s_n$$
$$\mu(e) = \frac{1}{2}, \quad \mu(g_1) = \frac{1}{4}, \quad \mu(g_2) = \frac{1}{4}$$
$$\mu \mapsto \frac{1}{2}e + \frac{1}{4}g_1 + \frac{1}{4}g_2.$$

If $\mu_1 \neq \mu_2$ then for some $s_i,\ \mu_i(s_i) \neq \mu_2(s-i)$

$$\Rightarrow \quad \psi(\mu_1) \neq \psi(\mu_2)$$

Thus ψ is one to one. Let $v \in P,\ v = a_1 s_1 + \cdots + a_n s_n,\ \sum_{i=1}^{n} a_i = 1$. Then if $\mu(s_1) = a_1 \ldots \mu(s_n) = a_n,\ \mu \in \mathrm{Prob}(S)$ and $\psi(\mu) = v$. Thus ψ is a one to one and onto P.

Now since $\mu_1, \mu_2 \in \mathrm{Prob}(S)$ and

$$\mu_1 * \mu_2(S) = \sum_{s_i s_j = S} \mu_1(s_i)\mu_2(s_j)$$

then

$$\psi(\mu_1 * \mu_2) = [\mu_1 * \mu_2(s_1)]s_1 + [\mu_1 * \mu_2(s_2)]s_2 + \cdots + [\mu_1 * \mu_2(s_n)]s_n.$$

Now let $B = [B_{ij}]$ be an $n \times n$ matrix. Then $\hat{T}'(B)$ is the set of decompositions of B in the semigroup algebra. We will also call the formal sum,

$$a_1 T(s_1) + \cdots + a_n T(s_n)$$

a decomposition of B, when $T : A(S) \to M[n]$ a representation.

The following Theorems are relevant:

Theorem 3.1 (Birkoff). Let B be an $n \times n$ doubly stochastic matrix. Then there exists a unique convex decomposition of B into permutation matrices.

Theorem 3.2. Let B be a stochastic matrix. Then there exists a convex decomposition of B into binary stochastic matrices.

Now if $T : S \to M[n]$ is a semigroup representation of S by $n \times n$ matrices. Then for $v \in A(S)$, $v = a_1 s_1 + \cdots + a_n s_n$, define $\hat{T}$ by $\hat{T} : A(S) \to M[n]$

$$\hat{T}(v) = a_1 T(s_1) + \cdots + a_n T(s_n).$$

Then $\hat{T}[\psi(\mu)]$ is an $n \times n$ matrix.

Homework. However, even if T is a semigroup isomorphism, $\hat{T}$ may not be an algebra isomorphism. $\hat{T}$, however, is always a homomorphism.

Example 3.5.

$S = \mathscr{F}_X \quad X = \{1,2\}$
$S = \{f_1, f_2, f_3, f_4\}$

	f_1			f_2	
1	$\to$	1	1	$\searrow$	1
2	$\to$	2	2	$\nearrow$	2

	f_3			f_4	
1	$\to$	1	1	$\searrow$	1
2	$\nearrow$	2	2	$\to$	2

Then $T : S \to M[2]$.

$$v_1 = \frac{1}{2} f_1 + \frac{1}{2} f_2$$
$$v_2 = \frac{1}{2} f_3 + \frac{1}{2} f_4$$
$$\hat{T}(v_1) = \frac{1}{2} T(f_1) + \frac{1}{2} T(f_2)$$
$$= \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

etc.

3.3 Convergence of Convolution Powers

Start with definition of topological semigroup. Given countable discrete example.

Now consider the case when (X_k) is an i.i.d. sequence of random variables. Thus $\mu_j = \mu$ (say) for all j.

Suppose

$$\mu^n \to \nu \text{ as } n \to \infty.$$

(This is what we mean by stochastic stability.) Then $\mu * \mu^n = \mu^{n+1}$, letting $n \to \infty$

$$\begin{aligned}
&\Rightarrow \quad \mu * \nu = \nu \\
&\qquad \text{using the continuity of convolution} \\
&\Rightarrow \quad \mu^k * \nu = \nu \text{ for all } k \\
&\Rightarrow \quad \nu * \nu = \nu.
\end{aligned}$$

Thus ν is an idempotent measure.

Thus idempotents will play a key role in our study of the convergence of convolution powers of a probability measure.

One of the most important concepts is that of the support of a measure.

Definition 3.2. Let $\mu \in \text{Prob}(S)$. Then $S(\mu) = \{y \in S \mid \mu(y) > 0\}$. $S(\mu)$ is called the support of μ.

Proposition 3.1. Let $\mu, \nu \in \text{Prob}(S)$. Then

$$S(\mu * \nu) = S(\mu) \cdot S(\nu).$$

Proof. Let $y \in S(\mu * \nu)$

$$\Rightarrow \quad \mu * \nu(y) > 0$$

but

$$\mu * \nu(y) = \sum_{s_1 s_2 = y} \mu(s_1)\nu(s_2) > 0$$

$$\begin{aligned}
&\Rightarrow \quad \mathscr{F} s_1, s_2 \text{ with } \mu(s_1)\nu(s_2) > 0 \\
&\Rightarrow \quad s_1 \in S(\mu),\ s_2 \in S(\nu) \\
&\Rightarrow \quad y = s_1 \cdot s_2 \in S(\mu) \cdot S(\nu).
\end{aligned}$$

□

Homework. Prove the converse.

Notice the above proposition. Show that if $\mu^2 = \mu$, $S(\mu) = S(\mu * \mu) = S(\mu) \cdot S(\mu)$. Thus the support of an idempotent measure is a semigroup.

Question. What kind of semigroup??

Notice the following. Let $f : s \to \mathbb{R}$ be any real-valued function defined on S and suppose $\nu = \nu * \mu$. Then for each $z \in S$,

$$\nu(z) = \sum_{st=z} \nu(s)\mu(t).$$

Thus

$$\begin{aligned}
f(z)\nu(z) &= f(z) \sum_{st=z} \nu(s)\mu(t) \\
\sum_{t \in S} f(z)\nu(z) &= \sum_{z \in S} \left[\sum_{st=z} f(z)\nu(s)\mu(t) \right] \\
&= \sum_{z \in S} \left[\sum_{st=z} f(st)\nu(s)\mu(t) \right]
\end{aligned}$$

let

$$G(z) = \{(s,t) \mid st = z\} \subset S \times S$$
$$\bigcup_{z \in S} G(z) = S \times S.$$

Notice $G(z_1) \cap G(z_2) = \emptyset$, $z_1 \neq z_2$. But $(s_1, s_2) \in S \times S$ then if $s_1 s_2 = s$, $(s_1, s_2) \in G(s)$. Clearly, $\bigcup G(z) \subset S \times S$. Thus,

$$\begin{aligned}\sum_{z \in S} f(z)\nu(z) &= \sum_{z \in S} \left[\sum_{(s,t) \in G(z)} f(st)\nu(s)\mu(t) \right] \\ &= \sum_{S \in S} \sum_{t \in S} f(st)\nu(s)\mu(t).\end{aligned}$$

In fact, this is how convolution is defined in the general case via the Riesz representation theorem.

Lemma 3.1. Let $\mu, \nu \in \mathrm{Prob}(S)$ such that $\nu = \nu * \mu = \mu * \nu$. Then for any $B \subset S$, $x \in S(\nu)$, $y \in S(\mu)$,

$$\nu(Bx^{-1}y^{-1}) = \nu(Bx^{-1})$$
$$\nu(y^{-1}x^{-1}B) = \nu(x^{-1}B).$$

Proof. Let $x \in S(\nu)$, $B \subset S$

$$\nu(Bx^{-1}) = a$$
$$A = \{s \in S \mid \nu(Bs^{-1}) \leq a\}.$$

Now $x \in A \Rightarrow \nu(A) > 0$.

Define $g(s) = \max(\nu(Bs^{-1}) - a, 0)$. Then $\sum_t \left[\sum_s g(st)\mu(s) - g(t) \right] \nu(t) = 0$. Now

$$\begin{aligned}\nu(Bt^{-1}) - 1 &= \sum_s \left[\nu(Bt^{-1}s^{-1}) - a \right] \mu(s) \\ &= \sum_j \nu[B(st)^{-1} - a]\mu(s) \\ &\leq \sum_s g(st)\mu(s)\end{aligned}$$

but

$$\sum_s g(st)\mu(s) \geq 0.$$

Thus

$$\max\left(\nu(Bt^{-1}) - a, 0\right) \leq \max\left(\sum_s g(st)\mu(s), 0 \right)$$
$$\Rightarrow \quad g(t) \leq \sum_s g(st)\mu(s).$$

Thus

$$g(t) = \sum_s g(st)\mu(s), \quad t \in S(\nu).$$

Now for $t \in A$, $g(t) = 0$

$$\begin{aligned} &\Rightarrow \quad g(st) = 0 \text{ for } s \in S(\mu), \quad t \in A \cap S(\nu) \\ &\Rightarrow \quad \nu\left[B(t^{-1})s^{-1}\right] \leq a, \quad t \in A \cap S(\nu),\ S \in S(\mu). \end{aligned}$$

Now $x \in A \cap S(\nu)$, $\nu(Bx^{-1}) = a$

$$\Rightarrow \quad \nu(Bx^{-1}s^{-1}) \leq \nu(Bx^{-1}) \text{ for all } s \in S(\mu).$$

Now since $\nu = \nu * \mu$

$$\nu(Bx^{-1}) = \sum_{s \in S(\mu)} \nu(Bx^{-1}y^{-1})\mu(y)$$

$$\sum_y \left[\nu(Bx^{-1}) - \nu(Bx^{-1}y^{-1})\right]u(y) = 0 \Rightarrow \nu(Bx^{-1}) = \nu(Bx^{-1}y^{-1}) \qquad \square$$

Now let w_G be the uniform measure on a finite group G, $|G| = n$. Let $x \in G$ and consider the delta measure at x, δ_X and let $B \subset G$. Then

$$\begin{aligned} w_G * \delta_X(B) &= \sum w_G(Bh^{-1})\delta_X(h) \\ &= w_G(Bx^{-1}) \\ &= |B|\frac{1}{n} \\ &= w_G(B). \end{aligned}$$

We want to generalize this notion of translation invariance to semigroups.

Definition 3.3. Let $\mu \in \text{Prob}(S)$. Then μ is called r^*-invariant if for $B \subset S$, $\mu = \mu * \delta_X\left[\mu(B) = \mu(Bx^{-1})\right]$ if $\mu = \delta_X * \mu$.

Homework. Show that if μ is r^*-invariant or 1^*-invariant, then $\mu^2 = \mu$.

Notice that if μ is r^*-invariant then

$$S(\mu) = S(\mu * \delta_X) = S(\mu) \cdot x.$$

1^*-invariant

$$S(\mu) = S(\delta_X * \mu) = x \cdot S(\mu).$$

Lemma 3.2. If μ is r^*-invariant, then for $x \in S(\mu)$, $x \cdot S(\mu)$ is left cancellative ($zx = zy \Rightarrow x = y$).

Proof. Let μ be a r^*-invariant measure on S. Then for any $s \in S(\mu)$, $S(\mu) \cdot s = S(\mu)$. Take arbitrary elements in $x - S(\mu)$, xy, xz, xw.

Suppose $(xy)(xz) = (xy)(xw)$. Now $S(\mu) \cdot (xyx) = S(\mu)$. Thus $t(xyx) = x$ for some $t \in S(\mu)$. Then,

$$t(xyx)z = t(xyx)w$$
$$\Rightarrow \quad xz = xw$$

Thus $x \cdot S(\mu)$ is left-cancellative. $\square$

Homework. Show that if μ is 1^*-invariant $S(\mu) \cdot x$ is right-cancellative.

Lemma 3.3. Let $\mu \in \text{Prob}(S)$ be both r^* and 1^*-invariant. Then $S(\mu)$ is a group and μ is the uniform measure on $S(\mu)$.

Proof. Since $\mu = \mu^2$, then $S(\mu) = S(\mu) \cdot S(\mu)$ and $S(\mu)$ is a semigroup. Now for $x \in S(\mu)$,

$$S(\mu) \cdot x = S(\mu) \text{ by } r^*\text{-invariance}$$
$$x \cdot S(\mu) = S(\mu) \text{ by } 1^*\text{-invariance.}$$

Thus $S(\mu)$ is both right and left simple and is thus a group.

Let $y, z \in S(\mu)$ and suppose $z \in yx^{-1}$ for $x \in S(\mu)$. Since $\mu = \mu * \delta_X$

$$\begin{aligned}\mu(y) &= \mu * \delta_X(y) = \mu(yx^{-1}) \\ &= \mu(yx^{-1} \cap S(\mu)) = \mu(z)\end{aligned}$$

where $\{z\} = yx^{-1} \cap S(\mu)$ is a single element since $S(\mu)$ is a group.

Thus $\mu(y) = \mu(z)$ for any pair $y, z \in S(\mu)$ which show μ is uniform on $S(\mu)$.

$\square$

Theorem 3.3. Suppose for $\mu \in \text{Prob}(S)$, $\mu = \mu^2$. Then $S(\mu)$ is a completely simple semigroup of S.

Proof. Now $S(\mu) \cdot S(\mu) = S(\mu)$ (since $\mu * \mu = \mu$). Thus

$$S(\mu) = \bigcup_{x \in S(\mu)} S(\mu) \cdot x.$$

If we can show that each $S(\mu) \cdot x$ is a minimal left ideal, then $S(\mu)$ will be completely simple.

Now, anytime $S \cdot S = S$, $S = \bigcup_{x \in S} S \cdot x$. But if S is not completely simple, then for some $x \in S$, $S \cdot x$ is not minimal.

Now, to show that $S(\mu) \cdot x$ is left simple we need to show that $(S(\mu) \cdot x) \cdot L = S(\mu) \cdot x$ for all subsets $L \subset S(\mu) \cdot x$. Clearly, it is sufficient to show this for each element $z \in S(\mu) \cdot x$.

Call $\mu * \delta_x = \mu_x$ and let $z \in S(\mu) \cdot x$. Then $z = y \cdot x$ for some $y \in S(\mu)$.

$$\begin{aligned}\mu * \delta_x * \delta_2(B) = \mu * \delta_x(Bz^{-1}) = \mu_x(Bz^{-1}) &= \mu_x(B(y\cdot x)^{-1}) \\ &= \mu_x B(x^{-1}y^{-1}) \\ &= \mu\, B(x^{-1}y^{-1}x^{-1}) \\ &= \mu\left[Bx^{-1}(y)^{-1}\right] \\ \text{by } xy \in S(\mu) \\ &= \mu(Bx^{-1}) = \mu_x(B) = \mu * \delta_x(B).\end{aligned}$$

Thus

$$\begin{aligned}&\mu * \delta_x * \delta_z = \mu * \delta_x \\ \Rightarrow\quad &S(\mu) * \delta_x * \delta_z = S(\mu * \delta_x) \\ \Rightarrow\quad &[S(\mu)\cdot x]\cdot z = S(\mu)\cdot x\end{aligned}$$

for all $z \in S(\mu)\cdot x$

$$\Rightarrow \quad S(\mu)\cdot x \text{ is left simple.}$$

Thus each $S(\mu)\cdot x$ is a minimal left ideal and

$$S(\mu) = \bigcup_{x\in S(\mu)} S(\mu)\cdot x$$

is a union of minimal left ideals and is completely simple. □

Theorem 3.4. Let $\mu^2 = \mu$ and let $S(\mu) \cong x \times G \times Y$. Then there exists $\mu_1 \in \text{Prob}(X)$, $\mu_2 \in \text{Prob}(G)$, $\mu_3 \in \text{Prob}(Y)$ such that $\mu = \mu_1 * \mu_2 * \mu_3$ where μ_2 is the uniform measure on G. Conversely, if $\mu_1, \mu_3 \in \text{Prob}(S)$ and μ_2 is the uniform measure on a subgroup G of S, with $S(\mu_3)\cdot S(\mu_1) \subset G$, then $\mu_1 * \mu_2 * \mu_3$ is idempotent.

Proof. Now, by the previous theorem, $S(\mu)$ is completely simple. Let $e = e^2 \in S(\mu)$. Consider $\delta_e * \mu * \delta_e$. By the previous theorem,

$$(\delta_e * \mu * \delta_e) * \delta_{ese} = \delta_e * \mu * \delta_e$$

and

$$\delta_{ese} * (\delta_e * \mu * \delta_e) = \delta_e * \mu * \delta_e.$$

Thus $\delta_e * \mu * \delta_e$ is both r^*- and ℓ^*-invariant, and thus is the uniform measure on its support, $e\cdot S(\mu)\cdot e$.

Also, notice that for $B \subset S$,

$$\begin{aligned}\mu * \delta_e * \mu(B) &= \sum \mu * \delta_e(Bx^{-1})\mu(x) \\ &= \sum \mu(Bx^{-1}e^{-1})\mu(x) \\ &= \sum \mu(Bx^{-1})\mu(x) \\ &= \mu * \mu(B) = \mu(B).\end{aligned}$$

Thus $\mu * \delta_e * \mu = \mu$ and $(\mu * \delta_e) * (\delta_e * \mu) = \mu$. Now, let $X = E[S(\mu) \cdot e]$, $G = e \cdot S(\mu) \cdot e$, $Y = E[e \cdot S(\mu)]$.

Define $\mu_1(A) = \mu * \delta_e\big(\Phi\big[A \times G \times \{e\}\big]\big)$. Notice $\mu_1 \in \mathrm{Prob}(X)$ and $\mu_2(B) = \mu * \delta_e\big(\Phi\big[X \times B \times \{e\}\big]\big)$. Now $A \subset X = E(S(\mu) \cdot e)$ and $S(\mu) = S(\mu * \delta_e) = S(\mu) \cdot e$. Also $B \subset G = eS(\mu)e \subset E[S(\mu) \cdot e] \cdot eS(\mu)e \subset S(\mu) \cdot e$.

Thus both μ_1, μ_2 are r^*-invariant, but since $S(\mu_2) = G$, then μ_2 must be the Haar measure on G. If $A \subset X$, $B \subset G$,

$$\begin{aligned}\mu_1 * \mu_2(AB) = \mu_1 * \mu_2(A \cdot B \cdot e) &= \sum_{y \in G} \mu_1\big(\{ABe\}y^{-1}\big)\mu_2(y) \\ &= \sum_{y \in G} \mu_1\big[\{ABe\}y^{-1} \cap X\big]\mu_2(y),\end{aligned}$$

but $\{ABe\}y^{-1} = \{z \mid zy \in ABe,\ y \in G\}$. If $zy \in ABe$, $z \in X$, $y \in G$, then $zye \in ABe$. However, recall that $\Phi : X \times G \times Y \to X \cdot G \cdot Y$ is an isomorphism. Since $zye \in X \cdot G \cdot Y$

$$\Phi^{-1}(zye) = (z, y, e)$$

and for each $y \in B$, $\big[\{ABe\}y^{-1}\big] = A$ and otherwise $[ABe]y^{-1} = \emptyset$. Thus

$$\begin{aligned}\mu_1 * \mu_2(AB) &= \sum_{y \in G} \mu_1\big[\{ABe\}y^{-1}\big]\mu_2(y) \\ &= \sum_{y \in B} \big[\mu_1\{ABe\}y^{-1}\big]\mu_2(y) \\ &\qquad + \sum_{y \in G - B} \big(\mu_1\{ABe\}y^{-1}\big)\mu_2(y) \\ &= \mu_1(A) \sum_{y \in B} \mu_2(y) \\ &= \mu_1(A)\mu_2(B).\end{aligned}$$

Similarly, if $\mu_3(C) = \delta_e * \mu\big(\Phi[e \times G \times C]\big)$ for $C \subset E[e \cdot S(\mu)] = Y$, $\mu_2 * \mu_3(BC) = \mu_2(B)\mu_3(C)$.

Since $\mu * \delta_e$ is r^*-invariant, it is idempotent and $(\mu * \delta_e) * (\mu * \delta_e) = \mu * \delta_e$. Thus if A, B are as above,

$$\begin{aligned}\mu * \delta_e\big(\Phi[A \times B]\big) &= (\mu * \delta_e) * (\mu * \delta_e)\big[\Phi(A \times B)\big] \\ &= \sum_{x \in S(\mu) \cdot e} \mu * \delta_e\big(x^{-1}\Phi[A \times B \times \{e\}]\big)\mu * \delta_e(x).\end{aligned}$$

Now

$$\Phi^{-1}[S(\mu) \cdot e] = \Phi^{-1}[XG(Y \cdot e) \cdot e] = X \times G \times \{e\}.$$

Thus

$$x \in S(\mu) \cdot e \Rightarrow \Phi^{-1}(x) \in X \times G \times \{e\}.$$

Also

$$X^{-1}\Phi[A \times B \times \{e\}] = \{z \mid xz \in \Phi[A \times B \times e]\}$$

$$\begin{aligned}
&\Rightarrow \quad \Phi^{-1}(xz) \in A \times B \times \{e\} \\
&\Rightarrow \quad \Phi^{-1}(x)\Phi^{-1}(z) \in A \times B \times \{e\} \\
&\Rightarrow \quad (x_1, g_1, e)(z_1, z_2, z_3) \in A \times B \times \{e\} \\
&\Rightarrow \quad (x_1, g_1, (ez_1)z_2, z_3) \in A \times B \times \{e\} \\
&\Rightarrow \quad x_1 \in A, \quad z_2 \in g_1^{-1}B.
\end{aligned}$$

Finally,

$$\begin{aligned}
\mu * \delta_e\big(\Phi[A \times B]\big) &= \sum_{x \in S(\mu)\cdot e} \mu * \delta_e\big(x^{-1}\Phi[A \times B \times \{e\}]\big)\mu * \delta_e(x) \\
&= \sum_{(x_1,g_1,e)\in A\times G\times\{e\}} \mu * \delta_e\big(\Phi[X \times g_1^{-1}B \times \{e\}]\big)\mu * \delta_e(x)\big[\Phi(x_1, g_1, e)\big] \\
&= \sum_{(x_1,g_1,e)\in A\times G\times\{e\}} \mu_2(g_1^{-1}B)\mu * \delta_e\big[\Phi(x_1, g_1, e)\big] \\
&= \mu_2(B) \sum_{(x_1,g_1,e)\in A\times G\times\{e\}} \mu * \delta_e\big[\Phi(x_1, g_1, e)\big] \\
&= \mu_1(A)\mu_2(B).
\end{aligned}$$

Now since $\mu_1 * \mu_2(AB) = \mu_1(A)\mu_2(B)$ and $\mu * \delta_e(AB) = \mu_1(A)\mu_2(B)$. Then $\mu_1 * \mu_2 = \mu * \delta_e$.

Similarly, $\mu_2 * \mu_3 = \delta_e * \mu$.

$$(\mu_1 * \mu_2 * \mu_3) = (\mu_1 * \mu_2) * (\mu_2 * \mu_3) = (\mu * \delta_3) * (\delta_e * \mu) = \mu. \qquad \square$$

Midterm.

$$(\mu_1 * \mu_2 * \mu_3(ABC)) = \mu_1(A) * \mu_2(B) * \mu_3(C)$$

3.4 Trees, unicycles, and the stationary distribution

For an irreducible equally weighted two-out graph, count parallel outgoing edges separately. Let s_i count in-directed spanning trees rooted at i, and put $S = \sum_i s_i$. The Markov chain tree theorem gives $\pi_i = s_i/S$. Let $\nu(C)$ count spanning functional subgraphs having the single limiting cycle C, including their attached trees and selected edge multiplicities. Adding either outgoing edge at the root of an i-tree, or deleting the edge out of i on a unicycle containing i, gives the bijection

$$2s_i = \sum_{C \ni i} \nu(C).$$

Hence

$$2S = \sum_C |C|\nu(C), \qquad \pi = \sum_C \frac{|C|\nu(C)}{2S}\frac{\chi_C}{|C|}.$$

This decomposes stationary mass into uniform cycle measures. The weights refer to complete spanning unicycles, not only bare directed cycles. It is a one-particle identity; it does not by itself identify the kernel partition/range measure. For weighted ladders, use weighted tree counts or edge-counting conventions appropriate to the probabilities.

Part II
Kernel geometry and coloring

Chapter 4
Tensor Squares and Equipartition

The tensor-square proof establishes equipartition before the expectation operators are introduced.

4.1 Setup and statement

Let $Q = \{1, \dots, d\}$, and let R and B be transformation matrices on Q: every row contains exactly one entry equal to 1, all other entries being zero. We use right-action composition,

$$x(fg) = (xf)g.$$

Thus a transformation f has matrix entry $f_{xy} = 1$ exactly when $xf = y$. Its matrix rank equals $|\operatorname{im} f|$.

Fix $0 < p < 1$ and set

$$T = pR + (1-p)B, \qquad S = \langle R, B \rangle.$$

Assume T is irreducible. Let $\boldsymbol{\pi}$ be its stationary probability row vector, so that

$$\boldsymbol{\pi} T = \boldsymbol{\pi}, \qquad \boldsymbol{\pi}\mathbf{1} = 1.$$

Write

$$r = \min_{s \in S} \operatorname{rank}(s), \qquad K = \{s \in S : \operatorname{rank}(s) = r\}.$$

The set K is the minimal ideal (kernel) of this finite transformation semigroup. In particular, the word "kernel" here denotes a semigroup ideal, whereas a transformation's fibers form its kernel partition.

Define the linear operator

$$\Phi(X) = pRXR^{\mathsf{T}} + (1-p)BXB^{\mathsf{T}}$$

on real $d \times d$ matrices. Identifying X with its array of entries indexed by ordered pairs, this is the operator with transition matrix

$$P_2 = p(R \otimes R) + (1-p)(B \otimes B).$$

Both particles use the same randomly chosen letter; this is not the independent-particle transition matrix $T \otimes T$.

Theorem 4.1. The Cesàro limit

$$H = \lim_{N \to \infty} \frac{1}{N} \sum_{\ell=0}^{N-1} \Phi^{\ell}(I_d)$$

exists and satisfies

$$H \succeq 0, \qquad H_{xx} = 1, \qquad \Phi(H) = H.$$

Moreover,

$$H\pi^{\mathsf{T}} = \frac{1}{r}\mathbf{1}, \qquad \pi H \pi^{\mathsf{T}} = \frac{1}{r}. \tag{4.1}$$

For every $k \in K$ and every fiber F of k,

$$\pi(F) := \sum_{x \in F} \pi_x = \frac{1}{r}. \tag{4.2}$$

Consequently the kernel rank is determined by the tensor-square action:

$$\boxed{r = \frac{1}{\pi H \pi^{\mathsf{T}}}.}$$

Equation (4.2) is the equal-stationary-mass conclusion associated with Friedman's work on road coloring [9]. The proof below is self-contained.

4.2 The collision matrix

For a word w in the two letters, write W for its transformation matrix, and let q_w be the product of its letter probabilities. Thus $q_w > 0$. Summing over words (not merely over distinct resulting matrices),

$$\Phi^{\ell}(X) = \sum_{|w|=\ell} q_w W X W^{\mathsf{T}}, \qquad T^{\ell} = \sum_{|w|=\ell} q_w W. \tag{4.3}$$

At length zero, the word matrix is I_d.

Lemma 4.1. The matrix H in Theorem 4.1 exists, is positive semidefinite, has diagonal entries one, and satisfies $\Phi(H) = H$. If distinct states x, y cannot be merged by any word, then $H_{xy} = 0$.

Proof. The matrix P_2 is a finite stochastic matrix. Its Cesàro powers converge, so their action on the vectorized identity also converges. This proves existence of H.

Each WW^{T} is positive semidefinite, and

$$(WW^{\mathsf{T}})_{xy} = \begin{cases} 1, & xW = yW, \\ 0, & xW \neq yW. \end{cases}$$

In particular its diagonal entries equal one. Formula (4.3) and passage to the limit prove the corresponding assertions for H. The fixed-point equation follows either from the Cesàro projection or from the telescoping identity

$$\Phi\left(\frac{1}{N}\sum_{\ell=0}^{N-1}\Phi^{\ell}(I_d)\right) - \frac{1}{N}\sum_{\ell=0}^{N-1}\Phi^{\ell}(I_d) = \frac{\Phi^{N}(I_d) - I_d}{N}.$$

The right side tends to zero because all its entries are bounded in absolute value by $1/N$. Finally, a nonmergeable pair has zero entry in every matrix being averaged, and hence in H.

Remark 4.1 (Probabilistic interpretation). The entry $(\Phi^{\ell}(I_d))_{xy}$ is the probability that two particles started at x, y agree after ℓ common random letters. Once they agree, they remain together. These probabilities are therefore nondecreasing in ℓ and bounded by one. Thus, in this particular construction, the ordinary limit $\Phi^{\ell}(I_d) \to H$ also exists, and H_{xy} is the probability of eventual coalescence. Cesàro averaging provides a direct finite-Markov-chain construction that does not require aperiodicity.

4.3 Positive semidefiniteness and stationarity

Lemma 4.2. For every word matrix $W \in S$,

$$HW^{\mathsf{T}}\pi^{\mathsf{T}} = H\pi^{\mathsf{T}}. \tag{4.4}$$

Furthermore, if $c = \pi H \pi^{\mathsf{T}}$, then

$$H\pi^{\mathsf{T}} = c\mathbf{1}. \tag{4.5}$$

Proof. Fix a word length $\ell \geq 1$. Since $\Phi^{\ell}(H) = H$ and $\pi T^{\ell} = \pi$, the word averages satisfy

$$\sum_{|w|=\ell} q_w \pi W = \pi, \qquad \sum_{|w|=\ell} q_w (\pi W) H (\pi W)^{\mathsf{T}} = c.$$

Expanding the squared deviations therefore gives

$$\sum_{|w|=\ell} q_w (\pi W - \pi) H (\pi W - \pi)^{\mathsf{T}} = 0. \tag{4.6}$$

Every summand is nonnegative because $H \succeq 0$, and every q_w is positive. For a positive semidefinite matrix, $vHv^{\mathsf{T}} = 0$ implies $Hv^{\mathsf{T}} = 0$. Hence

$$H(\pi W - \pi)^{\mathsf{T}} = 0$$

for every word of length ℓ. As ℓ was arbitrary, this proves (4.4) for every word.

Set $h = H\pi^{\mathsf{T}}$. Applying the fixed-point equation and the length-one cases of (4.4), we obtain

$$\begin{aligned} h &= pRHR^{\mathsf{T}}\pi^{\mathsf{T}} + (1-p)BHB^{\mathsf{T}}\pi^{\mathsf{T}} \\ &= pRh + (1-p)Bh = Th. \end{aligned}$$

For an irreducible finite stochastic matrix, the right fixed vectors are the constant vectors. Thus $h = \alpha \mathbf{1}$. Multiplying by π gives $\alpha = \pi h = c$, proving (4.5).

Remark 4.2. To obtain (4.4) for arbitrary words, the variance identity is applied separately at every word length. The length-one identities alone do not justify a direct iteration without a further invariance argument.

4.4 Minimum-rank ranges and equal fiber masses

Lemma 4.3. For every $k \in K$,

$$kHk^{\mathsf{T}} = kk^{\mathsf{T}}. \tag{4.7}$$

Proof. Let $U = \operatorname{im} k$, so $|U| = r$. No word can merge two distinct points of U: otherwise its composition after k would have rank below r. Lemma 4.1 consequently gives

$$H_{uv} = \delta_{uv} \qquad (u, v \in U).$$

For any $x, y \in Q$, therefore,

$$(kHk^{\mathsf{T}})_{xy} = H_{xk,yk} = \begin{cases} 1, & xk = yk, \\ 0, & xk \neq yk, \end{cases} = (kk^{\mathsf{T}})_{xy}.$$

Proof (Proof of Theorem 4.1). The initial properties of H were proved in Lemma 4.1. For $k \in K$, Lemmas 4.2 and 4.3 give

$$\begin{aligned} kk^{\mathsf{T}}\pi^{\mathsf{T}} &= kHk^{\mathsf{T}}\pi^{\mathsf{T}} \\ &= kH\pi^{\mathsf{T}} \\ &= ck\mathbf{1} = c\mathbf{1}, \end{aligned}$$

where $c = \pi H \pi^{\mathsf{T}}$. On the other hand,

$$(kk^{\mathsf{T}}\pi^{\mathsf{T}})_x = \sum_{y:\, yk=xk} \pi_y.$$

This is precisely the stationary mass of the fiber containing x. Thus all fibers of k have mass c. There are r such fibers and they partition Q, so $rc = 1$. This proves all the remaining assertions.

4.5 Consequences and scope

Corollary 4.1 (A coloring-independent arithmetic restriction). Suppose $\pi_x = w_x/W$, where the w_x are positive integers and $W = \sum_x w_x$. Then every kernel block has integer weight W/r; in particular r divides W. If π is uniform, every kernel block has cardinality d/r, so r divides d.

Proof. Multiply the equal-mass identity by W, or by d in the uniform case.

Corollary 4.2 (Synchronization). The following are equivalent:

1. S contains a constant transformation;
2. $r = 1$;
3. $H = \mathbf{1}\mathbf{1}^{\mathsf{T}}$;
4. $\pi H \pi^{\mathsf{T}} = 1$.

Proof. The equivalence of the first two statements follows from the definition of rank, and their equivalence with the fourth follows from the rank formula. A reset word occurs eventually with probability one among independent random letters, for instance by considering disjoint blocks of its length. It therefore forces eventual coalescence of every pair, so the first statement implies the third. The third implies the fourth because $\pi\mathbf{1} = 1$.

Remark 4.3 (Application to road colorings). For a 0–1 adjacency matrix A having two ones per row and a coloring $A = R + B$, take $p = 1/2$. Then $T = A/2$ and its stationary distribution do not depend on the coloring, whereas P_2 and H generally do. Thus every coloring obeys the same stationary-weight restrictions, while its particular tensor-square fixed point determines its kernel rank. Irreducibility and aperiodicity are the standing assumptions for the

road-coloring existence problem; only irreducibility is used in the equal-mass theorem proved here.

Remark 4.4 (More than two generators). The entire proof applies to finitely many transformation matrices $M_1, \ldots, M_s$ with positive probabilities p_i summing to one: replace T by $\sum_i p_i M_i$ and $\Phi(X)$ by $\sum_i p_i M_i X M_i^{\mathsf{T}}$. The same conclusions hold whenever T is irreducible.

Chapter 5
Kernel Expectations and Their Spectra

5.1 Basic observables and the kernel e-projections

Let K be the minimum-rank ideal of a finite transformation semigroup, with rank r, and let λ be its limiting kernel measure. Denote its partition and range marginals by α_i and β_j. Write e_{ij} for the idempotent with partition $\mathscr{P}_i$ and range $\mathscr{R}_j$. Each range is a transversal of every kernel partition. All matrices below use right-action conventions.

5.1.1 The two basic expectation operators

Our convention is

$$\boxed{N = \mathbb{E}_\lambda[KK^{\mathsf{T}}], \qquad M = \mathbb{E}_\lambda[K^{\mathsf{T}}JK].}$$

Here the K inside the expectation denotes a random kernel transformation, and $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}$. Set $C_i = e_{ij}e_{ij}^{\mathsf{T}}$, independently of j. Its entries are one precisely when their indices lie in the same block of $\mathscr{P}_i$. Then

$$N = \sum_i \alpha_i C_i, \qquad N_{xy} = \Pr\{xK = yK\}.$$

For $b(K) = \mathbf{1}^{\mathsf{T}}K$, the entries of this row are the fiber sizes at the image vertices, and

$$M = \mathbb{E}[b(K)^{\mathsf{T}}b(K)].$$

Thus N describes partitions and M describes weighted range data. Both are symmetric positive semidefinite, but neither is generally a projection. The pair-survival matrix is $J-N$, not another operator named N. The rank of the average N need not equal the kernel rank r.

5.1.2 Column projections: hold the range fixed

Put $\rho_j = \mathrm{diag}(\mathbf{1}_{\mathscr{R}_j})$ and define

$$P_j = \sum_i \alpha_i e_{ij}.$$

Idempotents in a fixed kernel column form a left-zero semigroup: $e_{ij}e_{\ell j} = e_{ij}$. Therefore

$$P_j^2 = \sum_{i,\ell} \alpha_i \alpha_\ell e_{ij} = P_j.$$

Since $C_i\rho_j = e_{ij}$, we obtain the useful reconstruction formula

$$\boxed{P_j = N\rho_j.}$$

On row vectors, P_j projects onto vectors supported on $\mathscr{R}_j$: its output is supported there and it fixes every coordinate row on that range. In particular, $\mathrm{rank}\, P_j = r$.

5.1.3 Row projections: hold the partition fixed

Define

$$Q_i = \sum_j \beta_j e_{ij}.$$

Idempotents in a fixed kernel row form a right-zero semigroup: $e_{ij}e_{i\ell} = e_{i\ell}$, so $Q_i^2 = Q_i$. For formulas involving the stationary row, assume the range-marginal identity supplied by equipartition in the irreducible stochastic setting:

$$\sum_j \beta_j \rho_j = r\omega, \qquad \omega = \mathrm{diag}(\pi).$$

This is a stated probabilistic hypothesis here, not a blanket assumption on all finite semigroups. The PtC application must justify it in its own setting when using these formulas. Averaging $C_i\rho_j = e_{ij}$ gives

$$\boxed{Q_i = rC_i\omega = re_i e_i^{\mathsf{T}}\omega,}$$

where e_i is any idempotent with partition $\mathscr{P}_i$. On column functions, Q_i projects onto functions constant on each block:

$$(Q_i f)_x = r\sum_{y\in C} \pi_y f_y \quad (x \in C \in \mathscr{P}_i).$$

Equipartition gives $\pi(C) = 1/r$, making this a weighted block average. It is not generally the Euclidean block-average projection. These partition projections and the range projections above should also be kept distinct from the higher-level limiting projections Ω_k.

5.1.4 The average of the distinguished idempotents

Averaging the distinguished idempotents with the product marginal law gives

$$\boxed{\overline{e} = \sum_{i,j} \alpha_i \beta_j e_{ij} = \sum_i \alpha_i Q_i = \sum_j \beta_j P_j = rN\omega.}$$

Although every P_j and Q_i is idempotent, this double average is not asserted to be idempotent in general. These identities retain the actual kernel e-projections; they are not replacements by unrelated orthogonal projections.

Notation throughout this chapter.

N and M are the expectation operators defined above. The pair-survival and centered operators are written explicitly as $J-N$ and $M-\tau\omega$. The symbols N_0 and M_0 are not used in this chapter. Later inherited chapters may still use their locally announced historical conventions.

5.2 Basic Terminology and Notations

A state vector for a given set is the indicator of that set, i.e., it is one at the indices which belong to the given set, and zero otherwise. We denote by χ_U the state vector for a set U. The vector u is $1 \times n$, all entries equal to 1. $J = u^\dagger u$ is the all-ones matrix.

If $\mathscr{A} = R + B$ is the adjacency matrix of the graph, $A = \frac{1}{2}\mathscr{A}$ is stochastic with invariant distribution π. The corresponding Cesaro limit matrix is $\Omega = u^\dagger \pi$. The diagonal matrix ω has diagonal entries $\omega_{ii} = \pi_i$, the invariant distribution. Set $\Delta = \frac{1}{2}(R-B)$.

The semigroup $\mathscr{S}$ is generated by R and B. The kernel of $\mathscr{S}$ is $\mathscr{K}$. The idempotents of $\mathscr{K}$ are $\mathscr{E}(\mathscr{K})$. For $k \in \mathscr{K}$, the corresponding range vector is $\rho(k) = \chi_{\text{range}(k)}$, the state vector for the range of k. With r the rank of $\mathscr{K}$, $\rho(k)u^\dagger = r$ for every $k \in \mathscr{K}$.

We denote by α the measure on the idempotents induced from the partitions of $\mathscr{K}$. Denote by β the measure on the ranges of the kernel $\mathscr{K}$.

We use the summation convention wherein Greek indices are summed over regardless of position.

5.3 Expectation operators and their fixed-point equations

Throughout this chapter, $A = (R+B)/2$ is irreducible, $\pi A = \pi$, $\pi u^{\mathsf{T}} = 1$, $\Omega = u^{\mathsf{T}}\pi$, and $\omega = \operatorname{diag}(\pi)$. The letters are selected independently with equal probabilities. Kernel expectations use the limiting Cesaro measure λ; aperiodicity is not required for this convention. Thus $\mathbb{E}_\lambda[K] = \Omega$. Write

$$N = \mathbb{E}_\lambda[KK^{\mathsf{T}}], \qquad M = \mathbb{E}_\lambda[K^{\mathsf{T}}JK], \qquad \tau = \operatorname{tr} M.$$

The complementary operator is always written $J-N$, and the centered operator is always written $M - \tau\omega$. In particular, neither is silently denoted by N or M.

Set $\Delta = (R-B)/2$. The common-letter fixed-point equations are

$$AXA^{\mathsf{T}} + \Delta X\Delta^{\mathsf{T}} = X, \qquad X \in \{N, J, J-N\}. \tag{5.1}$$

$$A^{\mathsf{T}}YA + \Delta^{\mathsf{T}}Y\Delta = Y, \qquad Y \in \{M, \omega, M-\tau\omega\}. \tag{5.2}$$

Indeed, the left sides are respectively $(RXR^{\mathsf{T}} + BXB^{\mathsf{T}})/2$ and $(R^{\mathsf{T}}YR + B^{\mathsf{T}}YB)/2$. Stationarity of λ under left and right multiplication by a random letter gives the equations for N and M. The identities for J and ω follow from stochasticity and $\pi A = \pi$; linearity gives the remaining two equations.

If e_i is the idempotent associated with partition i in any fixed kernel column, then

$$N = \sum_i \alpha_i e_i e_i^{\mathsf{T}}.$$

Moreover,

$$JM = n^2\Omega, \qquad \operatorname{tr}(JM) = n^2.$$

All five matrices N, M, $J-N$, $M-\tau\omega$ and J are symmetric. The two expectation operators N and M are positive semidefinite; this assertion does not extend automatically to their complementary or centered counterparts.

5.4 Basic data and properties

The identities below use equipartition, proved in Section 5.6. Put $r' = 1 - 1/r$ and $\tau' = n^2 - \tau$. Then

$$\pi N = \frac{1}{r}u, \qquad \pi(J-N) = r'u, \qquad (J-N)\pi^{\mathsf{T}} = r'u^{\mathsf{T}}, \qquad \Omega(J-N) = r'J.$$

For an idempotent e, the ith component of uee^{T} is the size of the block containing i. Thus $b = uN$ is the expected block-size row: b_i is the expected size of the partition block containing i. If a kernel partition has block sizes $s_1, \ldots, s_r$, then

$$\|uk\|^2 = \sum_{a=1}^{r} s_a^2, \qquad \tau = \mathbb{E}\left[\sum_{a=1}^{r} s_a^2\right] = \mathrm{tr}(JN).$$

In particular,

$$\tau' = \mathrm{tr}\big(J(J-N)\big) = \mathrm{tr}\big(J(M-\tau\omega)\big).$$

The centered-operator identities are

$$u(M-\tau\omega) = \tau'\pi, \qquad (M-\tau\omega)u^{\mathsf{T}} = \tau'\pi^{\mathsf{T}}, \qquad J(M-\tau\omega) = \tau'\Omega.$$

For comparison, the uncentered identities are $uM = n^2\pi$ and $JM = n^2\Omega$. Finally, for every word $w \in \mathscr{S}$,

$$\pi we = \pi e = \frac{1}{r}\rho(e), \qquad \pi wN = \frac{1}{r}u, \qquad \pi w(J-N) = r'u.$$

5.5 Norms

Build up a series of words in $\mathscr{S}$, by concatenation of R or B at each step. Denote w_i the ith word, of length i, the total number of R's and B's appearing in it. Then

Proposition 5.1. The squared norms $\|uw_i\|^2 = uw_iw_i^*u^\dagger$ are nondecreasing. I.e.,

$$\|uw_i\|^2 \le \|uw_{i+1}\|^2$$

Proof. Say $w_{i+1} = w_iR$. Recall that for any word w, ww^* has ones on the diagonal. In particular, $RR^* - I$ has nonnegative, 0 or 1, entries. Thus

$$\begin{aligned}\|uw_{i+1}\|^2 - \|uw_i\|^2 &= uw_iRR^*w_i^*u^\dagger - uw_iw_i^*u^\dagger \\ &= uw_i(RR^* - I)w_i^*u^\dagger\end{aligned}$$

is nonnegative. □

Once w_i is in the kernel, the squared norm is maxed out, and for all succeeding w_{i+k}, $k \ge 0$, equals the sum of the squares of the block sizes of the partition for w_i.

5.6 Equipartitioning

We show basic results based on the fact due to BM-F that $\pi k = (1/r)\rho(k)$, for every $k \in \mathscr{K}$. Given this, it follows that, with $e \in \mathscr{E}(\mathscr{K})$ such that $ek = ke = k$, $\pi kk^* = \pi ee^* = (1/r)u$, and averaging yields

$$\pi N = (1/r)u$$

Now suppose that f is any idempotent of $\mathscr{K}$ such that $\rho(k) = \rho(f)$. Then

$$\pi kf^* = (1/r)\,\rho(k)f^* = (1/r)\,\rho(f)f^* = \pi ff^* = (1/r)u$$

Taking $k = we$, we have the useful form

$$\pi wee^* = \frac{1}{r}\,u$$

for any $w \in \mathscr{S}$, and any $e \in \mathscr{E}(\mathscr{K})$, since $we \in \mathscr{K}$ has the same range as e.

We now show these results directly.

Proposition 5.2. For every $e \in \mathscr{E}(\mathscr{K})$, $\pi\Delta e = 0$. Equivalently, $\pi Re = \pi Be = \pi e$.

Proof. Multiplying eq. (5.1), for N, on the left by Ω and on the right by Ω^*, we have, using $\Omega A = A\Omega = \Omega$,

$$\Omega N\Omega^* + \Omega\Delta N\Delta^*\Omega^* = \Omega N\Omega^*$$

Hence,

$$\Omega\Delta N\Delta^*\Omega^* = 0 = \alpha_\mu \Omega\Delta e_\mu e_\mu^*\Delta^*\Omega^*$$

Taking traces, all terms are nonnegative so that $\Omega\Delta e = 0$ for every e. From the definition of Δ the alternative formulation results after averaging $\pi Re = \pi Be$ to $\pi Ae = \pi e$. □

Corollary 5.1. For any $k \in \mathscr{K}$, $\pi\Delta k = 0$.

This follows from Proposition 5.2 by writing $k = ek$, where e is the identity of the local group containing k, so $ek = ke = k$. As well

Corollary 5.2. We have $\pi\Delta\left(J - N\right) = 0$.

Proposition 5.3. For every $w \in \mathscr{S}$, and $e \in \mathscr{E}(\mathscr{K})$, $\pi we = \pi e$.

Proof. This follows by induction on l, the length of w. Proposition 5.6 is the result for $l = 1$. Let w_{l+1} be a word of length $l+1$, then $w_{l+1} =$ one of Rw_l, Bw_l, where w_l has length l. Since $w_l e \in \mathscr{K}$ for $e \in \mathscr{K}$, Corollary 5.1 yields

$$\begin{aligned} \pi w_{l+1}e &= \pi Rw_l e \\ &= \pi Bw_l e \end{aligned}$$

$$\begin{aligned} &= \pi w_l e && \text{(averaging)} \\ &= \pi e && \text{(by induction)} \end{aligned} \qquad \square$$

Proposition 5.4. For any $e \in \mathscr{E}(\mathscr{K})$, $w \in \mathscr{S}$,

$$\pi we = (1/r)\,\rho(e) \qquad \text{and} \qquad \pi wee^* = (1/r)\,u$$

Proof. From Proposition 5.6, multiplying on the right by e^* , we have $\pi\Delta ee^* = 0$. Averaging over e yields $\pi\Delta N = 0$. Equivalently, $N\Delta^*\pi^\dagger = 0$. Hence, multiplying equation (5.1) on the right by $\pi^\dagger$, using $A^*\pi^\dagger = \pi^\dagger$, we find

$$AN\pi^\dagger = N\pi^\dagger$$

So $N\pi^\dagger$ is fixed by A, hence is a constant vector. Transposing back, say $\pi N = au$ for some constant a. Now let $e \in \mathscr{E}(\mathscr{K})$ and take the e_i with the same range as e in the summation defining N. Then from Proposition 5.6 we have $\pi ee_i = \pi e_i$. Since the idempotents of a column are a left-zero semigroup, $\pi ee_i = \pi e$. Thus on the one hand $\pi eNe^* = \pi Ne^*$ and on the other $\pi eNe^* = \pi ee^*$. Or $\pi ee^* = \pi Ne^* = aue^* = au$ for any idempotent e. Thus, $\pi e = a\rho(e)$ and $\pi eu^\dagger = 1 = ar$ yields $a = 1/r$, where r is the rank of the kernel. So, for any $w \in \mathscr{S}$, $\pi we = \pi e = (1/r)\rho(e)$ and $\pi wee^* = (1/r)u$, as required. $\square$

Averaging the second relation in the above Proposition

Corollary 5.3. For any $w \in \mathscr{S}$, $\pi wN = \frac{1}{r}\,u$. And $\pi w\big(J-N\big) = r'\,u$.

5.7 Algebra generated by the expectation operators

Proposition 5.5. The product $(J-N)(M-\tau\omega)$ commutes with Ω.

Proof. Put $r' = 1 - 1/r$ and $\tau' = n^2 - \tau$. The identities of Section 5.4 give

$$\Omega(J-N)(M-\tau\omega) = r'\tau'\Omega.$$

By symmetry, $(J-N)\pi^{\mathsf{T}} = r'u^{\mathsf{T}}$ and $(M-\tau\omega)u^{\mathsf{T}} = \tau'\pi^{\mathsf{T}}$. Hence

$$(J-N)(M-\tau\omega)\Omega = r'\tau'\Omega.$$

Theorem 5.1. If A is doubly stochastic, the matrices M, N and J generate a commutative algebra. The same is true of $M-\tau\omega$, $J-N$ and J.

Proof. Here $\pi = u/n$, $\Omega = J/n$ and $\omega = I/n$. The equipartition identity $\pi kN = u/r$ for every kernel element k yields $ukN = nu/r$. Therefore

$$MN = \mathbb{E}[K^{\mathsf{T}}u^{\mathsf{T}}uKN] = \frac{n}{r}\mathbb{E}[K^{\mathsf{T}}u^{\mathsf{T}}]u = \frac{n}{r}J.$$

Since M and N are symmetric, transposition gives $NM = MN$. Also $JM = nJ = MJ$ and $JN = (n/r)J = NJ$. Subtracting multiples of I and forming $J - N$ preserves commutativity.

Further question.

Without double stochasticity, the structure of the commutator $MN - NM$ requires separate analysis. No general commutativity assertion is made here.

Chapter 6
Delta, Canonical Colorings, and Recoloring

6.1 Questions about a fixed graph and its colorings

In this chapter we impose the road-coloring hypotheses stated at the opening. The normalized matrix $A = (R+B)/2$ and its stationary row π are fixed while the outgoing red/blue choices vary. Three questions must be distinguished: whether a CC coloring exists; what kernel rank a given coloring has; and whether a linear-algebraic test certifies synchronization.

The target-pair graph H_A is defined in Definition 1.2. It is unchanged by recoloring. We first relate its components to the ranks of A and Δ, then use bipartiteness to characterize CC existence. We next study consequences for kernel rank and the vector spaces generated by the semigroup. Sufficient ranked tests are developed in the following rank-test chapter. The full sixteen-row four-state run is collected in Section C.1, while the CC construction itself is illustrated locally in Figure 6.1.

Definition 6.1. For a coloring semigroup $S = \langle R, B \rangle$, the kernel K is its minimal nonempty two-sided ideal, equivalently its minimum-rank transformations. Their common rank is the kernel rank. A synchronizing instruction is a word whose transformation has rank one.

6.2 The matrix Δ

We will use the following notation. All matrices are $n \times n$, where n is the number of vertices in the graph. The normalized adjacency matrix $T = \frac{1}{2}(R+B)$, while the coloring difference matrix $\Delta = \frac{1}{2}(R-B)$.

Here $T = A = \mathscr{A}/2$ in the normalized notation used elsewhere. The remarks below use the road-coloring hypotheses: the original graph is simple and two-out, and A is irreducible and aperiodic (and traceless).

Remark 6.1 (Ranks and the auxiliary target-pair graph). Let H_A be the target-pair graph of Definition 1.2. Different sources may give the same auxiliary edge; its multiplicity does not affect the statements below. Let b and o be the numbers of bipartite and nonbipartite connected components, respectively, and put $c = b + o$. Then

$$\dim\ker A = b, \qquad \dim\ker \Delta = c, \qquad \operatorname{rank} A - \operatorname{rank} \Delta = o.$$

Indeed, $\Delta v = 0$ says that $v_{xR} = v_{xB}$, so v is constant on each component. The equation $Av = 0$ says that $v_{xR} = -v_{xB}$; it has one free parameter on each bipartite component and forces zero on each nonbipartite component. In particular,

$$\operatorname{rank} \Delta \le \operatorname{rank} A,$$

with equality if and only if H_A is bipartite, equivalently if and only if the graph admits a CC coloring. Also $\operatorname{rank} \Delta = n-1$ if and only if H_A is connected. These ranks depend only on the uncolored graph, not on the choice of red/blue coloring.

Remark 6.2 (Bounds given the rank of A). Put $a = \operatorname{rank} A$ and $d = \operatorname{rank} \Delta$. Irreducibility ensures that every vertex occurs as a destination, so H_A has no isolated vertices. Simplicity of the original graph ensures that H_A has no loops. Thus each bipartite component has at least two vertices and each nonbipartite component at least three. Consequently

$$n \ge 2b + 3o = 2(n-a) + 3(a-d),$$

and therefore

$$\boxed{\left\lceil \frac{n+a}{3} \right\rceil \le d \le \min\{a, n-1\}.}$$

The component-count identities are exact; the bounds use only n and a.

Remark 6.3 (Invertibility does not force rank $n-1$). If A is invertible, every component of H_A is nonbipartite. Hence

$$\operatorname{rank} \Delta = n - c, \qquad \left\lceil \frac{2n}{3} \right\rceil \le \operatorname{rank} \Delta \le n-1.$$

Equality on the right holds exactly when H_A is connected; invertibility alone does not imply this connectedness. For example,

$$R = [245114], \qquad B = [366235]$$

give $\det A = 1/16$ and $\operatorname{rank} \Delta = 4 = n-2$. Their auxiliary graph is the disjoint union of triangles on $\{1,2,3\}$ and $\{4,5,6\}$. The original directed graph is simple, loopless and strongly connected. Its cycles $1 \to 2 \to 4 \to 1$ and $1 \to 2 \to 6 \to 5 \to 1$ have lengths three and four, so it is aperiodic.

Since the graph is strongly connected and aperiodic, $\lim_{n\to\infty} T^n = \Omega$ exists, where Ω is a positive stochastic matrix with identical rows. Each row of Ω is a probability vector which we will refer to as π. In what follows, important structures will be the real vector space spanned by S, denoted vec(S), and the span of the kernel K, denoted vec(K).

Any eventual solution to the conjecture must find a systematic method for analyzing the possible recolorings of the graph. To this end, let ϕ_i be the diagonal matrix with -1 in the ith diagonal position, and 1's along the rest of the diagonal. We refer to ϕ_i as the ith elementary recoloring matrix, where an elementary recoloring of vertex i is a recoloring that switches the red and blue labels for the edges leading from i, and leaves all other labels in the graph unchanged. Notice that given the coloring difference matrix $\Delta = \frac{1}{2}(R-B)$, then $\phi_i\Delta$ is the coloring difference matrix induced by an elementary recoloring of i. It is clear any possible difference matrix Δ' is of the form $\Delta' = \phi_{i_1}\phi_{i_2}\ldots\phi_{i_j}\Delta$ where $\phi_{i_1}\phi_{i_2}\ldots\phi_{i_j}$ are elementary recoloring matrices. Because all possible coloring differences matrices are of this form, the following proposition is immediate.

Proposition 6.1. All coloring difference matrices have the same rank.

6.3 Existence of canonical colorings from the transition matrix

We use the convention of the Delta notes: a CC coloring has disjoint red and blue ranges. Its subsequent canonical recoloring is called an RCC. In this road-coloring discussion we impose the road-coloring assumptions: $\mathscr{A}$ is the adjacency matrix of a finite simple two-out directed graph, and $A = \mathscr{A}/2 = (R+B)/2$ is stochastic, traceless, irreducible and aperiodic. For source x, its two destinations $i = xR$ and $j = xB$ satisfy $i \neq j$, $i \neq x$, and $j \neq x$. All nullspaces below are right nullspaces over the real numbers. The criterion remains valid under weaker hypotheses, but no such weakening changes the setting of this discussion.

Construct the undirected target-pair graph H_A by joining the two destinations in each row of A. The two destinations are distinct, so H_A has no loops. Different sources may yield the same auxiliary edge; retain its multiplicity or suppress duplicates, since neither changes bipartiteness. Let c be the number of its connected components, and let b be the number of components that are bipartite, with bipartiteness determined by the absence of odd cycles. Its component partition is the graph-intrinsic canonical partition, not in general a kernel partition or a red/blue target partition.

Theorem 6.1. The following conditions are equivalent:

1. The graph admits a CC coloring.
2. There exists $s \in \{+1,-1\}^n$ with $As = 0$.

3. H_A is bipartite.
4. $\dim\ker A = c$.
5. For any coloring $A = (R+B)/2$ and $\Delta = (R-B)/2$, $\dim\ker A = \dim\ker \Delta$.

In fact $\dim\ker A = b$ and $\dim\ker \Delta = c$.

Proof. For a row with destinations i, j, the equation $As = 0$ is $s_i + s_j = 0$. A sign solution is therefore precisely a bipartition of H_A. Color its positive-target edge red and its negative-target edge blue. The resulting ranges are disjoint and cover all vertices, since irreducibility ensures that every vertex occurs as a destination. Conversely a CC coloring assigns signs to its two target sets and supplies a sign solution.

For an arbitrary real column x, the equation $Ax = 0$ requires $x_i = -x_j$ across every edge of H_A. A bipartite component contributes one free parameter; an odd cycle, including a loop, forces every value on its component to vanish. Thus $\dim\ker A = b$. The equation $\Delta x = 0$ instead requires $x_i = x_j$ across every edge, independently of the coloring. Its solutions are exactly the columns constant on components, so $\dim\ker \Delta = c$.

6.3.1 Constructing and counting the CC colorings

Given a sign solution, put $g = \mathrm{diag}(s)$. Then

$$R_{\mathrm{CC}} = A(I+g), \qquad B_{\mathrm{CC}} = A(I-g), \qquad \Delta_{\mathrm{CC}} = Ag, \qquad A = \Delta_{\mathrm{CC}} g.$$

Each row of these matrices selects one of the original outgoing edges. If H_A is bipartite, its c components can be oriented independently, giving exactly 2^c labeled CC colorings or 2^{c-1} up to global red/blue interchange. Every target is incident to an auxiliary edge, so distinct sign choices give distinct colorings. Changing the signs on a target component induces row swaps at its predecessor source vertices, which need not be the vertices of the component itself. This distinction matters when implementing recolorings.

A breadth-first or depth-first bipartiteness test either constructs the signs or returns an odd-cycle obstruction. Thus existence is decidable without enumerating all row recolorings.

6.3.2 A four-state illustration of the target-pair test

Consider $R = [4312]$ and $B = [3443]$. The two targets at each source give

$$\{4,3\}, \quad \{3,4\}, \quad \{1,4\}, \quad \{2,3\}.$$

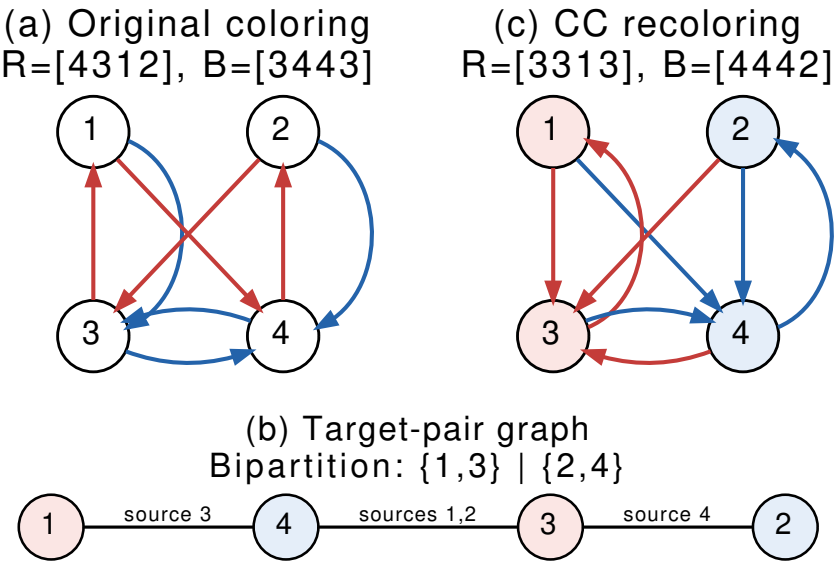


Fig. 6.1 The four-state CC test. Top: (a) the original coloring and (c) its CC recoloring, with red R-arrows and blue B-arrows. Bottom: (b) the undirected target-pair graph; its edge $\{3,4\}$ comes from both sources 1 and 2. Tinted vertices indicate the chosen red and blue target types. The directed graph is unchanged; only the outgoing colors at sources 1 and 4 are interchanged.

Ignoring the repeated edge, the auxiliary graph is the path $1-4-3-2$. It is bipartite, with sides $\{1,3\}$ and $\{2,4\}$. These sides are target types, not kernel blocks. The auxiliary graph is connected, so its component partition has just one block.

Choose $\{1,3\}$ as the red target side. At each source, color the edge ending on that side red and the other edge blue. This swaps the outgoing colors at sources 1 and 4, producing

$$R_{\mathrm{CC}} = [3313], \qquad B_{\mathrm{CC}} = [4442].$$

Their ranges are disjoint: $\mathrm{Im}\,R_{\mathrm{CC}} = \{1,3\}$ and $\mathrm{Im}\,B_{\mathrm{CC}} = \{2,4\}$. The original coloring is not CC, since its red and blue ranges overlap. Thus bipartiteness tests existence of a CC recoloring, not whether the supplied coloring is already CC. Swapping the two target types gives the only other labeled CC choice. The swaps here construct a CC coloring; they are not the subsequent CC-to-RCC canonical recoloring operation.

6.3.3 A test involving only matrix operations on A

Define

$$L_A = \operatorname{diag}(A^{\mathsf{T}}\mathbf{1}) - A^{\mathsf{T}}A.$$

Proposition 6.2. For every coloring,

$$L_A = \Delta^{\mathsf{T}}\Delta.$$

Consequently L_A is positive semidefinite, $\dim\ker L_A = c$, and

$$\text{CC exists} \iff \operatorname{nullity} A = \operatorname{nullity} L_A.$$

Proof. Let $u = e_i$, $v = e_j$ be the two destination columns for one row. That row contributes

$$\tfrac{1}{2}(uu^{\mathsf{T}} + vv^{\mathsf{T}}) - \tfrac{1}{4}(u+v)(u+v)^{\mathsf{T}} = \tfrac{1}{4}(u-v)(u-v)^{\mathsf{T}}$$

to L_A. This is also its contribution to $\Delta^{\mathsf{T}}\Delta$, regardless of which edge is red. The identity holds when $i = j$ as well. Summing proves it, and the nullspace assertions follow from $\ker(\Delta^{\mathsf{T}}\Delta) = \ker\Delta$ and Theorem 6.1.

The identity uses the equally weighted two-out hypothesis. It should not be applied unchanged to an arbitrary stochastic matrix.

6.3.4 Singularity is necessary but not sufficient

An invertible A rules out CC colorings. A singular A need not admit one: some components of H_A may be bipartite while others are not. For example, take

$$R = [24111], \qquad B = [35233].$$

The directed graph is irreducible and aperiodic, while H_A consists of a triangle on $\{1,2,3\}$ and an edge on $\{4,5\}$. Therefore

$$\ker A = \operatorname{span}\{(0,0,0,1,-1)^{\mathsf{T}}\}, \qquad \dim\ker\Delta = 2.$$

The transition matrix is singular, but no sign solution exists and there is no CC coloring. If H_A is connected, on the other hand, singularity of A is sufficient as well as necessary.

6.3.5 Existence versus kernel rank

These tests decide existence and construct all CC colorings; they do not force synchronization. Our nine-state graph has two CC choices up to color interchange, of kernel ranks three and one. Their associated RCCs have ranks six and two. The six-state CC example $[312344],[666565]$ has a unique CC choice up to interchange; its RCC has partition $(1234 \mid 56)$ and rank two.

6.4 The matrix Δ annihilates only rank one kernels

Denote the all 1's (column) vector by $\mathbf{u}$. We say that a word in S has "length n" if it can be written as a product of n R's and B's.

Lemma 6.1. Let the vector $\mathbf{y}$ satisfy $\Delta w\mathbf{y} = 0$ for all words $w \in S$. Then $\mathbf{y}$ is a constant vector, i.e., a multiple of $\mathbf{u}$.

Proof. Notice that $\Delta\mathbf{y} = 0$, implies that $R\mathbf{y} = B\mathbf{y}$. Thus, $T\mathbf{y} = \frac{1}{2}(R+B)\mathbf{y} = R\mathbf{y} = B\mathbf{y}$. Inductively, assume that any word w_n of length n satisfies $w_n\mathbf{y} = T^n\mathbf{y}$. Now $Rw_n\mathbf{y} = Bw_n\mathbf{y} = Tw_n\mathbf{y} = T^{n+1}\mathbf{y}$ implies that any word w_{n+1} of length $n+1$ satisfies $w_{n+1}\mathbf{y} = T^{n+1}\mathbf{y}$.

Now let e be an idempotent in the kernel, of length l. Then

$$e\mathbf{y} = T^l\mathbf{y} = e^2\mathbf{y} = T^{2l}\mathbf{y} = \cdots = T^{nl}\mathbf{y}$$

for all positive integers n. Letting $n \to \infty$ we get $\Omega\mathbf{y} = c\mathbf{u}$, where the constant $c = \pi \cdot \mathbf{y}$. In other words, for every i in the range of e, $y_i = c$. Since every i occurs in the range of some idempotent in the kernel (this follows from strong connectivity), we get $\mathbf{y} = c\mathbf{u}$. □

Corollary 6.1. Let the vector $\mathbf{y}$ satisfy $\Delta T^n\mathbf{y} = 0$ for all integer $n \geq 0$. Then $\mathbf{y}$ is a constant vector, i.e., a multiple of $\mathbf{u}$.

Proof. As above, from $\Delta\mathbf{y} = 0$, we get $R\mathbf{y} = B\mathbf{y} = T\mathbf{y}$. Proceeding inductively as in the above proof $w_n\mathbf{y} = T^n\mathbf{y}$ for all words of length n implies that $\Delta w_n\mathbf{y} = 0$ or

$$Rw_n\mathbf{y} = Bw_n\mathbf{y} = Tw_n\mathbf{y} = T^{n+1}\mathbf{y}.$$

Hence, $\Delta w\mathbf{y} = 0$ for all words w and by the preceding Lemma, $\mathbf{y}$ is constant. □

Corollary 6.2. $\Delta k = 0$ for all $k \in K$ if and only if K is rank one.

Proof. Suppose $\Delta k = 0$ for all $k \in K$. Let e be an idempotent in K. Then for all words w, $we \in K$, so that $\Delta we = 0$, for all w. By the Lemma, the columns of e are constant vectors. Since any row has exactly one nonzero entry, one column is all ones and the rest are all zeros. The converse is immediate. □

Corollary 6.3. $\Delta e = 0$ for all idempotents in K if and only if K is rank one.

Proof. For any $k \in K$, since K is completely simple and, therefore, a union of isomorphic groups, there is an idempotent such that $k = ek$. Hence $\Delta k = \Delta ek = 0$ and by the previous Corollary, K is rank one. □

6.5 Special rank one matrices in Vec(S)

In this section we will focus on the vector spaces generated by either the matrix semigroup or its kernel. Any rank-one matrix with identical rows in vec(S) will be referred to as rank-one special, or ROSP. A ROSP in vec(K) will be a KROSP. Note, that the ROSP's form a vector space of dimension at most n. In general, we will identify a ROSP with its first row. Observe that if M is any ROSP, then MN is a ROSP for any matrix N in vec(S). In addition, if w is any word in S, $wM = M$. That is, M is a right zero for elements in S.

A range vector is a vector consisting of 0's and 1's with 1's corresponding to the range of an idempotent e in K, that is, a vector whose components indicate the non-zero columns of e. A range matrix is a KROSP whose rows equal some range vector. If there is no chance of confusion, for simplicity, we identify the range vector with its corresponding KROSP.

Proposition 6.3. The vector space spanned by the KROSPs equals:

1. Span of $\{\Omega w : w \in S\}$.
2. Span of $\{Mk : k \in K\}$, where M is any ROSP in vec(S) satisfying $M\mathbf{u} \neq 0$.
3. Span of the range matrices.

Proof. For #1, first note that Ω is an element of vec(K), and therefore a KROSP (see the proof of Theorem [7] in [4]). Since K is an ideal, $kw \in K$, for any $w \in S$. So $\Omega w \in$ vec(K) and hence a KROSP. Equality will follow from #2.

For #2, for $k \in K$, $Mk \in$ vec(K) and so is a KROSP. If the set $\{Mk : k \in K\}$ does not span the KROSPs, then there is a KROSP, X, with first row $\mathbf{x}$ such that

$$Mk\mathbf{x}^t = 0, \quad \text{for all} \quad k \in K.$$

Let $M\mathbf{u} = c\mathbf{u}$, with $c \neq 0$. Since X itself is a KROSP, $cXX^t = MXX^t = 0$. So $X = 0$ and the result follows.

For #3, by the proof of Theorem 7 in [4], πk is a multiple of a range vector for every $k \in K$. So the result follows from #2, taking $M = \Omega$. □

The constant functions play a special role in any investigation of the road coloring problem. Let C_i denote the matrix of the constant function i; that is for C_i, the ith column is all ones. If S contains the constants C_i, they must be in K due to minimality of the rank of the kernel. Clearly, any element in

the linear span of the constants C_i, $i = i, \ldots, n$ is a KROSP. So the KROSPs have dimension n if and only if vec(K) contains the constants C_i. In fact, we can say more.

Proposition 6.4. The KROSPs have dimension n if and only if the rank of the kernel is one.

Proof. One direction is immediate. For the other, assume that the KROSPs have dimension n. By the above, vec(K) contains the constants C_i.

Now $C_i = \sum_{k \in K} \alpha_k k$. Choose an idempotent $e \in K$. Then $eC_ie = \sum_{k=K} \alpha_k eke$. Since K is a completely simple semigroup (see [7]), then $eC_ie = C_j$ is in vec(G) where $G = eKe$ is the subgroup for which e is the local identity element (again [7]), and $ie = j$.

Suppose the rank of K is r. Then G is a group of block permutation matrices of rank r. In particular, the matrices of G have r nonzero columns. Notice $\Omega C_j = C_j$. On the other hand, $\Omega C_j = \sum_{k=K} \alpha_k \Omega eke = C_j$. But, as stated above, the proof of Theorem 7 in [4] shows that, for each $k \in K$, Ωeke is a fixed ROSP whose rows are multiples of the range vector for the group G. Thus $C_j = \Omega eke \sum_{k=K} \alpha_k$ where Ωeke is a ROSP with r non-zero rows. But this is only possible if $r = 1$. □

6.6 The Rank Condition

Recall that π denotes the invariant distribution, the row vector satisfying $\pi T = \pi$, with $\pi \cdot \mathbf{u} = 1$.

Definition 6.2. Let d denote the rank of Δ. The rank condition holds for Δ if the span of $\{\pi\Delta, \pi\Delta^2, \ldots, \pi\Delta^d\}$ is a d-dimensional space, in other words, these vectors span the range (row space) of Δ.

As a consequence of the rank condition, any vector $\mathbf{y}$ satisfying $\Omega\Delta^i\mathbf{y} = 0$, $1 \le i \le d$, satisfies $\Delta\mathbf{y} = 0$ as well.

Once again, the proof of Theorem 7 in [4] shows that πk is a multiple of a range vector for every $k \in K$. Specifically, if k is in K, then πk is a multiple of the range vector corresponding to the nonzero columns of k, and that multiple depends only on the collection of non-zero columns, that is, the range of k. Observe that for any $k \in K$, Rk and Bk are elements of K with the same range. Hence, $\Omega Rk = \Omega Bk$, or $\Omega\Delta k = 0$ for all $k \in K$.

Assume the rank condition. Let e be any idempotent in K. Let w be any word in S. Then Rwe and Bwe are elements of K with the same range. Hence, $\Omega Rwe = \Omega Bwe$ is a multiple of the range matrix corresponding to e. In other words, $\Omega\Delta we = 0$ for any w in S. But then $\Omega\Delta^i e = 0$, for all $i \ge 0$, since

powers of Δ are in $\mathrm{vec}(S)$. From the rank condition, we have $\Delta e = 0$, for all idempotents $e \in K$. By Corollary 3.4, K is rank one.

We thus have the following

Theorem 6.2. The rank condition implies that the kernel is rank one.

The following Corollary is immediate, and is stated here only for completeness.

Corollary 6.4. Let G be a strongly connected, aperiodic graph. Suppose G admits a coloring semigroup $S = \langle R, B \rangle$ such that the rank condition holds for the difference matrix $\Delta = \frac{1}{2}(R - B)$ associated with the coloring. Then S contains a synchronizing instruction.

6.7 Coloring and recoloring

The rank condition not holding means that for every coloring ϕ, there is a polynomial p_ϕ, of degree at most d, with $p_\phi(0) = 0$, such that $\Omega p_\phi(\phi\Delta) = 0$.

Introduce variables ε_i, $\varepsilon_i^2 = 1$. Then generically

$$\phi = \mathrm{diag}(\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n) = \begin{pmatrix} \varepsilon_1 & 0 & \cdots & 0 \\ 0 & \varepsilon_2 & \cdots & 0 \\ & & \ddots & \\ 0 & 0 & \cdots & \varepsilon_n \end{pmatrix}$$

We want the matrices $\{\Omega\Delta, \Omega\Delta^2, \ldots, \Omega\Delta^d\}$ to be linearly independent. Since these are all KROSPS, we form a matrix H whose i^{th} row is $\pi(\phi\Delta)^i$. Then the rank of H is d if and only if $\det HH^* \neq 0$.

Remark 6.4. Note that Ω is always linearly independent of the $\Omega\Delta^i$, just apply to the all 1's vector. So if $d = n-1$, you can include π as the first row of H and get an $n \times n$ matrix. Then you only need to check $\det H$.

Procedure

Initialize $\phi_0 = \mathrm{diag}(1, 1, \ldots, 1)$.

1. Form HH^*.
2. Calculate $D(\phi) = \det HH^*$.
3. If $D(\phi_0) \neq 0$, DONE else continue.
4. Compute $\nabla_\varepsilon D(\phi_0)$.

5. If any component of $\nabla_\varepsilon D(\phi_0)$ is nonzero, recolor the corresponding vertex, DONE else continue.
6. If all components of the gradient are zero, randomly pick a new ϕ_0. Repeat from Step 3.

6.8 Main Result

In the remainder of the paper we will make the assumption that the graph G is 2-out. This assumption simply allows us to minimize notation. The argument can be extended to the d-out case.

Theorem 6.3. Let A be the adjacency matrix of a strongly connected, 2-out digraph. Suppose $\mathscr{S} = \langle r, b \rangle$, where $A = r + b$, is a minimal coloring semigroup with kernel K. Suppose $K = G \times Y$ is a right group, where rank $(K) = t$. Then A is periodic of order t.

Proof. Since K is a right group, for any two idempotents e_1, e_2, both $e_1 e_2 = e_2$ and $e_2 e_1 = e_1$. From this it follows that the partition of each function in K is constant. That is, for all $k_1, k_2 \in K$,

$$\pi_{k_1} = \pi_{k_2} = \pi_K \text{ where}$$

$$\pi_K = \{Q_1, \ldots, Q_t\}$$

and

$$Q_i = k^{-1}(v_i), \qquad v_i \in \text{Range(k)}, \quad i = 1, \ldots, t.$$

Now A is periodic of some order, say p. Thus, there is a partition of the vertices $\{P_1, \ldots, P_p\}$ such that the graph can be written as in Figure 6.2. That is, an edge goes from vertex v to w iff $v \in P_i$ and $w \in P_{i+1}$ $(P_{p+1} \equiv P_1)$. The partition π_k is a refinement of $\{P_1, \ldots, P_p\}$. That is, for each i, $Q_i \subset P_j$ for some j. This is clear, since if $k \in K$, $Q_i k = \{v\}$. Thus, there is a path connecting every vertex of Q_i to vertex v. Thus, $Q_i \subset P_j$, for some j.

Fig. 6.2 The periodic partition of A.

Next we show that both r, b are 1 to 1 functions on the sets $\{Q_i : i = 1, \ldots, t\}$. Suppose not. Then for some $v_1 \in Q_{i_1}$, $v_2 \in Q_{i_2}$,

Fig. 6.3 The graph of the function r.

$$v_1 r \in Q_j$$
$$v_2 r \in Q_j,$$

but then if $k \in K$,

$$v_1 rk \in Q_j k$$
$$v_2 rk \in Q_j k,$$

but since k is an element of the kernel, $Q_j k$ is a single vertex. Since v_1, v_2 get mapped to the same vertex by rk (also a member of the kernel), they must have started from the same element in the partition. Thus $Q_{i_1} = Q_{i_2}$ and r is a one-to-one set function. Of course, the same argument is valid for b.

Notice that this shows that there must be exactly the same number of the Q_i's in each of the periodic classes $P_1, \ldots, P_p$. This is true, since if there are n of the Q_i's in P_1, $Q_1, \ldots, Q_n$, then each of these are mapped by r into the n Q-classes in P_2. Then n Q-classes in P_3 are then images of these, etc. (See Figure 6.3.) Observe, the above argument shows $p|t$. Also in Figure 2, we have renumbered the Q-classes, Q_{ij}, $i = 1, \ldots, p$, $j = 1, \ldots, n$ with

$$(Q_{ij})r \subset Q_{i+1,j}.$$

Since we have fixed the order of the Q-classes in P_1, the mapping from P_p to P_1 is some arbitrary permutation, σ, that is

$$(Q_{p,1})r \subset Q_{1,\sigma(1)}$$

Next we show that if $r = b$ as set functions, then $p = t$.

Suppose $r = b$. Choose Q_{11} as the "initial state" and consider the cycle induced by powers of r, b

$$Q_{11} r \subset Q_{21} \quad (\text{also, } Q_{11} b \subset Q_{21})$$
$$Q_{21} r \subset Q_{31}, \quad \text{etc.}$$

Suppose the above cycle is of length $\ell < t$. Then since r, b are permutations (as set functions on the Q-classes) they are the product of disjoint cycles. In

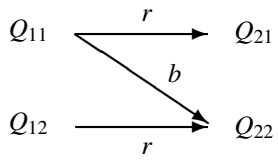
this case the graph would be disconnected, producing a contradiction. Thus, $\ell = t$.

Clearly, if c is any simple cycle within the graph, then $|c| = mt$ for some integer m. But since, if $c_1, \ldots, c_j$ is the set of all simple cycles, and

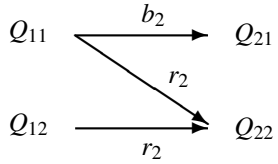
$$p = \gcd\{|c_1|, |c_2|, \ldots, |c_j|\},$$

then $t \leq p$. Thus we have shown that if $r = b$, then the graph is periodic of order t.

Now suppose that $r \neq b$. We must have the following. (WLOG)



Now recolor in the following way.



For all other edges, $r_2 = r$, $b_2 = b$.

Let $e = g_1 \cdot g_2 \cdot \ldots \cdot g_n$ be any idempotent in K, where $g_i = r$ or b. Then for some $v_{11} \in Q_{11}$, $Q_{11}e = v_{11}$.

Form $g_1^1 \cdot g_2^1 \cdot \ldots \cdot g_n^1 \in S_2 = \langle r_2, b_2 \rangle$ in the following way. If $g_1 = r$, then $g_1^1 = b_2$ and if $g_1 = b$, then $g_1^1 = r_2$. For $2 \leq i \leq n$,

$$g_i^1 = \begin{cases} r_2 & \text{if } g_i = r \text{ and } Q_{11} g_1^1 \cdot \ldots \cdot g_{i-1}^1 \notin Q_{11} \\ b_2 & \text{if } g_i = b \text{ and } Q_{11} g_1^1 \cdot \ldots \cdot g_{i-1}^1 \notin Q_{11} \\ r_2 & \text{if } g_i = b \text{ and } Q_{11} g_1^1 \cdot \ldots \cdot g_{i-1}^1 \in Q_{11} \\ b_2 & \text{if } g_i = r \text{ and } Q_{11} g_1^1 \cdot \ldots \cdot g_{i-1}^1 \in Q_{11} \end{cases}$$

Thus, as in the second diagram above, change the color only after mapping into Q_{11}. Then $Q_{11} g_1^1 \cdot g_2^1 \cdot \ldots \cdot g_n^1 = v_{11}$ and we have collapsed the set Q_{11} to a single vertex.

Now $Q_{12} g_1^1 \cdot g_2^1 \cdot \ldots \cdot g_n^1 \subset Q_{1j}$ for some j. Then $Q_{1j}e = v_{1j} \in Q_{1j}$. As above, construct $g_1^2 \cdot \ldots \cdot g_n^2$ from e by recoloring it if Q_{1j} is mapped into Q_{11}. Now $Q_{12} g_1^1 \cdot g_2^1 \cdot \ldots \cdot g_n^1 \cdot g_1^2 \cdot \ldots \cdot g_n^2 = v_{1j}$ and we have collapsed all of Q_{12} to a single vertex.

Continue in this manner for each element Q_{ij} of the partition, each time recoloring e after mapping into Q_{11}, if this occurs. Since e collapses each element in the partition to a single vertex, at the end of this process we have constructed a function $h = g_1^1 \cdot \ldots \cdot g_n^1 \cdot g_1^2 \cdot \ldots \cdot g_n^2 \cdot \ldots \cdot g_1^t \cdot \ldots \cdot g_n^t$ such that $\operatorname{rank}(h) \leq t$.

Suppose $\text{rank}(h) = t$. Then either each vertex in the range of h is in a distinct Q class or not. If there is a Q-class with more than one vertex from the range of h, then appending e to h, once again recoloring as before, will collapse these vertices into a single vertex and reduce the rank to below t, thus contradicting minimality. If each Q-class has a single vertex in the range of h, appending first r_2 then e (appropriately recolored) to h will reduce the rank below t.

Thus, we have shown that r must equal b as set functions, and therefore the graph is periodic of order t. □

Chapter 7
Rank Tests, Cycles, and Potential Operators

The target-pair graph controls the fixed ranks of A and Δ, but does not determine the kernel rank or whether the positive powers $\pi\Delta^i$ span the row space of Δ. We now develop sufficient ranked criteria. Section C.1 separates these invariants from the changing ranks in a full recoloring run.

Notation in inherited proofs.

The inherited source sections below write N for the separation operator $J-N$ of the current convention, and N_0 for the current collision operator N. Their proofs must be read with this crosswalk until the formula-by-formula notation revision is complete.

7.1 Ranking

Definition 7.1. Given a vector v and matrix T, of rank d, define the span matrix, or, cycle matrix,

$$\Sigma(v,T) = \begin{pmatrix} — & vT & — \\ — & vT^2 & — \\ & \vdots & \\ — & vT^d & — \end{pmatrix}$$

with rows $\{vT, vT^2, \ldots, vT^d\}$.

Definition 7.2. Let w be any word in $\mathscr{S}$. Say that w is ranked if the set $\{\pi w, \pi w^2, \ldots, \pi w^d\}$ is linearly independent, where $d = \mathsf{rank}(w)$.

In other words, w is ranked if $\mathsf{rank}\,\Sigma(\pi, w) = \mathsf{rank}\,w$.

Theorem 7.1. If w is ranked, then the kernel, $\mathscr{K}$, has rank one.

Proof. Let $x_i = \pi w^i$ denote the ith row of $\Sigma(\pi, w)$, the corresponding cycle matrix. Let e be any idempotent in $\mathscr{K}$. Then by BM-F, we have

$$x_i e e^* = \frac{1}{r} u$$

Since every row of w is a linear combination of the x_i, every row of wee^* is constant. Since w is a function, a row of wee^* is some row of ee^*. Since $\pi wee^* = (1/r)u$, wee^* is not identically zero. So some row of ee^* is nonzero and constant. Hence that row consists of all 1's, which means that that vertex is in the same block of a partition of $\mathscr{K}$ as every other vertex. In other words, $\mathscr{K}$ is rank one. □

Let R have a single limiting cycle, C, of length c. Let η_R be the idempotent for the kernel of the semigroup generated by R. Define the closure of R, $\bar{R} = \eta_R R$. So $\bar{R}$ has rank c.

Let $w = \bar{R}$ in the above Theorem. Let Σ' be the result of deleting zero columns from $\Sigma(\pi, \bar{R})$. The condition that $\bar{R}$ be ranked becomes that the determinant of Σ' be nonzero. Σ' has the form CP, where C is a circulant and P is a permutation. We thus find, after a suitable renumbering of the first row of M', the required condition is that its finite Fourier transform never vanishes.

7.1.1 Ranking with Δ

Instead of working only with words from the semigroup, we can consider as well $\Sigma(\pi, \Delta)$. We say Δ is ranked if $\operatorname{rank}\Sigma(\pi, \Delta) = \operatorname{rank}\Delta$.

Theorem 7.2. If Δ is ranked, then the kernel, $\mathscr{K}$, has rank one.

Proof. Corollary 5.2 says $\pi\Delta N = 0$. In fact, Corollary 5.3 implies further that $\pi\Delta wN = 0$ for all $w \in \mathscr{S}$. Thus, $\pi\Delta^i N = 0$ for all $i \geq 1$. If Δ is ranked, then $\Delta N = 0$, and eq. (5.1) reduces to

$$ANA^* = N$$

Now Proposition 8.5 yields $N = 0$, i.e., $N_0 = J$, which means the kernel is rank one. □

7.1.2 Ranking with $\mathbf{A} + \varepsilon\Delta$

Consider mixtures of R and B other than 50-50. Taking $p, q \geq 0$, $p + q = 1$, we consider $pR + qB$. Take $|\varepsilon| \leq 1$ and set $p = (1+\varepsilon)/2, q = (1-\varepsilon)/2$. Then we

set

$$A_\varepsilon = pR + qB = A + \varepsilon\Delta$$

recovering R when $\varepsilon = 1$ and B when $\varepsilon = -1$. We say that A_ε is ranked if $\mathsf{rank}\,\Sigma(\pi, A_\varepsilon) = \mathsf{rank}\,A_\varepsilon$. Note that $\varepsilon = 0$ gives back A, which cannot be ranked. We have

Theorem 7.3. For $\varepsilon \neq 0$, if A_ε is ranked, then the kernel, $\mathscr{K}$, has rank one.

Proof. Calculate

$$\pi A_\varepsilon N = \pi(A + \varepsilon\Delta)N = \pi N = r'u$$

Similarly, expanding A_ε^i, for $i > 1$, apply π on the left, using $\pi A = \pi$. Except for A^i, we get terms of the form $\pi\Delta x N$ where x is in the semigroup algebra. Hence, all terms vanish except $\pi A^i N = \pi N = r'u$. If A_ε is ranked, then every row of $(A + \varepsilon\Delta)N$ is constant. I.e., $(A + \varepsilon\Delta)N = bJ$, where b is a diagonal matrix. Let e_i be the standard basis row vector. So the ith row is

$$e_i(A + \varepsilon\Delta)N = b_i u$$

Applying this to $\pi^\dagger$, using $N\pi^\dagger = r'u^\dagger$, yields $r'e_i u^\dagger = b_i$. I.e., $b = r'I$ or

$$(A + \varepsilon\Delta)N = r'J$$

Writing A_ε in terms of R and B, the ij entry of $A_\varepsilon N$ is

$$\underline{A_\varepsilon N}_{ij} = pN_{iR,j} + qN_{iB,j}$$

These are constant as j varies. Taking $j = iR$, $j = iB$ and recalling the symmetry of N, we have

$$pN_{iR,iB} = qN_{iR,iB}$$

Hence, if $p \neq q$, $N_{iR,iB} = 0$. So in row iR, the entries all vanish. By symmetry, $N_{iB,iR} = 0$ and the entries in row iB vanish. But every vertex i' is iR or iB for some i, so N vanishes identically, i.e., the kernel is rank one. □

7.2 Divisibility

Let R have a single limiting cycle, C, of length c. Let $\bar{R}$ denote the closure of R. The vector u_c is $1 \times c$ with all entries equal to 1.

Lemma 7.1. The sum $\bar{R} + \bar{R}^2 + \cdots + \bar{R}^c = u^\dagger \chi_C$.

Proof. Immediate since every vertex is either one step away from the cycle C or is in the cycle so visits each point of the cycle exactly once. □

Theorem 7.4. Let r denote the rank of the kernel, $\mathscr{K}$. Then $r|c$.

Proof. Let $\Sigma = \Sigma(\pi, \bar{R})$. Let e be any idempotent in $\mathscr{K}$. Then by BM-F, we have

$$\pi w e e^* = \frac{1}{r} u$$

for any word in $\mathscr{S}$. Hence,

$$u_c \Sigma e e^* = u_c (\frac{1}{r} u_c{}^\dagger u) = \frac{c}{r} u$$

On the other hand, by the Lemma,

$$u_c \Sigma e e^* = \pi u^\dagger \chi_C e e^* = \chi_C e e^*$$

which has integer entries. Hence the result. □

7.3 Applications

7.3.1 Prime Cycle

If c, the length of a single limiting cycle for R, is prime, then the Fourier condition means that that the first row of $\Sigma(\pi, \bar{R})'$ is not constant. Using BM-F, we know there is an aperiodic vertex. If not already synchronized, recoloring there guarantees that the probabilities on the cycle C are nonconstant. Hence there is a kernel which is rank one.

7.3.2 Friedman

Let W denote the Friedman weight. I.e., rationalize π to $\tilde{\pi}$ which has integer entries with gcd equal to 1. These are the weights. The Friedman weight is the sum of those weights. From BM-F it immediately follows that r, the rank of the kernel, divides W. If c and W are relatively prime, then since r divides both c and W, the kernel $\mathscr{K}$ has rank one.

7.4 A potential operator and changes of letter probabilities

Let R, B be stochastic matrices and suppose $A = (R+B)/2$ is irreducible and aperiodic, with $\Omega = \mathbf{1}\pi$. Set

$$\Delta = (R-B)/2, \qquad Z = (I - A + \Omega)^{-1}, \qquad V = \Delta Z.$$

Since $\Delta\mathbf{1} = 0$, we have $V\mathbf{1} = 0$, $V\Omega = 0$, and

$$V = \lim_{N\to\infty} \Delta \sum_{j=0}^{N} A^j.$$

Indeed $\rho(A-\Omega) < 1$ and $\Delta\Omega = 0$. For an irreducible periodic A, retain the algebraic definition and the Abel limit $V = \lim_{t\uparrow 1} \Delta(I-tA)^{-1}$, rather than this spectral-gap argument.

Proposition 7.1 (Real-parameter invertibility). Under the standing hypotheses,

$$I - tV \text{ is invertible for every } t \in \mathbb{R} \text{ with } |t| < 1.$$

This is the replacement for Proposition 4.2 of V-Operator.pdf; no bound on the moduli of the nonreal eigenvalues of V is asserted.

Proof. Put $A_t = A + t\Delta$. For real $|t| < 1$, this is an irreducible stochastic matrix, since its positive entries have the same support as those of A. Suppose a row x satisfies $x(I-tV) = 0$. Since $V\mathbf{1} = 0$, we have $x\mathbf{1} = 0$, hence $x\Omega = 0$. Multiplying by $Z^{-1} = I - A + \Omega$ gives

$$x(I - A + \Omega) = tx\Delta,$$

so $xA_t = x$. The stationary row space of A_t is one-dimensional; its normalized stationary row has sum one. Therefore its only zero-sum member is zero, and $x = 0$. Thus $I - tV$ is invertible.

Theorem 7.5. For real $-1 < \tau < 1$, put $A_\tau = A + \tau\Delta$. Then

$$\Omega_\tau = \Omega(I - \tau V)^{-1}, \qquad \pi_\tau = \pi(I - \tau V)^{-1}.$$

At $\tau = 1$, $I - V$ is invertible exactly when R has a unique stationary distribution. At $\tau = -1$, $I + V$ is invertible exactly when B does. In these cases the corresponding formula holds at the endpoint, using the Cesaro projection of the endpoint matrix.

Proof. For interior real τ, A_τ has the same support as A, hence a unique stationary distribution. Also

$$(I - \tau V)Z^{-1} = I - A_\tau + \Omega.$$

If a row x annihilates the matrix on the right, multiplying by $\mathbf{1}$ gives $x\mathbf{1} = 0$, after which $xA_\tau = x$ forces $x = 0$. Thus the matrices are invertible. The stationary equations give $\pi_\tau - \pi = \tau\pi_\tau V$, proving the formula. At an endpoint, the same argument works precisely when the stationary row space has dimension one. If it has higher dimension it contains a nonzero zero-sum row, which gives a left nullvector of $I - A_\tau + \Omega$.

7.4.1 A necessary correction concerning the complex spectrum

Invertibility for real $|\tau| < 1$ does not imply $\rho(V) \le 1$. The spectral-radius assertion in Proposition 4.2 of the supplied V-Operator.pdf must not be used as stated. For

$$R = [23451], \qquad B = [51234],$$

A is irreducible and aperiodic, while exact calculation gives

$$\det(\lambda I - V) = \frac{\lambda}{5}(5\lambda^4 + 10\lambda^2 + 1), \qquad \rho(V) = \cot(\pi/5) > 1.$$

The proof in the source uses stochastic uniqueness and therefore applies to real parameters only. A Neumann series in τV is justified locally for $|\tau|\rho(V) < 1$, not on the entire unit disk in general. Nevertheless Theorem 7.5 holds throughout the real interval. In particular,

$$\left.\frac{d^k \pi_\tau}{d\tau^k}\right|_{\tau=0} = k!\pi V^k.$$

7.4.2 Endpoint cycle measures and a wave operator

If R has a unique stationary distribution, define

$$U = (I+V)(I-V)^{-1}.$$

Then

$$\Omega_B U = \Omega_R, \qquad U(I-R) = I - B.$$

The endpoint identity $\Omega_B(I+V) = \Omega$ holds even when B has more than one recurrent class, so the first formula does not require uniqueness for B. For a deterministic generator with one limiting cycle C, its Cesaro projection is $\Omega_R = \mathbf{1}\chi_C/|C|$. For any k in the kernel, $k\Omega_R = \Omega_R$, and the Cesaro averages of kR^j belong to the convex hull of the finite kernel. Thus Ω_R belongs to that convex hull, hence to its linear span. The equal intersection counts $|C \cap F| = |C|/r$ for kernel blocks F connect this construction to the cycle-divisibility results already developed in this chapter.

7.4.3 An explicit partial-circulant potential

For distinct indices a, b, take $R = M_a$, $B = M_b$ and use the unique recurrent core. The potential has rank one and satisfies

$$V^2 = \lambda V, \qquad \lambda = \frac{b-a}{a+b}.$$

Consequently

$$(I-\tau V)^{-1} = I + \frac{\tau}{1-\tau\lambda} V, \qquad U = I + \frac{2}{1-\lambda} V.$$

To see the scalar, write $V = e_1 w$, so $V^2 = w_1 V$. The ladder formula gives

$$\pi_\tau(j) = \frac{(1+\tau)\mathbf{1}_{j\le a} + (1-\tau)\mathbf{1}_{j\le b}}{(1+\tau)a + (1-\tau)b}.$$

On the other hand the rank-one inverse gives $\pi_\tau(1) = \pi(1)/(1-\tau w_1)$. Comparing denominators yields $w_1 = (b-a)/(a+b)$. The formula is zero beyond $\max(a,b)$; transient states are included in the operator but carry no stationary mass. For periodic cores use the algebraic or Abel version of the potential.

7.4.4 Adjugate and notation qualifications

The adjugate identity in the source requires eigenvalue 1 to be algebraically simple, not merely to have a one-dimensional eigenspace. Under that hypothesis, if $p(t) = \det(I - tX)$ and P_X is the spectral projection, then $\operatorname{adj}(I-X) = -p'(1)P_X$. For a stochastic matrix with a unique stationary distribution the hypothesis holds. The source's final $N_0 = \mathbb{E}[EE^{\mathsf{T}}]$ is the collision operator denoted N in our present convention.

Part III
Higher representations and constructions

Chapter 8
Tensor, Symmetric, and Zeon Hierarchies

8.1 From particle configurations to the zeon quotient

Let V have basis $e_1, \dots, e_n$ over $\mathbb{Q}$. A deterministic transformation f sends e_i to e_{if} and acts on tensor powers by

$$(e_{i_1} \otimes \cdots \otimes e_{i_k}) f^{\otimes k} = e_{i_1 f} \otimes \cdots \otimes e_{i_k f}.$$

This is simultaneous common-transformation motion of ordered particles. Passing to $\mathrm{Sym}^k(V)$ forgets order but retains multiplicity: repeated factors remain valid configurations.

8.1.1 The invariant collision filtration

Let F_ℓ be the span of degree-k monomials with at most ℓ distinct occupied vertices. Deterministic transformations cannot increase the number of distinct images, so

$$0 \subseteq F_1 \subseteq \cdots \subseteq F_k = \mathrm{Sym}^k(V)$$

is an invariant filtration. The collision-free span itself need not be invariant: distinct particles may collide. For $1 \le k \le n$, the quotient F_k/F_{k-1} has basis the squarefree monomials e_I, indexed by k-subsets, with action

$$e_I f^{\vee k} = \begin{cases} e_{If}, & |If| = k, \\ 0, & |If| < k. \end{cases}$$

This is the degree-k zeon representation. Thus zeons isolate the collision-free layer of the symmetric-power action rather than introduce a different deterministic dynamics. Lost cardinality cannot be restored, which proves $(fg)^{\vee k} = f^{\vee k} g^{\vee k}$. An equivalent set-theoretic construction is the quotient of

the module on subsets of size at most k by its submodule on subsets of size at most $k-1$. This is not an exterior power: permutation signs are absent.

8.1.2 Averages and limiting projections

For deterministic generators M_i chosen with probabilities p_i, put

$$A_k = \sum_i p_i M_i^{\vee k}, \qquad \Omega_k = \lim_{m\to\infty} \frac{1}{m} \sum_{t=0}^{m-1} A_k^t.$$

Each summand preserves the collision filtration before taking the quotient. One must average these induced actions, not substitute the averaged one-particle matrix into a symmetric-power formula. A general stochastic linear map need not preserve the collision subspace of the symmetric power. The independent-particle permanent construction is treated separately.

The basic observables remain visible in the full tensor construction. Writing

$$\Phi(X) = \sum_i p_i M_i X M_i^{\mathsf{T}}, \qquad \Psi(X) = \sum_i p_i M_i^{\mathsf{T}} X M_i,$$

the limiting kernel measure gives

$$N = \lim_{m\to\infty} \frac{1}{m} \sum_{t=0}^{m-1} \Phi^t(I), \qquad M = \lim_{m\to\infty} \frac{1}{m} \sum_{t=0}^{m-1} \Psi^t(J).$$

At zeon level two, $u_2 = \Omega_2 \mathbf{1}$ records survival of a distinct pair, and its symmetric, zero-diagonal matrix realization is

$$\boxed{\mathrm{Mat}(u_2) = J - N.}$$

The full projection Ω_2 acts on pair space; $J-N$ is reconstructed from its right field and is not that full projection. The P_j, Q_i projections of Section 5.1 act on the original state space and have yet another role. The following sections develop the higher fields and their compatible inclusion maps within this framework.

Inherited operator convention.

The inherited blocks in this chapter retain their source notation: $N_0^{\mathrm{old}} = N$, $N^{\mathrm{old}} = J - N$, $M_0^{\mathrm{old}} = M$, and $M^{\mathrm{old}} = M - \tau\omega$, where $\tau = \mathrm{tr} M$. The superscripts in this crosswalk are editorial only; the inherited formulas below still print the old symbols. A token-level mathematical notation revision, not a global text substitution, remains necessary.

8.2 Zeon algebra and structure of the kernel

8.2.1 Introduction

The notation $\mathbf{diag}(v)$ denotes the diagonal matrix with entries v_i of the vector $\mathbf{v}$.

The theta symbol for pairs of single indices

$$\theta_{ij} = \begin{cases} 1, & \text{if } i < j \\ 0, & \text{otherwise} \end{cases}$$

We indicate the component of a vector by underlining and then subscripting, e.g. $\underline{Tv}_i$ is the ith component of the result of the matrix T acting on the vector v.

$\mathscr{V}$ is the vector space isomorphic to $\mathbb{Q}^n$. Denote by $\mathbf{Sym}_n(\mathscr{V})$ the space of symmetric matrices acting on $\mathscr{V}$.

Denote by $\mathscr{Z}$ the algebra of zeons, generated by n commuting variables e_i such that $e_i^2 = 0$, $1 \le i \le n$. Then we identify $\mathscr{V}$ with the $\mathbb{Q}$-span of the zeons and generate an algebra, $\mathscr{Z}$. The kth zeon tensor power of $\mathscr{V}$ is denoted $\mathscr{V}^{\vee k}$. Effectively $\mathscr{V}^{\vee k}$ is the space of symmetric tensors with no repeated factors. Elements of $\mathscr{V}^{\vee k}$ are indexed by k-tuples of positive integers all distinct. I.e., the space has dimension $\binom{n}{k}$. We will denote multi-indices by roman capital letters $\mathrm{I} = (i_1, i_2, \ldots, i_k)$, J, K, etc. A linear operator W on $\mathscr{V}$ extends to the operator $W^{\vee k}$ acting on $\mathscr{V}^{\vee k}$. The zeon Fock space,

$$\mathscr{F} = \mathbb{Q} \oplus (\bigoplus_{k \ge 1} \mathscr{V}^{\vee k})$$

is finite dimensional, like the exterior algebra, and the second quantization of W is the extension of W to $\mathscr{F}$. Of course, one could work over a different field such as $\mathbb{C}$ for quantum mechanics with zeons.

The IJ component of $W^{\vee k}$ is the permanent of the corresponding submatrix of W with rows indexed by I and columns by J. For dimension 2, we will use indexes ij instead of (i_1, i_2), as usual.

For given n, X, Y, etc., are vectors in $\mathscr{V}^{\vee 2} \approx \mathbb{Q}^{\binom{n}{2}}$. The mapping $\mathbf{Mat}\colon \mathscr{V}^{\vee 2} \to \mathbf{Sym}_n(\mathscr{V})$ is a linear embedding taking the vector $X = (x_{ij})$ to the symmetric matrix $\hat{X}$ with off-diagonal components $\hat{X}_{ij} = x_{ij} = \hat{X}_{ji}$, for $i < j$ and zeros on the diagonal, $\hat{X}_{ii} = 0$, $1 \le i \le n$. We will use the explicit notation $\mathbf{Mat}(X)$ as needed for clarity.

We equip $\mathscr{V}^{\vee 2}$ with the inner product

$$\langle X, Y \rangle = X_{\lambda\mu} Y_{\lambda\mu} = \mathrm{tr}\hat{X}\hat{Y} \ .$$

8.2.2 Basic Identities

Multiplying X with u, we observe that

$$Xu^\dagger = \frac{1}{2}\operatorname{tr}\hat{X}J = uX^\dagger \tag{8.1}$$

Observe also that if D is diagonal, then $\operatorname{tr}D = \operatorname{tr}DJ$.

Proposition 8.1. (Basic Relations) We have

1. $\operatorname{Mat}(XA^{\vee 2}) = A^*\hat{X}A - D^+$, where D^+ is a diagonal matrix satisfying $\operatorname{tr}D^+ = \operatorname{tr}A^*\hat{X}A$.
2. $\operatorname{Mat}(A^{\vee 2}X^\dagger) = A\hat{X}A^* - D^-$, where D^- is a diagonal matrix satisfying $\operatorname{tr}D^- = \operatorname{tr}A\hat{X}A^*$.
3. If A and X have nonnegative entries, then D^+ and D^- have nonnegative entries. In particular, in that case, vanishing trace for $D^\pm$ implies vanishing of the corresponding matrix.

Proof. The components of $XA^{\vee 2}$ are

$$\begin{aligned}\underline{XA^{\vee 2}}_{ij} &= \theta_{ij}\theta_{\lambda\mu}(x_{\lambda\mu}A_{\lambda i}A_{\mu j} + x_{\lambda\mu}A_{\mu i}A_{\lambda j}) \\ &= \theta_{ij}x_{\lambda\mu}A_{\lambda i}A_{\mu j} = \theta_{ij}\underline{A^*\hat{X}A}_{ij}\end{aligned}$$

noting that the diagonal terms of $\hat{X}$ vanish anyway. And $A^*\hat{X}A$ will be symmetric if $\hat{X}$ is. Since the left-hand side has zero diagonal entries, we can remove the theta symbol and compensate by subtracting off the diagonal, call it D^+. Taking traces yields #1. And #2 follows similarly.

Remark 8.1. Observe that D^+ and D^- may be explicitly given by

$$\begin{aligned}D^+_{ii} &= x_{\lambda\mu}A_{\lambda i}A_{\mu i} \\ D^-_{ii} &= x_{\lambda\mu}A_{i\lambda}A_{i\mu}\end{aligned}$$

where for D^+ the A elements are taken within a given column, while for D^-, the A elements are in a given row.

Proposition 8.2. Let X and A be nonnegative. Then

$$\begin{aligned}\hat{X} = A^*\hat{X}A &\Rightarrow XA^{\vee 2} = X \\ \hat{X} = A\hat{X}A^* &\Rightarrow A^{\vee 2}X^\dagger = X^\dagger\end{aligned}$$

Proof. Start with

$$D^+ = A^*\hat{X}A - \operatorname{Mat}(XA^{\vee 2})$$

with D^+ vanishing if $\mathrm{tr}D^+ = \mathrm{tr}A^*\hat{X}A$ vanishes. If $\hat{X} = A^*\hat{X}A$, then since $\hat{X}$ has vanishing trace, $\mathrm{tr}A^*\hat{X}A = 0$. Thus, $\mathrm{tr}D^+ = 0$ and $\hat{X} = A^*\hat{X}A = \mathrm{Mat}(XA^{\vee 2})$. The second implication follows similarly.

8.2.3 Trace Identities

Using equation (8.1), we will find some identities for these quantities.

Proposition 8.3. We have

1. $XA^{\vee 2}u^\dagger = \frac{1}{2}\,\mathrm{tr}\,(\hat{X}A(J-I)A^*)$.
2. $uA^{\vee 2}X^\dagger = \frac{1}{2}\,\mathrm{tr}\,(\hat{X}A^*(J-I)A)$.
3. If A is stochastic, then $XA^{\vee 2}u^\dagger = \frac{1}{2}\,\mathrm{tr}\,(\hat{X}(J-AA^*))$.

Proof. We have, using equation (8.1) and Basic Relation 1,

$$\begin{aligned} XA^{\vee 2}u^\dagger &= \tfrac{1}{2}\,\mathrm{tr}\,\mathrm{Mat}(XA^{\vee 2})J \\ &= \tfrac{1}{2}\,\mathrm{tr}\,(A^*\hat{X}AJ - D^+J) \\ &= \tfrac{1}{2}\,\mathrm{tr}\,(A^*\hat{X}AJ - A^*\hat{X}A) \end{aligned}$$

and rearranging terms inside the trace yields #1. Then #3 follows since A stochastic implies $AJ = J = JA^*$. And #2 follows similarly, using the second Basic Relation in the equation $uA^{\vee 2}X^\dagger = \frac{1}{2}\,\mathrm{tr}\,\mathrm{Mat}(A^{\vee 2}X^\dagger)J$. □

Using equation (8.1) directly for X, we have

$$X(I-A^{\vee 2})u^\dagger = \tfrac{1}{2}\,\mathrm{tr}\,(\hat{X}(J-AJA^*+AA^*)) \tag{8.2}$$

$$u(I-A^{\vee 2})X^\dagger = \tfrac{1}{2}\,\mathrm{tr}\,(\hat{X}(J-A^*JA+A^*A)) \tag{8.3}$$

8.2.3.1 Stochastic case

For stochastic A, equation (8.2) yields

$$X(I-A^{\vee 2})u^\dagger = \tfrac{1}{2}\,\mathrm{tr}A^*\hat{X}A \tag{8.4}$$

The above formula is "integration-by-parts for zeons".

We denote an invariant distribution for A by π, i.e. $\pi A = \pi = (p_1, \ldots, p_n)$. We now have a converse result to Proposition 2.

Proposition 8.4. Let A be stochastic and X nonnegative.

1. $XA^{\vee 2} = X \Rightarrow \hat{X} = A^*\hat{X}A$.
2. If A has a strictly positive invariant distribution, π, then

   $A^{\vee 2}X^\dagger = X^\dagger \Rightarrow \hat{X} = A\hat{X}A^*$.

Proof. If $XA^{\vee 2} = X$, then Proposition 8.1, eq. (5.1), reads

$$\hat{X} = A^* \hat{X} A - D^+$$

Multiply by $u^\dagger$ on the right and by u on the left, using $Au^\dagger = u^\dagger$ and $uA^* = u$ to get

$$u\hat{X}u^\dagger = u\hat{X}u^\dagger - uD^+u^\dagger$$

So $uD^+u^\dagger = 0$ and taking traces yields $\mathrm{tr}\,(D^+J) = 0$, thus $D^+ = 0$.

If $A^{\vee 2}X^\dagger = X^\dagger$, then Proposition 8.1, eq. (5.2), reads

$$\hat{X} = A\hat{X}A^* - D^-$$

Multiply by $\pi^\dagger$ on the right and by π on the left, using $\pi A = \pi$ and $A^*\pi^\dagger = \pi^\dagger$ to get

$$\pi\hat{X}\pi^\dagger = \pi\hat{X}\pi^\dagger - \pi D^-\pi^\dagger$$

So $\pi D^-\pi^\dagger = 0$ and taking traces yields $\mathrm{tr}\,(D^-\pi^\dagger\pi) = 0$. I.e., $\sum_i D^-_{ii} p_i^2 = 0$, so $D^- = 0$. □

With A stochastic, irreducible and aperiodic, $\lim_{n\to\infty} A^n = \Omega = u^\dagger\pi$, satisfying $\Omega^2 = \Omega = A\Omega = \Omega A$. Observe that

$$\begin{aligned}\Omega\Omega^* &= u^\dagger\pi\pi^\dagger u = (\textstyle\sum p_i^2)J \\ \Omega^*\Omega &= \pi^\dagger uu^\dagger\pi = n\,\pi^\dagger\pi\end{aligned}$$

both of these having all positive entries. Thus

Proposition 8.5. Let A be stochastic, irreducible and aperiodic. Then, for nonnegative X, either of $\hat{X} = A\hat{X}A^*$ or $\hat{X} = A^*\hat{X}A$ implies $X = 0$.

Proof. If $\hat{X} = A\hat{X}A^*$, then, inductively for positive integers k,

$$\hat{X} = A^k\hat{X}(A^*)^k$$

Letting $k \to \infty$, we have $\hat{X} = \Omega\hat{X}\Omega^*$. Taking traces yields $\mathrm{tr}\,\hat{X}\Omega^*\Omega = 0$. Since $\hat{X}$ is nonnegative and $\Omega^*\Omega$ is a matrix with positive entries, we have $X = 0$. The proof for $\hat{X} = A^*\hat{X}A$ is similar. □

Now we have

Theorem 8.1. Let $A^{\vee 2} \neq 0$. Then
A is aperiodic and irreducible if and only if $\det(I - A^{\vee 2}) \neq 0$.

Remark 8.2. Note that the hypothesis is saying that A is not the identity matrix, i.e., that the chain moves at least one state to some other one.

Proof. If A is reducible, let $\{C_0,\dots,C_{s-1}\}$ denote the closed classes partitioning the states. Define X by $x_{ij}=0$ if there is a k such that $i,j\in C_k$, 1 otherwise, i.e.,

$$X = 1-\sum_k \chi_{C_k\times C_k}$$

We will show that $A^{\vee 2}X^\dagger = X^\dagger$.

Step 1. First check that the diagonal vanishes, i.e., $A_{i\lambda}x_{\lambda\mu}A_{i\mu}=0$. By assumption, if $i\in C_k$, then $\lambda,\mu\in C_k$. But then $x_{\lambda\mu}=0$.

Step 2. For $i\neq j$, $\underline{A^{\vee 2}X^\dagger}_{ij} = A_{i\lambda}x_{\lambda\mu}A_{j\mu}$. If $i\in C_k$, then the sum is over $\lambda\in C_k$. If $j\in C_k$, then the sum is over $\mu\in C_k$ as well, $x_{\lambda\mu}=0$, and the result is zero. If $j\in C_{k'}$, $k'\neq k$, then $x_{ij}=1$. Writing out the sum, we have

$$\sum_{l\in C_k, m\in C_{k'}} A_{il}x_{lm}A_{jm}$$

For fixed m, $x_{lm}=1$, and the sum over $l\in C_k$ gives all choices that i could map to, while for $l\notin C_k$, $A_{il}=0$, so $\sum_{l\in C_k} A_{il}=1$. Similarly, summing over m gives 1. Hence, we get 1 in case $x_{ij}\neq 0$ and 0 otherwise, as required.

If A is periodic, with period p, say, let $\{C_0,\dots,C_{p-1}\}$ denote the classes partitioning the states such that $i\in C_k$ implies $iA\in C_{k+1}$, with indices modulo p. Define X by $x_{ij}=1$ if there exists k such that $i\in C_k$ and $j\in C_{k+1}$, 0 otherwise. We check that $A^{\vee 2}X^\dagger = X^\dagger$. The steps are similar to that for the case of reducibility:

Step 1. First check that the diagonal vanishes, i.e., $A_{i\lambda}x_{\lambda\mu}A_{i\mu}=0$. By periodicity, if $i\in C_{k-1}$, then $\lambda\in C_k$ and hence $\mu\in C_{k+1}$ by definition of X. But then $A_{i\mu}$ vanishes since $\mu\in C_{k+1}$ requires $i\in C_k$.

Step 2. For $i\neq j$, $\underline{A^{\vee 2}X^\dagger}_{ij} = A_{i\lambda}x_{\lambda\mu}A_{j\mu}$. If $i\in C_{k-1}$, then $\lambda\in C_k$, $\mu\in C_{k+1}$ requires $j\in C_k$ so $x_{ij}\neq 0$. Now, writing out the sum, we have

$$\sum_{l\in C_k, m\in C_{k+1}} A_{il}x_{lm}A_{jm}$$

For fixed m, $x_{lm}=1$, and the sum over l gives all choices that i could map to, i.e., $\sum_{l\in C_k} A_{il}=1$. And summing over m, gives 1 as well. Hence, we get 1 in case $x_{ij}\neq 0$ and 0 otherwise, as required.

Suppose A is aperiodic and irreducible. By Perron-Frobenius the top nonnegative eigenvalue has a nonnegative eigenvector, X. Proposition 8.4 says that a solution to $XA^{\vee 2}=X$ satisfies $\hat{X}=A^*\hat{X}A$. Then Proposition 8.5 implies that $X=0$.

8.2.4 Kernel structure

Let $A_2 = A^{\vee 2} + \Delta^{\vee 2} = \frac{1}{2}(R^{\vee 2} + B^{\vee 2})$.

For Δ we have, from Proposition 8.3,

$$X\Delta^{\vee 2}u^\dagger = \tfrac{1}{2}\operatorname{tr}(\hat{X}(\Delta J\Delta^* - \Delta\Delta^*))$$

which becomes

$$X\Delta^{\vee 2}u^\dagger = -\tfrac{1}{2}\operatorname{tr}\Delta^*\hat{X}\Delta$$

since $\Delta J = 0$. And Proposition 8.3, #2, yields

$$u\Delta^{\vee 2}X^\dagger = \tfrac{1}{2}\operatorname{tr}(\hat{X}(\Delta^*(J-I)\Delta)\ .$$

8.2.4.1 **M** and **N**

Note that $\Omega = \langle K\rangle$, the average element of the kernel. As well, $\Omega = \beta_\lambda \rho(k_\lambda)$, where k_i runs through any row, so that each range is represented once.

Let ω be the diagonal matrix with π on the diagonal, i.e. $\omega = \operatorname{diag}(\pi)$.

Proposition 8.6. $\langle K^*K\rangle = n\,\omega$.

Proof. For #1, start by observing that $\operatorname{diag}(v) = \operatorname{diag}(\operatorname{diag}(v)\mathrm{u}^\dagger)$. Thus, since k^*k is diagonal for every k, we check $\langle K^*Ku^\dagger\rangle = \langle K^*u^\dagger\rangle = \Omega^*u^\dagger = n\,\pi^\dagger$.

If Ω_2 is the limiting matrix of A_2, define $u_2 = \Omega_2 u^\dagger$ and $\pi_2 = u\Omega_2$. Then we have the following relations.

Proposition 8.7. For any X,

$$X\Omega_2 u^\dagger = Xu_2{}^\dagger = \langle X, u_2\rangle = \tfrac{1}{2}\operatorname{tr}\hat{X}\hat{u}_2 = \tfrac{1}{2}\operatorname{tr}\hat{X}N$$

where $N = \langle J - KK^*\rangle$, taken over the kernel.
And

$$u\Omega_2 X^\dagger = \pi_2 X^\dagger = \langle \pi_2, X\rangle = \tfrac{1}{2}\operatorname{tr}\hat{\pi}_2\hat{X} = \tfrac{1}{2}\operatorname{tr}\hat{X}\tilde{M}$$

where $\tilde{M} = \langle K^*(J-I)K\rangle$, taken over the kernel.

Proof. Using the homomorphism property of the map $\cdot^{\vee 2}$, we have $\Omega_2 = \langle K^{\vee 2}\rangle$ taken over the kernel. Thus, using Proposition 8.3 for stochastic matrices,

$$X\Omega_2 u^\dagger = \langle XK^{\vee 2}u^\dagger\rangle = \tfrac{1}{2}\operatorname{tr}\langle\hat{X}(J - KK^*)\rangle = \tfrac{1}{2}\operatorname{tr}\hat{X}N$$

Similarly,

$$u\Omega_2 X^\dagger = \langle uK^{\vee 2}X^\dagger\rangle = \tfrac{1}{2}\operatorname{tr}\langle\hat{X}K^*(J-I)K)\rangle = \tfrac{1}{2}\operatorname{tr}\hat{X}\tilde{M}\ . \qquad \square$$

Let $N_0 = \langle KK^* \rangle = \langle EE^* \rangle$, where K denotes a random element of $\mathscr{K}$, and E denotes a random idempotent, element of $\mathscr{E}(\mathscr{K})$. Define $N = J - N_0$. Since $\langle J \rangle = J$, we set $u_2 = \Omega_2 u^\dagger$, satisfying $A_2 u_2 = u_2$, and identify N as $\langle J - KK^* \rangle = \mathrm{Mat}(u_2)$. Note that $N = \alpha_\mu e_\mu e_\mu^*$ where the e_i are (all of) the idempotents with a given range, i.e., they can be chosen as the idempotents from a single column in the kernel.

Let $M_0 = \langle K^* J K \rangle$. Define $\tau = \mathrm{tr} M_0$ and $M = M_0 - \tau\omega$. Since M and $\tilde{M}$ agree on off-diagonal elements, we set $\pi_2 = u\Omega_2$, satisfying $\pi_2 A_2 = \pi_2$, and identify M as $\mathrm{Mat}(\pi_2)$.

Note that M_0, N_0, M, N as well as J are all symmetric matrices.

Define the mappings, $F, G\colon \mathrm{Sym}_n(\mathscr{V}) \to \mathrm{Sym}_n(\mathscr{V})$ by

$$F(Y) = AYA^* + \Delta Y \Delta^* \qquad \text{and} \qquad G(Y) = A^* Y A + \Delta^* Y \Delta \ .$$

Note that

$$F(Y) = \tfrac{1}{2}\left(RYR^* + BYB^*\right) \qquad \text{and} \qquad G(Y) = \tfrac{1}{2}\left(R^* Y R + B^* Y B\right) \ .$$

First, some observations

Proposition 8.8. For any vector v, $G(\mathrm{diag}(v)) = \mathrm{diag}(vA)$.

Proof. Let $Y = G(\mathrm{diag}(v))$. Then $2Y_{ij} = R_{\lambda i} v_\lambda R_{\lambda j} + B_{\lambda i} v_\lambda B_{\lambda j}$. A term like R_{li} means that R maps l to i. Since R is a function, l can only map to both i and j if $i = j$. So Y is diagonal. If $i = j$, then we get, using $R_{ij}^2 = R_{ij}$, $B_{ij}^2 = B_{ij}$,

$$2Y_{ii} = R_{\lambda i} v_\lambda + B_{\lambda i} v_\lambda = 2 v_\lambda A_{\lambda i}$$

as required.

For example, taking $v = u$, we see that

$$G(I) = I \text{ if and only if } A \text{ is doubly stochastic.}$$

Proposition 8.9. $F(J) = J$ and $G(\omega) = \omega$.

Proof. First, we have $F(J) = AJA^* + \Delta J \Delta^*$, with $AJ = J = JA^*$ and $\Delta J = 0$, this reduces to J. Second, from the previous Proposition, we have $G(\omega) = G(\mathrm{diag}(\pi))) = \mathrm{diag}(\pi A) = \mathrm{diag}(\pi) = \omega$.

Note that equation (8.4) applied to R and B and then averaged yields

$$X(I - A_2)u^\dagger = \tfrac{1}{2}\,\mathrm{tr}\, G(\hat{X}) \tag{8.5}$$

The next theorem shows the invariance of N and M under F and G respectively as well as the general relation between fixed points of A_2 and fixed points of F and G. Note that we are not restricting a priori to nonnegative solutions except in case 2 of the following theorem.

Theorem 8.2.

1. $F(\hat{X}) = \hat{X}$ if and only if $A_2 X^\dagger = X^\dagger$.
2. For nonnegative solutions X, $G(\hat{X}) = \hat{X}$ if and only if $XA_2 = X$.

Proof. First, from Proposition 8.1, applying the Basic Relations for R and B and averaging, we get

$$\mathrm{Mat}(XA_2) = G(\hat{X}) - \tfrac{1}{2}(D_R^+ + D_B^+) \tag{8.6}$$

$$\mathrm{Mat}(A_2 X^\dagger) = F(\hat{X}) - \tfrac{1}{2}(D_R^- + D_B^-) \tag{8.7}$$

where the subscripts on the D's indicate the diagonals corresponding to R and B respectively. Recalling the remark following Proposition 8.1, note that for D_R^-, say, we have terms like $R_{i\lambda} x_{\lambda\mu} R_{i\mu}$. Since R is a function, we must have $\lambda = \mu$, but then $x_{\lambda\mu} = 0$. In other words, D_R^- and D_B^- vanish so that, in fact, $\mathrm{Mat}(A_2 X^\dagger) = F(\hat{X})$. And #1 follows immediately.

For #2, if $G(\hat{X}) = \hat{X}$, then the trace of the diagonal terms in equation (8.6) vanish, so under the assumption of nonnegativity, they must vanish. And replacing $G(\hat{X})$ by $\hat{X}$, we have $XA_2 = X$.

Assume $XA_2 = X$. Taking traces in equation (8.6), the trace of the left-hand side vanishes and the trace of the diagonal terms equals $\mathrm{tr}\, G(\hat{X})$. Now, use the integration-by-parts formula for A_2, equation (8.5). The left-hand side vanishes, hence $\mathrm{tr}\, G(\hat{X}) = 0$, implying the vanishing of the terms D_R^+ and D_B^+ as well. Then equation (8.6) reads $\mathrm{Mat}(X) = G(\hat{X})$ as required. □

From this Theorem we have

With $A_2 = A^{\vee 2} + \Delta^{\vee 2} = \frac{1}{2}(R^{\vee 2} + B^{\vee 2})$, we have Ω_2 as the Cesàro limit of A_2^k. Then with u here denoting the all-ones vector of dimension $\binom{n}{2}$, $u_2 = \Omega_2 u^\dagger$ and $\pi_2 = u\Omega_2$. Then

1. $N = \langle J - KK^* \rangle = \mathrm{Mat}(u_2)$ satisfies $ANA^* + \Delta N \Delta^* = N$. It is a nonnegative solution with $\mathrm{tr}\, N = 0$. $N_0 = \langle KK^* \rangle$ and $J = u^\dagger u$ are also nonnegative solutions, with trace equal to n.

2. $M = \langle K^*(J - (\tau/n)I)K \rangle = \mathrm{Mat}(\pi_2)$ satisfies $A^* M A + \Delta^* M \Delta = M$. It is a nonnegative solution with $\mathrm{tr}\, M = 0$. $M_0 = \langle K^* J K \rangle$ and $\omega = \mathrm{diag}(\pi)$ are also nonnegative solutions, with trace equal to τ and 1 respectively.

Note that in Proposition 8.7 we get $\tilde{M} = \langle K^*(J - I)K \rangle$. However, $\mathrm{tr}\, \tilde{M}$ will not be zero. To cancel off the diagonal of M_0, we need to subtract off a multiple of ω. From Proposition 8.6, we see that an appropriate multiple of K^*K will work.

8.2.4.2 Determinants and traces

Let $X = e_1 \operatorname{adj}(I - A^{\vee 2})$ in equation (8.4). Since $X(I - A^{\vee 2})u^{\dagger} = \det(I - A^{\vee 2})$, we get

$$\det(I - A^{\vee 2}) = \tfrac{1}{2} \operatorname{tr} A^* \hat{X} A$$

Similarly, let $X = e_1 \operatorname{adj}(I - A_2)$ in equation (8.5). We get

$$\det(I - A_2) = \tfrac{1}{2} \operatorname{tr} G(\hat{X}) = \tfrac{1}{2} \operatorname{tr} (\hat{X}(AA^* + \Delta\Delta^*))$$

Note that if $\det(I - A_2) = 0$, then, if nonvanishing, $X = e_1 \operatorname{adj}(I - A_2)$ will be a nontrivial solution to $XA_2 = X$. That is, if the solution space is one-dimensional, this is a multiple of π_2 and we scale to $\hat{X} = M$. Then the vanishing of the trace will be equivalent to the vanishing of the determinant, i.e., the existence of solutions to $XA_2 = X$.

8.3 Levels

Let π be the invariant measure for $A = \frac{1}{2}(R + B)$. Denote the components of π by $(p_1, \ldots, p_n)$.

r always denotes the rank of the kernel, $\mathscr{K}$, of the semigroup $S = \langle R, B \rangle$.

Indices, even though denoted as κ-tuples are sets, i.e., unordered, distinct elements, denoted using capitals. [Note: read κ as "kay".]

All vectors are row vectors. Transposes will be denoted by superscript plus, so that a typical column vector is v^+. We use roman type for the identity matrix.

For $1 \le \kappa \le r$, let $R_1 = R$, R_κ denotes the mapping on κ-tuples induced by R. I.e., for κ-tuples $\mathrm{I} = (i_1, \ldots, i_\kappa)$, $\mathrm{J} = (j_1, \ldots, j_\kappa)$,

$$R_\kappa(\mathrm{I}, \mathrm{J}) = \begin{cases} 1, & \text{if } R \text{ maps the set of vertices I to the set J} \\ 0, & \text{otherwise} \end{cases}$$

Similarly for B_κ. Then, we have

$$A_\kappa = \tfrac{1}{2}(R_\kappa + B_\kappa), \qquad \Delta_\kappa = \tfrac{1}{2}(R_\kappa - B_\kappa)$$

For each κ, R_κ, B_κ, and A_κ are substochastic matrices.

Note that $r \le \min(\operatorname{rank}(R), \operatorname{rank}(B))$ and is the maximal index κ such that the semigroup $S_\kappa = \langle R_\kappa, B_\kappa \rangle$ does not contain the zero matrix.

Generally, denoting by w_κ the action of $w \in S$ induced on κ-tuples, we have, for any two words, $w, w' \in S$, $(ww')_\kappa = w_\kappa w'_\kappa$, i.e., the map $w \to w_\kappa$ is a homomorphism of semigroups $S \to S_\kappa$.

8.4 Distributions

At level 1, we have left and right eigenvectors of A: $\pi_1 = \pi$ and $u_1^+ = u^+$, respectively. All of the components of u are equal to 1. And $\Omega = \Omega_1 = u^+\pi$. At every level κ, $1 \le \kappa \le r$, we have an abel limit

$$\Omega_\kappa = \lim_{s\uparrow 1}(1-s)(I - sA_\kappa)^{-1}$$

When π_κ and u_κ are unique, we have $\Omega_\kappa = u_\kappa^+\pi_\kappa$.

At level r, as at level 1, we have $\Omega_r = u_r^+\pi_r$.

8.4.1 Interpretation

Start at level r. Then the nonzero entries of π_r are in the columns corresponding to the ranges of the kernel, $\mathscr{K}$. The nonzero entries of u_r^+ are in rows corresponding to cross-sections of the partition classes of $\mathscr{K}$. I.e., for each partition class, find all possible cross-sections weighted by one, take the total, then the corresponding entries of u_r^+ will be at least that. In particular, for a right group, all cross-sections appear with the same weight, which can be scaled to 1.

At each level, if unique, π_κ and u_κ give you the recurrent class and corresponding cross-sections.

To get a unique sequence of pairs (π_κ, u_κ), construct them inductively from (π_r, u_r) as consecutive marginals. Thus, for $\kappa < r$, let

$$\pi_\kappa(I) = \sum_{J \supset I} \pi_{\kappa+1}(J)$$

Note that in the sum each J differs from I by a single vertex. Starting from level r, we could thus set

$$\pi_\kappa(I) = \sum_{\substack{J \supset I \\ |J \setminus I| = r-\kappa}} \pi_r(J)$$

Thus, for each κ there is a unique π_κ.
For u's, proceed one step at a time:

$$u_\kappa(I) = \sum_i p_i\, u_{\kappa+1}(I \cup \{i\})$$

where the weights p_i are the components of the invariant measure π.

With this construction, one recovers at level one, a multiple of $\pi_1 = \pi$ and a constant vector u_1, a multiple of u.

8.5 Absorbing states

Note that the kernel $\mathscr{K}$ is rank one if and only if the semigroup $S_2 = \langle R_2, B_2 \rangle$ contains zero. In this case, Ω_2 is the zero matrix. More generally, r is the maximum level κ such that $\Omega_\kappa \neq 0$.

By introducing an absorbing state at each level, R_κ, B_κ, and A_κ, extended accordingly, are stochastic matrices as at level one.

8.6 Inclusion operators and normalized compatible fields

Write α for the kernel partition distribution and β for its range distribution. A subset is split by a partition if its vertices lie in different blocks. For $1 \leq \ell \leq r$, define

$$u_\ell(I) = \Pr_\alpha\{I \text{ is split}\}, \qquad \eta_\ell(I) = \binom{r}{\ell}^{-1} \Pr_\beta\{I \subseteq \mathscr{R}\}.$$

Thus η_ℓ is a probability row, with $\eta_r = \beta$. For adjacent levels the normalized range marginal is

$$\eta_\ell(I) = \frac{1}{\ell+1} \sum_{j \notin I} \eta_{\ell+1}(I \cup \{j\}).$$

Indeed every $(\ell+1)$-set contributes to exactly $\ell+1$ of its ℓ-subsets. Equivalently, count the $r-\ell$ extensions inside a fixed range and use the binomial coefficients.

Equipartition gives the exact right-field normalization

$$u_\ell(I) = \frac{r}{r-\ell} \sum_{j \notin I} \pi_j u_{\ell+1}(I \cup \{j\}), \qquad \ell < r.$$

For a fixed kernel partition, an already split I occupies ℓ blocks; the union of the unoccupied blocks has stationary mass $(r-\ell)/r$. If I is not split, no extension is split. Average this identity over α to obtain the formula.

The incidence matrix $E_{\ell+1,\ell}(J,I) = \mathbf{1}_{I \subset J}$ expresses the first recurrence in operator form. The common-letter lift maps each kernel range injectively to another range, so its top stationary range law descends to stationary η_ℓ at every level. The right field is $u_\ell = \Omega_\ell \mathbf{1}$, with the Abel projection Ω_ℓ. One must not infer that every intermediate projection has rank one, or identify a normalized row $\mathbf{1}^{\mathsf{T}} \Omega_\ell$ with the chosen top-range marginal without an argument. The full inclusion-compatibility identities and their proofs have their primary source in kernelHierarchy.tex; this section fixes the normalizations used in the organized edition.

Chapter 9
Permanental Markov Chains

9.1 Permanents as killed independent-particle transitions

For a stochastic matrix P and k-subsets I,J, define

$$Q_k(I,J) = \operatorname{per} P[I,J].$$

This is the probability that k independently moving labeled particles, initially at distinct states I, occupy the distinct unordered states J after one step. The permanent sums the assignments of labels to J. The matrix is substochastic: the missing row mass is the probability of a collision. A cemetery state makes it stochastic; repeated powers track survival without collisions at every intermediate step. In general $Q_k^t \neq (P^t)^{\vee k}$, since the latter permits intermediate collisions followed by separation. General permanental lifts are not multiplicative. They are multiplicative for deterministic transformations.

9.2 A positivity hypothesis in the converse

Suppose P has a strictly positive stationary distribution. Then

Theorem 9.1. The following are equivalent: P is irreducible and aperiodic; $\rho(Q_2) < 1$; and $\det(I - Q_2) \neq 0$.

Proof. If P is primitive, choose t with every entry of P^t positive. For every distinct starting pair the independent particles have a uniformly positive probability of meeting at time t. The probability of surviving without any collision through time t is therefore uniformly less than one. Hence $\|Q_2^t\|_\infty < 1$ and $\rho(Q_2) < 1$. If P is reducible, strict positivity of a stationary row excludes transient states. There are then at least two closed classes; a particle in each cannot collide. If P is irreducible but periodic, two particles in different cyclic classes also cannot collide. In either case the finite killed chain has a nonempty

closed stochastic class and Q_2 has eigenvalue one. Finally a nonnegative substochastic matrix has spectral radius at most one, and if it equals one, one itself is an eigenvalue. This proves the determinant equivalence.

Without strict positivity the converse fails. For example $P = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 0 \end{smallmatrix}\right)$ has $Q_2 = 0$ but is not irreducible. Thus uniqueness of a stationary measure alone is not the hypothesis of this theorem.

9.3 Independent versus common-letter motion

For $P = \sum_i p_i M_i$, distinguish

$$P^{\vee k} \quad \text{from} \quad \sum_i p_i M_i^{\vee k}.$$

The first uses independent choices and kills collisions; the second uses one common random transformation. They agree for PtC because every generator differs only in row 1 and a submatrix contains that row at most once. The detailed proof is in the laboratory chapter. This equality is special, not a general averaging rule.

Chapter 10
Coupled Semigroups and Opposite Letters

10.1 Opposite-letter tensor coupling and its kernel

Let $S = \langle R, B \rangle$ be a finite transformation semigroup, with kernel K of rank r. For a word w, let $\bar{w}$ exchange every R and B. Define

$$T_{-} = \langle (R,B), (B,R) \rangle \subseteq S \times S.$$

Its action on ordered pairs is represented by $R \otimes B$ and $B \otimes R$, and a word gives $(w, \bar{w})$.

Theorem 10.1. The kernel of the opposite-letter semigroup is

$$K_{-} = T_{-} \cap (K \times K),$$

and every element of K_{-} has transformation rank r^2 on $X \times X$. Both coordinate projections $K_{-} \to K$ are surjective.

Proof. The rank of (f,g) on $X \times X$ is $\mathrm{rank}(f)\,\mathrm{rank}(g) \geq r^2$. Choose a word $u \in K$. The word $u\bar{u}$ gives the pair $(u\bar{u}, \bar{u}u)$; both coordinates lie in K by the ideal property. Thus minimum rank is r^2, attained exactly when both coordinates lie in K. The minimum-rank ideal of a finite transformation semigroup is its kernel, proving the formula. Each coordinate projection of T_{-} is onto S. Its image of K_{-} is a nonempty ideal of S contained in K, and hence equals K.

The product fibers are $F \times G$ and the product ranges are $A \times B$, but only the pairs compatible with complementary words occur. Local groups are subdirect products of the two corresponding local groups: restriction to an idempotent cell is a subgroup of their product, and sandwiching preimages by that idempotent proves surjectivity on each factor.

10.1.1 The original kernel alone is insufficient

Letter exchange need not define a map on S: equality $w = v$ need not imply $\bar{w} = \bar{v}$. If it does define an automorphism θ, then T_- is its graph and $K_- = \{(k, \theta(k)) : k \in K\}$. In general the required extra information is complementary-word compatibility. For a concrete counterexample, the pairs

$$[312344], [666565] \qquad \text{and} \qquad [112345], [612345]$$

have the same concrete base kernel, all six constant maps, but their opposite-letter kernels have respectively 16 and 31 elements. For the second pair, R^5B^5 is a reset word on the product state space; its target is $(2,1)$ and its orbit has 31 points. For the first pair, R^2B resets to $(6,4)$, whose orbit has 16 points. Once one constant product map is obtained, its orbit gives every constant product map, so these finite orbit counts determine the two kernels.

10.1.2 The running examples

Exact enumeration gives

example	$\lvert T_- \rvert$	$\lvert K_- \rvert$	rank	partitions	ranges	$\lvert G_- \rvert$
4 states	44	36	4	3	3	4
6 states	44	16	1	1	16	1
9 states	300	288	9	4	4	18

The examples are the previously studied pairs $[4312], [3443]$, $[312344], [666565]$, and $[444777111], [295838562]$. In the nine-state example the local group is

$$\{(\sigma, \eta) \in S_3 \times S_3 : \operatorname{sgn}\sigma = \operatorname{sgn}\eta\}.$$

Enumeration in an idempotent cell produces all 18 equal-parity pairs. Thus it is not the full product of the original local groups. For six states the kernel constants have targets precisely in

$$(\{1,2,3,4\} \times \{5,6\}) \cup (\{5,6\} \times \{1,2,3,4\}),$$

the opposite-type sector of the CC coloring.

10.1.3 Same-letter and opposite-letter chains

For $A = (R+B)/2$ and $\Delta = (R-B)/2$, put

$$P_+ = \tfrac{1}{2}(R\otimes R + B\otimes B) = A\otimes A + \Delta\otimes\Delta,$$

$$P_- = \tfrac{1}{2}(R\otimes B + B\otimes R) = A\otimes A - \Delta\otimes\Delta.$$

Use Cesaro or Abel limits unless convergence of ordinary powers is proved. The limiting kernel measure of T_- need not be the product of the two original kernel measures; both word and probability correlations matter.

If A is primitive, the midpoint $Q = A\otimes A$ is primitive with stationary projection $\Omega\otimes\Omega$. The preceding potential construction applies with

$$D = \Delta\otimes\Delta, \qquad V_{\text{pair}} = D(I - Q + \Omega\otimes\Omega)^{-1}.$$

It interpolates $Q + \tau D$ between P_- and P_+. Invertibility of $I \mp V_{\text{pair}}$ detects uniqueness of the stationary distribution at the respective endpoints. For P_+, uniqueness is equivalent to synchronization: the diagonal is a closed irreducible class; if it is the only recurrent class every pair can be merged, hence successive pair mergers synchronize the entire set. Conversely a reset word makes all off-diagonal states transient.

For a CC target-sign column s, we have $As = 0$, $\Delta s = \mathbf{1}$, and $\pi s = 0$. Therefore

$$Q(s\otimes s) = 0, \qquad D(s\otimes s) = \mathbf{1}.$$

If μ_τ is the stationary row of $Q + \tau D$, then

$$\mu_\tau(s\otimes s) = \tau.$$

At the endpoints this is the same-type sector for P_+ and the opposite-type sector for P_-. This is a direct connection between the potential operator, the tensor hierarchy, and canonical colorings.

10.1.4 A partial-circulant consequence

Let the base generators be M_a, M_b, with $d = \gcd(a,b)$. Their kernel has one partition, the residue classes modulo d, and local group C_d. The opposite-letter kernel also has one partition, namely the products of those residue classes. It is therefore a right group, of transformation rank d^2. Its local group is still C_d, rather than $C_d \times C_d$. Indeed each letter decreases both coordinates by one modulo d, so both coordinate phases of every kernel pair agree. In an idempotent cell the local group is contained in the diagonal C_d and projects onto it, hence equals it. The remaining problem is to characterize the compatible pairs of ranges, not to determine the local group. The relative residue $x - y \pmod d$ is preserved, explaining why the local group need not act transitively on all d^2 points of a product range. The example M_1, M_6 already shows nontrivial range compatibility when $d = 1$.

Chapter 11
Graph Lifts and the Double Construction

11.1 The edge-state lift

Order the $2n$ edge states by the red outgoing edge from each vertex, then the blue one; parallel edges remain distinct labeled edge states. The induced transformations and their average are

$$\widehat{R} = \begin{pmatrix} R & 0 \\ B & 0 \end{pmatrix}, \quad \widehat{B} = \begin{pmatrix} 0 & R \\ 0 & B \end{pmatrix}, \quad \widehat{A} = \tfrac{1}{2}\begin{pmatrix} R & R \\ B & B \end{pmatrix}.$$

If $\pi A = \pi$, then

$$\widehat{\pi} = (\pi R/2, \pi B/2).$$

Multiplication verifies the identity, since $\pi R + \pi B = 2\pi$. In general this is not $(\pi/2, \pi/2)$. Writing $U = \frac{1}{2}[R;B]$ and $W = [I,I]$ gives $\widehat{A} = UW$, $A = WU$, so

$$\det(\lambda I_{2n} - \widehat{A}) = \lambda^n \det(\lambda I_n - A).$$

11.2 Kernel rank and kernel elements under the lift

Assume every vertex has an incoming edge, as in the irreducible case. Let $H = [R;B]$ be the head map, and $J_R = [I,0]$, $J_B = [0,I]$ the two edge injections. Then $\widehat{L} = HJ_L$ and $J_L H = L$. For a nonempty word $w = L_1 \cdots L_t$,

$$\widehat{w} = H(L_1 \cdots L_{t-1})J_{L_t}.$$

The head map is onto and each J_L is injective, so the rank is the rank of the prefix. If the base kernel has rank r, the lifted kernel therefore has rank r and is precisely

$$\widehat{K} = \{HkJ_R, HkJ_B : k \in K\}.$$

If a prefix is empty its rank is n; it contributes to the kernel only when $r = n$, in which case the base semigroup is a group containing the identity. The two last-color ranges are disjoint and H is onto, so these representations are distinct and $|\widehat{K}| = 2|K|$. Kernel partitions pull back through H; kernel ranges are the two edge copies of the base ranges. Transport of all hierarchy fields is a further question, not an automatic identification of stationary laws.

11.3 A systematic source of CC examples

Every outgoing pair at a lifted source consists of edge vertices i and $i+n$, where i is its head. Since every head occurs, the auxiliary target-pair graph is exactly a matching of n pairs. Thus the double is CC and has 2^n labeled CC choices. The displayed lift is one such choice, with disjoint red and blue ranges. Graph doubling preserves the kernel rank in this coloring; canonical recoloring of that CC coloring doubles it. These are different operations. This distinction is essential when comparing the lift with the CC rank-doubling theorem.

Part IV
Synthesis

Chapter 12
Lie Methods and Lie Constants

Separate the proved solvability/right-group implication from the Lie-constant converse, which remains unproved in this project.

12.1 Completely Simple Semigroups, Lie Algebras, and the Road Coloring Problem

Studying the dynamics inherent in state-transition diagrams, either in the stochastic setting as Markov chains or in deterministic models as automata, continues to be a rich source of interesting mathematics and applications. Even for finite state systems, fascinating unsolved problems exist. Ĉerný's conjecture concerning the length of a synchronizing instruction for deterministic automata is one such example. Another open problem in this area is referred to as the "road coloring problem" and was the initial motivation for this work. The road coloring problem appeared in published form for the first time in a paper by Adler, Goodwyn, and Weiss [1] and can be stated very simply: Suppose you are given a strongly connected directed graph (or state-transition diagram). Is there a way to label the edges so that a synchronizing instruction exists? A precise definition of synchronizing instruction will be provided below. In [1], it was shown that aperiodicity was necessary for such an instruction to exist. The road coloring problem conjectures that aperiodicity is a sufficient condition as well.

12.1.1 Background

A finite automaton consists of a finite set V of states, a finite set of inputs A, and a state transition function $\delta : V \times A \to V$. We can associate with each $a \in A$ a corresponding transformation $R_a : V \to V$ defined by $R_a(v) = \delta(v,a)$, which

we write compactly as va. The semigroup S generated under composition by $\{R_a : a \in A\}$ is called the semigroup of the automaton. Given any automaton, one obtains a labelled digraph by taking the vertices of the graph to be the set of states V and for each $(v,a) \in V \times A$ defining a directed edge from v to va with label a. Note that the constructed directed graph has uniform outdegree, i.e., each vertex of V has the same number of edges leading from it.

If one considers an unlabelled digraph $G = (V, \mathscr{E})$ of uniform outdegree d, then any labelling of the edges with members of A, where $|A| = d$, such that each edge issuing from any given vertex has a distinct label is called a coloring of the digraph. Any such coloring uniquely determines an automaton $\delta : V \times A \to V$, where $R_a(v) = va$ is the terminal point of the directed edge with initial point v and label a. We henceforth assume that digraphs have uniform outdegree, identifying the coloring with the set $\mathscr{C} = \{R_a : a \in A\}$, and we refer to the semigroup $S = \langle \mathscr{C} \rangle$, generated by $\mathscr{C}$, as the coloring semigroup for the given labelling.

Definition 12.1. Let S be a coloring semigroup of a digraph $G = (V, \mathscr{E})$. The rank of $s \in S$ is the cardinality $|Vs|$ of the image of s. Any finite transformation semigroup S has a minimal ideal or kernel, which consists of the elments of minimal rank (see [11]). This common minimal rank is called the rank of the kernel.

A synchronizing instruction for an edge-labelled digraph is thus any transformation of rank one. The road coloring problem can now be stated in the following way.

Road Coloring Conjecture. Let $G = (V, \mathscr{E})$ be a d-out, strongly connected digraph. Let $\{S_i\}$ be the set of all coloring semigroups. If G is aperiodic, then for some i, S_i contains a synchronizing instruction.

The notion of period used here is the standard one from Markov chain theory, where for any vertex v in a directed graph, $\mathrm{per}(v) - \gcd\{|c_1|, \ldots, |c_n|\}$ where $c_1, \ldots, c_n$ are the simple cycles containing v. The assumption of strong connectivity ensures that all vertices have the same period.

Notation. We use rank and trace to denote the rank and trace respectively of a matrix. For a function, rank and trace refer to its representing matrix.

12.1.2 Semigroups and kernels

Let S be a coloring semigroup of a strongly connected digraph with kernel K. Let $E(S)$ be the set of idempotents of S. Impose on $E(S)$ the order $e \leq f$ iff $ef = fe = e$. Form the Rees product $x \times G \times Y$ of K in the usual manner choosing an idempotent, e_0, minimal with respect to this order and letting

$$x = E(Ke_0), \quad G = e_0Ke_0, \quad Y = E(e_0K)$$

with product

$$(x_1, g_1, y_1)(x_2, g_2, y_2) := (x_1, g_1(y_1x_2)g_2, y_2)\,.$$

Notice that $y_1x_2 \in G$. The sandwich function

$$\phi : Y \times X \to G, \qquad \phi(y,x) = yx$$

plays an important rôle in this paper. In particular, recall that if $\phi(y,x) = e_0$ for all $(y,x) \in Y \times X$, then $X \times G \times Y$ is a direct product.

Recall that if K is the kernel of a finite transformation semigroup S, the sets X, G, and Y can be further characterized. Suppose S is a semigroup of functions on the set $\{1,\ldots,n\}$. Then each idempotent $e \in X$ can be associated with a partition π_e of the set $\{1,\ldots,n\}$ in the standard way: $\pi_e = \{e^{-1}(j) : 1 \le j \le n\}$. Thus the set X can also be thought of as a collection of partitions of the set $\{1,\ldots,n\}$. Similarly each idempotent $f \in Y$ can be associated with its range $R_f = \{jf : 1 \le j \le n\}$, where we employ the convention of writing inputs to the transformation on the left. The group $G = e_0Ke_0$ can then be seen as the subgroup of transformations within K having the same partition and range as e_0.

The ideal structure of K will also play a rôle in our discussions. Each set of the form eK with $e \in X$ is a minimal right ideal consisting of the set of all transformations in K having partition π_e. Similarly, Kf with $f \in Y$ is a minimal left ideal consisting of the set of all transformations in K having range R_f. When $X = E(Ke_0) = \{e_0\}$, a single idempotent, K is, of course, a right group. Notice that when a coloring semigroup has a synchronizing instruction or rank one instruction, K is easily shown to be a right group.

The importance of right groups in the context of the road colroing problem has been shown in recent papers [2, 8] and this paper continues an exploration of their rôle. In fact, we have the following theorem.

Theorem 12.1. Suppose a strongly connected, apreiodic digraph G has a coloring semigroup S whose kernel K is a right group. Then G has a coloring semigroup that contains a synchronizing instruction.

Proof. Suppose the (unique) vertex partition induced by the right group K, is $\Pi = \{P_1, P_2 \ldots P_l\}$, where $\text{rank}(K) = t \ge 2$ (as we are done if t=1).

Notice that every element of S, and therefore the generators of S, act as permutations on the elements of Π. Suppose not. Then there are vertices $v_i \in P_i$, $v_j \in P_j$, and an element s in S such that

$$v_is \in P_m, \qquad v_js \in P_m\,.$$

Then, if k is any element of K,

$$v_isk \in P_mk, \qquad v_jsk \in P_mk\,.$$

But $P_m k = w$, since the elements of the kernel collapse the partition elements to a single vertex. Since sk is in the kernel, the $\{v : vsk = w\}$ is one of the partition elements. Thus $P_i = P_j$, showing that s is a permutation on partition elements.

Form the quotient graph induced by the coloring on the "vertices" $\Pi = \{P_1, P_2, \ldots P_l\}$. Since the generators act as permutations on the elements of Π, the resulting graph will be "Eulerian", having the indegree of each vertex equal to its outdegree. Now, we will use the algorithm introduced by Kari [15], to find a coloring that synchronizes this Eulerian graph. This is done in the following way.

Choose any element $w_0 = c_1 c_2 \cdots c_l$ from the original kernel K, each c_i being a generator of S. Note that $P_1 w_0$ is a single vertex in the original graph. Proceeding as in [15], choose a vertex P_{i_0}, in the quotient graph, such that for colors r and b,

$$P_{i_0} r \neq P_{i_0} b\,.$$

For step one, swap the labels of the r-edges leaving P_{i_0} with the b-edges leaving P_{i_0}. No other edge labels are changed.

For step two, we reassign colors in w_0 as follows. If $P_1 = P_{i_0}$ then change c_1. If c_1 is either r or b, swap r with b, otherwise leave it unchanged. Working through w_0, whenever $P_1 c_1 c_2 \cdots c_{j-1} \subset P_{i_0}$, change c_j accordingly. This results in a new word, $w_1 = c_1' c_2' \cdots c_l'$, with $P_1 w1$ remaining a single vertex.

Each stage of this procedure is performed in the same way. At stage k, Kari's algorithm guarantees that the new quotient graph is Eulerian with fewer vertices. Step one consists of finding a vertex in this new quotient graph and relabelling as described above. For step two, the word w_{k-1} is changed each time the vertices of P_1 map to that partition element that was changed by this kth relabelling, ensuring that $P_1 w_k$ is a single vertex.

This procedure terminates in a synchronizing instruction w' for the quotient graph induced by the kernel, i.e., w' maps every P_j to a single P_{j_0}. Since the graph is strongly connected at the quotient level, there exists an instruction w'' such that $P_{j_0} w'' \subset P_1$. If the procedure requires m steps then $w_{\text{sync}} = w' w'' w_m$ is a synchronizing instruction for the original graph. □

12.2 Solvability and right groups

Throughout this work, we consider Lie algebras over $\mathscr{C}$. In general, if $\mathfrak{g}$ is a Lie algebra, its derived algebra, $\mathfrak{g}' = [\mathfrak{g}, \mathfrak{g}]$, is the linear span of all products $[x, y]$, $x, y \in \mathfrak{g}$. This is an ideal of $\mathfrak{g}$. Continuing, one defines a sequence of ideals

$$\mathfrak{g} \supset \mathfrak{g}' \supset \mathfrak{g}'' \supset \cdots \supset \mathfrak{g}^{(n)} \supset \cdots$$

with $\mathfrak{g}^{(n+1)} = [\mathfrak{g}^{(n)}, \mathfrak{g}^{(n)}]$, which is the derived series of $\mathfrak{g}$. If this series terminates at 0 after finitely many steps, then $\mathfrak{g}$ is solvable. Note that the last

nonzero term of the series of a solvable algebra is abelian.

The main property of a solvable Lie algebra that we are using here is Lie's Theorem to the effect that (over an algebraically closed field) a solvable Lie algebra of matrices can be simultaneously upper-triangularized ([13, p. 50], also see [19, p. 21–26]). In this context the work by Radjavi & Rosenthal [19] is of particular interest. Especially, Radjavi's Theorem on permutable traces [19, p. 33] is part of the inspiration behind the proof.

There are three Lie algebras associated with a coloring $\mathscr{C}$ of a strongly connected digraph with coloring semigroup S and its kernel K. In general, we denote by $\mathfrak{g}(\cdot)$ the Lie algebra generated by transformations from a given set. In this context, we specify $\mathscr{S} = \mathfrak{g}(S)$ the Lie algebra generated by the full semigroup. The underlying set is the same as the semigroup algebra (without 1) with commutator as product. The Lie algebra $\mathscr{L} = \mathfrak{g}(\mathscr{C})$, generated by the transformations in $\mathscr{C}$, is a subalgebra of $\mathscr{S}$ and is our main object of interest. Finally, $\mathscr{K} = \mathfrak{g}(K)$, is a subalgebra of $\mathscr{S}$, that is, in fact, a Lie ideal of $\mathscr{S}$ since K is an ideal in the semigroup. The main feature is that all three Lie algebras are solvable if and only if the generators $\mathscr{C}$ are simultaneously triangularizable if and only if $\mathscr{L}$ is solvable. In that case, we call the graph Lie-solvable.

Theorem 12.2. If $\mathscr{L}$ is solvable, then the kernel K is a right group.

We start with a Lemma.

Lemma 12.1. If $\mathscr{L}$ is solvable, then the kernel K is isomorphic to a Rees product semigroup that is a direct product.

Proof. We want to show that

$$K \cong X \times G \times Y \quad \text{where if} \quad (x_1, g_1, y_1), (x_2, g_2, y_2) \in X \times G \times Y$$

then

$$(x_1, g_1, y_1)(x_2, g_2, y_2) = (x_1 x_2, g_1 g_2, y_1 y_2).$$

It is well known that this is equivalent to the property:

$$\text{for any two idempotents } e, f \in K,\ (ef)^2 = ef$$

In other words, the idempotents of K should form a band, a semigroup of idempotents.

Note that if we show that $\mathsf{rank}\,(ef) = \mathsf{tr}\,(ef)$, then we are done, since for a function

$$\mathrm{rank}\,(f) = \mathrm{tr}\,(f) \text{ iff } f \text{ is an idempotent.}$$

Let $e, f \in K$, then, since ef is contained in some maximal subgroup of K, $(ef)^p$ is an idempotent for some $p \geq 1$. Since $\mathscr{L}$ is solvable, the matrices of $\mathscr{L}$ can be made simultaneously upper triangular, thus $\mathrm{tr}\,((ef)^p) = \mathrm{tr}\,(e^p f^p)$. Also, since both $(ef)^p$ and ef are in K, $\mathrm{rank}\,((ef)^p) = \mathrm{rank}\,(ef)$. Thus we have $\mathrm{rank}\,(ef) = \mathrm{rank}\,((ef)^p) = \mathrm{tr}\,((ef)^p) = \mathrm{tr}\,(e^p f^p) = \mathrm{tr}\,(ef)$, since $e^p = e$ and $f^p = f$.
Therefore, ef is an idempotent and it follows that the Rees product is a direct product. □

Now we proceed with the proof of the theorem.

Proof. Consider two partitions π_1, π_2 of the vertices V, corresponding to minimal right ideals of K. Let $\pi_1 = \{P_1, P_2, \ldots, P_r\}$, $\pi_2 = \{Q_1, Q_2, \ldots, Q_r\}$, and suppose $R_1, R_2, \ldots, R_t$ are all possible ranges of the functions in K. (Note: r is the rank of the kernel.)

Label the idempotents $e_i = E\big(H(\pi_1, R_i)\big)$ and $f_i = E\big(H(\pi_2, R_i)\big)$ where $H(\pi_j, R_i)$ is the H-class determined by partition π_j and range R_i. Notice that since K is a direct product, $e_i f_j = e_j$, since $e_i f_j \in H(\pi_1, R_j)$ and is an idempotent.

Let $R_1 = \{x_1, x_2, \ldots, x_r\}$, with $x_i \in P_i$. Consider the sets $O_j = \{x_j e_i \mid i = 1, \ldots, t\}$. Each idempotent acts as a set identity function on its range, so that $x_j e_i \in P_j$ for $i = 1, \ldots, t$ and $j = 1, \ldots, r$. Hence $O_j \subset P_j$ for $j = 1, \ldots, r$. In particular, $O_{j_1} \cap O_{j_2} = \emptyset$, if $j_1 \neq j_2$.

Notice that if, for some $x \in P_j$, $xe_i = x$, then $x_j e_i = x$ will pick up $x \in O_j$. Thus,

$$V = \bigcup_{j=1}^{r} O_j$$

since $\bigcup_{i=1}^{t} R_i = V$ so that every element of V is in the range of some idempotent e_i. So, in fact, $O_j = P_j$ for $j = 1, \ldots, r$.

Similarly, $Q_j = \{x_j f_k \mid k = 1, \ldots, t\}$. But for each k, $1 \leq k \leq t$,

$$x_j f_k = x_j e_1 f_k = x_j e_k \in P_j.$$

Thus $Q_j \subset P_j$ for $j = 1, \ldots, r$. Then, since $V = \bigcup_{j=1}^{r} Q_j = \bigcup_{i=1}^{r} P_i$, it is clear that $Q_j = P_j$ for $j = 1, \ldots, r$ and thus, $\pi_1 = \pi_2$. Hence K is a right group and the theorem is proved. □

Combining this with Theorem 12.1, we have

Corollary 12.1. If the Lie algebra $\mathscr{L}$ of a coloring is solvable, then there exists a coloring semigroup of the graph that contains a synchronizing instruction. In other words, the road coloring conjecture holds for Lie-solvable graphs.

12.3 The constant ideal

Use right-action transformation matrices and the bracket $[X,Y] = XY - YX$. Let $\mathfrak{g} = \mathrm{Lie}_{\mathbb{Q}}(R,B)$, and let $C_j = \mathbf{1}e_j^{\mathsf{T}}$ be the constant transformation with value j. Then $C_iC_j = C_j$, and consequently

$$[C_i,C_j] = C_j - C_i.$$

The Lie constants in the supplied summary are the derived algebra of the constant transformations:

$$\mathfrak{c}_0 = \{\mathbf{1}v : v \in \mathbb{Q}^{1\times n},\ v\mathbf{1} = 0\} = \mathrm{span}\{C_j - C_1 : 2 \le j \le n\}.$$

This is an $(n-1)$-dimensional abelian algebra; indeed products of any two of its elements are zero. It is the differences of constants, not necessarily the individual C_j, that the conjecture requires.

Every $X \in \mathfrak{g}$ satisfies $X\mathbf{1} = c(X)\mathbf{1}$ for a scalar $c(X)$. Moreover

$$[X,\mathbf{1}v] = \mathbf{1}\big(c(X)v - vX\big),$$

so $\mathfrak{c}_0$ is an ideal in the ambient constant-row-sum Lie algebra. In particular $\mathfrak{g} \cap \mathfrak{c}_0$ is a module under commutation with R and B. Once one nonzero constant difference is obtained, its commutator orbit can produce further constant differences.

The derived algebras of $\mathrm{Lie}(R,B)$ and $\mathrm{Lie}(R-I,B-I)$ agree, since all brackets are unchanged by scalar identity shifts. This does not identify their complete linear spans without a separate argument.

12.4 An exact computational procedure

Start with an independent basis of $\mathrm{span}\{R,B\}$. For each new basis matrix X, adjoin $[X,R]$ and $[X,B]$ if independent of the current span. Repeat until no new direction occurs. The stable space is the generated Lie algebra: it contains the generators and is stable under their adjoint actions; Jacobi's identity then gives stability under adjoints of all Lie words. Its dimension is at most $n^2 - n + 1$ because of the constant-row-sum constraint. Rational row reduction avoids floating-point rank decisions.

If G is the matrix whose columns are vectorizations of the resulting basis, and C has columns $\mathrm{vec}(C_j - C_1)$, then

$$\dim(\mathfrak{g} \cap \mathfrak{c}_0) = \operatorname{rank} G + (n-1) - \operatorname{rank}[G\ C].$$

Thus membership of all Lie constants is an exact linear-algebra test. The coefficients in $Gz = \mathrm{vec}([X_i, X_j])$ give structure constants in this basis. Keeping the Lie-word expression attached to each adjoined basis matrix also produces explicit commutator witnesses.

12.5 The kernel-partition obstruction

Assume the irreducible stochastic setting of our equipartition results. For every semigroup word W and every kernel element k,

$$\pi W k = \pi k.$$

Every $X \in \mathfrak{g}$ is a linear combination of semigroup words, whose coefficient sum is $c(X)$. Hence

$$\pi X k = c(X)\pi k.$$

If $X = \mathbf{1}v \in \mathfrak{g} \cap \mathfrak{c}_0$, then $c(X) = 0$ and $\pi X = v$, so $vk = 0$ for every k in the kernel.

Proposition 12.1. Under these assumptions,

$$\mathfrak{g} \cap \mathfrak{c}_0 \subseteq \left\{ \mathbf{1}v : \sum_{x \in F} v_x = 0 \text{ for every fiber } F \text{ of every kernel partition} \right\}.$$

In particular, if the kernel rank is r, then

$$\dim(\mathfrak{g} \cap \mathfrak{c}_0) \le n - r.$$

Containment of all Lie constants therefore forces rank one.

Proof. For a transformation k, the coordinates of vk are the sums of v over its fibers. Fixing one rank-r partition imposes r independent linear conditions on $v \in \mathbb{Q}^n$, so the allowed dimension is at most $n - r$. Requiring the same for all partitions can only decrease it.

For partial circulants, on the recurrent core $n = m$, the allowed space is

$$\{\mathbf{1}v : \sum_{x \equiv a \pmod r} v_x = 0 \text{ for all } a\},$$

of dimension $m - r$. With transient states, the same bound holds for the Lie-constant intersection: invariance of the core forces $v_x = 0$ outside it, and the

core argument applies. This is a proved upper bound; equality in the examples below is computational evidence, not a general proof.

12.6 A short explicit extraction

Example 12.1. Take $R = M_2 = [212]$ and $B = M_3 = [312]$. Set

$$H = [R,B], \qquad F = \frac{1}{3}\big(H - [[H,R],R]\big).$$

Direct multiplication gives

$$F = C_2 - C_1 = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 1 & 0 \\ -1 & 1 & 0 \end{pmatrix}, \qquad [B,F] - F = C_3 - C_1.$$

Thus the generated Lie algebra contains all Lie constants in this example. These identities were checked exactly over the rationals. Its dimension is six and its derived-series dimensions are $6,5,5$, so it is not solvable, although the transformation kernel is a right group (in fact rank one).

12.7 Computations and the remaining problem

n	Generator indices	$\dim \mathfrak{g}$	$\dim(\mathfrak{g} \cap \mathfrak{c}_0)$
2	1,2	3	1
3	2,3	6	2
4	2,4	8	2
4	3,4	12	3
6	3,6	18	3
5	2,4	10	2
4	4	1	0

All ranks are exact rational computations. In the six-state row the kernel rank is three, and in the five-state row the largest index is four with one transient state. In every displayed case the intersection dimension attains the bound $m - r$. Proving or refuting this equality for arbitrary partial-circulant supports is a concrete intermediate problem. Its rank-one case would establish the proposed converse for this family.

The known solvability-to-right-group implication and the Lie-constant conjecture should not be conflated. Right groups need not have solvable generated Lie algebras, as the example above demonstrates. Moreover, in a strongly connected synchronizing automaton the associative algebra contains all constants (a reset word followed by suitable paths gives each constant),

but this does not automatically yield nested-commutator expressions for their differences. That associative-to-Lie step is the essential gap in the proposed general converse.

These notes establish definitions, obstructions, an algorithm and finite examples. They do not settle that converse or assert a general equality from the observed table.

Chapter 13
Partial Circulants: A Running Laboratory

Scope: an application of the general theory.

The partial-circulant family is an application of the general transformation semigroup and operator theory, not an exception to it. The additional assumptions used for road-coloring questions are not imposed in this chapter. The number of generators and their probabilities are specified locally. In the associated stochastic chain, loops, coincident outgoing targets, transient states or periodicity can occur. Stationary and limiting statements use the hypotheses stated with each result; road-coloring hypotheses must not be imported implicitly.

13.1 Definitions and statement

Let $\mathscr{N} = \{1,\dots,n\}$. For $1 \le i \le n$, define

$$xM_i = \begin{cases} i, & x = 1, \\ x-1, & x > 1. \end{cases}$$

We use right actions: $x(fg) = (xf)g$. Equivalently, transformation matrices have a 1 in position (x,xf), and composition agrees with ordinary matrix multiplication. Their matrix rank is the cardinality of their image.

Fix a nonempty set $I \subseteq \{1,\dots,n\}$ and write

$$S = \langle M_i : i \in I \rangle, \qquad m = \max I, \qquad d = \gcd\{i : i \in I\}, \qquad C = \{1,\dots,m\}.$$

Here S consists of nonempty words in the generators. The kernel of a finite semigroup means its unique minimal nonempty two-sided ideal; it is not to be confused with the fiber partition of a transformation.

For $r \in \mathbb{Z}/d\mathbb{Z}$, put

$$P_r = \{x \in \mathcal{N} : x \equiv r \pmod{d}\}, \qquad C_r = P_r \cap C,$$

and let

$$\mathcal{A} = \{A \subseteq C : |A \cap C_r| = 1 \text{ for every } r \in \mathbb{Z}/d\mathbb{Z}\}.$$

For $A \in \mathcal{A}$, denote its unique point in C_r by a_r. Define

$$xk_{A,t} = a_{[x]-t} \qquad (x \in \mathcal{N},\ t \in \mathbb{Z}/d\mathbb{Z}),$$

where $[x]$ denotes the residue class of x modulo d.

Theorem 13.1. The following statements hold.

1. The minimum rank of an element of S is d.
2. The kernel of S is exactly

$$K = \{s \in S : \mathrm{rank}(s) = d\} = \{k_{A,t} : A \in \mathcal{A},\ t \in \mathbb{Z}/d\mathbb{Z}\}.$$

3. Every element of K has fiber partition $\{P_r : r \in \mathbb{Z}/d\mathbb{Z}\}$. Its possible ranges are exactly the members of $\mathcal{A}$.
4. Multiplication is given by

$$k_{A,t}k_{B,u} = k_{B,t+u}.$$

   Consequently $K \cong C_d \times R_{\mathcal{A}}$, where C_d is the cyclic group of order d and $R_{\mathcal{A}}$ is the right-zero semigroup on $\mathcal{A}$, with product $AB = B$. In particular, K is a right group.
5. The number of ranges and the cardinality of the kernel are

$$|\mathcal{A}| = (m/d)^d, \qquad |K| = d(m/d)^d.$$

13.2 Congruence classes and elementary collapses

Lemma 13.1. If w is a word of length ℓ, then

$$xw \equiv x - \ell \pmod{d} \qquad (x \in \mathcal{N}).$$

Moreover $Cw \subseteq C$ and $|Cw| \geq d$. In particular, $\mathrm{rank}(w) \geq d$.

Proof. For every $i \in I$, we have $xM_i \equiv x - 1 \pmod{d}$: at $x = 1$ this uses $d \mid i$, and at all other points it follows from the definition. Induction proves the formula for words. Since every generator preserves C, so does every word. The set C meets every residue class modulo d, and the congruence formula shows that its image does as well.

Lemma 13.2. For each $i \in I$ with $i < m$, and each $a \in C$, the restriction semigroup $S|_C$ contains the transformation fixing every point other than a and sending a to $a + i$ modulo m.

Proof. On C, let $c = M_m|_C$. This is the permutation $x \mapsto x-1$ modulo m, so $c^{-1} = c^{m-1}$. The restriction of $M_i M_m^{m-1}$ fixes every $x > 1$ and sends 1 to $i+1$. Conjugation by powers of c gives all the asserted transformations. These conjugations are realized by words in the original generators, since both c and c^{-1} are restrictions of positive powers of M_m.

Lemma 13.3. For every $A \in \mathscr{A}$, there exists $v_A \in S$ such that

$$xv_A = a_r \quad (x \in C_r), \qquad xv_A \equiv x \pmod{d} \quad (x \in \mathscr{N}).$$

Proof. Form the directed graph on $\mathbb{Z}/m\mathbb{Z}$ with edges

$$a \longrightarrow a+i \qquad (i \in I,\ i < m).$$

Its strongly connected components are exactly the classes modulo d. Indeed, the increments generate the subgroup $d\mathbb{Z}/m\mathbb{Z}$. In a finite cyclic group the additive inverse of any increment is a nonnegative multiple of that increment, so directed reachability gives precisely the cosets of this subgroup.

In each component choose a directed spanning tree oriented toward the prescribed root a_r. Such a tree exists, for example by choosing from each nonroot vertex an edge along a shortest directed path to the root. For each nonroot vertex apply the elementary collapse to its parent, in decreasing order of distance from the root. A point first moves to its parent and is subsequently moved along the remaining path to the root. Thus the product of these collapses is the desired retraction on C.

Choose words realizing these elementary collapses as in Lemma 13.2, and concatenate them to obtain v_A. Each collapse preserves residues modulo d. By Lemma 13.1, a representing word for such a collapse must have length divisible by d (apply the congruence formula to any point of C). Hence that word preserves residues on all of $\mathscr{N}$, not only on C. The product v_A does too.

If there are no nonroot vertices, then $d = m$, $I = \{m\}$, and $A = C$. In this case take $v_A = M_m^m$, which has the required properties.

13.3 Proof of the theorem

Proof (Proof of Theorem 13.1). Choose a positive integer L divisible by d and satisfying $L \geq n - m$. Iteration of M_m moves every point outside C down by one until it enters C, and subsequently stays inside C. Therefore

$$\mathscr{N} M_m^L \subseteq C.$$

For $A \in \mathscr{A}$ and an integer representative $0 \leq t < d$, Lemma 13.3 now gives

$$M_m^{L+t} v_A = k_{A,t}.$$

Indeed the prefix sends x to a point of C congruent to $x-t$ modulo d, and v_A sends that point to $a_{[x]-t}$. Thus every $k_{A,t}$ belongs to S and has rank d. Lemma 13.1 proves that this is the minimum rank.

Conversely, suppose $w \in S$ has rank d, and choose a word representing it of length ℓ. Since $Cw \subseteq \mathscr{N}w$ and $|Cw| \geq d$, we have

$$\mathscr{N}w = Cw \subseteq C.$$

Its image therefore consists of exactly one point in each residue class modulo d. Call this transversal A. The congruence formula forces

$$xw = a_{[x]-[\ell]},$$

so $w = k_{A,[\ell]}$. This proves the classification of all minimum-rank elements and their fiber partitions and ranges.

For multiplication, observe that $a_r \equiv r \pmod{d}$, whence

$$xk_{A,t}k_{B,u} = a_{[x]-t}k_{B,u} = b_{[x]-t-u}.$$

Thus $k_{A,t}k_{B,u} = k_{B,t+u}$. The parametrization is injective: the image determines A, and the induced shift of residue classes determines t. It follows that $(t,A) \mapsto k_{A,t}$ is an isomorphism $C_d \times R_{\mathscr{A}} \to K$.

It remains to check that K is the minimal ideal, rather than merely a subsemigroup. Multiplication cannot increase transformation rank, and no element of S has rank below d. Hence K is a two-sided ideal. The multiplication formula also gives $kK = K$ for every $k \in K$. If J is any nonempty two-sided ideal, choose $s \in J$ and $k \in K$. Then $sk \in J \cap K$, and

$$K = (sk)K \subseteq J.$$

Thus K is the unique minimal nonempty two-sided ideal.

Finally, each of the d sets C_r has m/d points. Choosing one point independently from each gives $|\mathscr{A}| = (m/d)^d$, and $|K| = d|\mathscr{A}|$.

13.4 Further structure

Corollary 13.1. The idempotents of K are precisely $k_{A,0}$ for $A \in \mathscr{A}$. They are the retractions onto the possible ranges. For each fixed A,

$$H_A = \{k_{A,t} : t \in \mathbb{Z}/d\mathbb{Z}\}$$

is a maximal subgroup, cyclic of order d, with identity $k_{A,0}$. There is one Green $\mathscr{R}$-class in K, while its Green $\mathscr{L}$-classes (and hence its $\mathscr{H}$-classes) are the sets H_A indexed by the ranges.

Proof. All assertions follow from the direct-product multiplication. In particular, $k_{A,t}^2 = k_{A,2t}$ equals $k_{A,t}$ precisely when $t = 0$, and fixing A gives the group law of $\mathbb{Z}/d\mathbb{Z}$.

Corollary 13.2. The semigroup S contains a constant transformation if and only if $d = 1$. In that case its kernel is the right-zero semigroup consisting of the m constant maps with values in C.

Proof. A constant transformation has rank one. Apply the minimum-rank formula and the classification of kernel ranges.

Remark 13.1 (Composition convention). All right-group statements above use $x(fg) = (xf)g$. If transformations are multiplied with the opposite convention, the resulting semigroup is the opposite semigroup: the corresponding kernel is a left group, with the same rank, fiber partitions, and ranges.

13.5 The ladder chain and its stationary distribution

Assign positive probabilities p_i to the selected generators, with $\sum_{i\in I} p_i = 1$, and put $T = \sum_{i\in I} p_i M_i$. Write $\mu = \sum_{i\in I} i p_i$. Then

Proposition 13.1. The stationary distribution, extended by zero outside C, is

$$\pi_j = \frac{\sum_{i\in I,\ i\ge j} p_i}{\mu} \quad (1 \le j \le m), \qquad \pi_j = 0 \quad (j > m).$$

On C the chain is irreducible and its period is $d = \gcd(I)$.

Proof. The stationary equations read $\pi_j = \pi_1 p_j + \pi_{j+1}$ on the core, where $p_j = 0$ for unselected indices and $\pi_{m+1} = 0$. Backward substitution and normalization give the formula. The M_m cycle visits all of C, so the chain is irreducible there. A return from state 1 to itself consists of excursions of lengths in I, and each selected length occurs. Their greatest common divisor is therefore the period. States above m descend into C.

13.6 The spectrum of the collision operator

For a kernel-valued random transformation K, use the convention

$$N = \mathbb{E}[KK^{\mathsf{T}}], \qquad N_0 = J - N.$$

Every kernel transformation has the same partition, so KK^{T} is independent of K. Its (x,y) entry is one exactly when $x \equiv y \pmod{d}$. Consequently, after ordering the core by residue classes,

$$N|_{C\times C} = I_d \otimes J_q, \qquad q = m/d.$$

The nonzero eigenvalue on the core is q, of multiplicity d; zero has multiplicity $m-d$. The matrix N/q restricted to the core is the orthogonal projection onto vectors constant on each residue class. On the full state space the nonzero eigenvalues are the full residue-block sizes, which need not equal q when transient states are present.

13.7 Why permanental and common-letter hierarchies agree here

For a matrix X, define $X^{\vee k}$ by permanents of k-by-k submatrices, indexed by subsets. In general the permanental lift of an average is not the average of the lifts. Here, however,

Proposition 13.2. At every subset level,

$$T^{\vee k} = \sum_{i\in I} p_i M_i^{\vee k}.$$

Proof. All generators differ only in row 1. A submatrix either omits that row, in which case it is identical for every generator, or includes it once. A permanent is linear in each row separately. In either case averaging commutes with the submatrix permanent.

Thus the common-letter semigroup hierarchy is also the permanental hierarchy of the single ladder transition matrix. The equality is special to this shared-row structure, not a general identity for stochastic matrices. Identifying compatible fields, projection normalizations, and cyclic-pattern recurrent classes is the next extension of this chapter, using the separate hierarchy paper and the operator draft.

13.8 The rank condition: a hypothesis test

For distinct a,b, take $R = M_a$, $B = M_b$, on their irreducible core. Then

$$\Delta = \tfrac{1}{2} e_1 (e_a^{\mathsf{T}} - e_b^{\mathsf{T}}), \quad \pi\Delta = \tfrac{\pi_1}{2} (e_a^{\mathsf{T}} - e_b^{\mathsf{T}}).$$

Since $\pi_1 > 0$, the row span generated by $\pi\Delta$ already equals the one-dimensional row space of Δ. This is Chapter 5's rank condition. When $\gcd(a,b) = 1$, the core is aperiodic and the synchronization criterion applies. When the gcd exceeds one, the same rank condition holds but synchronization is impossible. Thus aperiodicity is indispensable in the surrounding criterion; a theorem statement separated from its standing hypotheses would be misleading. The example $a = 2, b = 4$ should be used as a routine hypothesis check.

13.9 A worked example and exercises

Take $n = m = 6$, $I = \{2,4,6\}$, and equal probabilities. Then

$$d = 2, \quad q = 3, \quad \pi = (1/4, 1/4, 1/6, 1/6, 1/12, 1/12).$$

The partition is $(135 \mid 246)$, there are nine ranges and eighteen kernel elements, and N has eigenvalues $3,3,0,0,0,0$. This chain is periodic, so ordinary powers need not converge; use Cesaro or Abel projections.

1. Verify each stationary equation and that each partition block has stationary mass $1/2$.
2. List the nine ranges and verify the kernel multiplication law.
3. Diagonalize the two J_3 blocks and interpret their eigenspaces.
4. Extend the example to $n = 8$ without changing I. Determine the transient states and the full-space spectrum of N.
5. Compare the common-letter and permanental level-two matrices in this example. Find a non-PtC pair for which the equality fails.
6. Work out the rank-one case $I = \{2,5\}$ on five states.

13.10 Research directions, not asserted theorems

The Lie-constant converse in the separate Lie notes remains open in this project. Exact computations suggest that the Lie-constant intersection on the recurrent core has dimension $m - d$; no proof for arbitrary PtC supports is supplied here. Unicycle enumeration, canonical recoloring, and the double construction are further tests of the general theory, with their original hypotheses and normalization conventions kept explicit.

Appendix A
Weak Convergence and Compact Semigroups

A.1 Weak Convergence of Probability Measures

Recall that we said a sequence of measures (μ_n) converges to a measure ν iff $\mu_n(x) \to \nu(x)$ for each $x \in S$. This notion is completely clear in the case when S is finite. What about when S is discrete, but countably infinite?

Definition A.1. A set in $\mathbb{R}^n$ is called compact if it is closed and bounded. A set is locally compact if every point is contained in some compact set.

Definition A.2. Let (μ_n) be a sequence of measures on a locally compact space. Then $\mu_n \to \nu$ weakly if

$$\int f\, d\mu_n \to \int f\, d\nu$$

For every bounded continuous function f. If

$$\int F\, \mu_n \to \int f\, d\nu$$

for every continuous function with compact support, then we say $\mu_n \to \nu$ vaguely or in the weak*-topology.

A topology, in some sense, is characterized by the convergence properties of sequences of elements within the space. Let us consider a few examples of sequences of measures and their convergence properties.

Some facts about integration:

(1) If you integrate the function that is identically 1 with respect to a measure, then you simply get the measure of the set you are integrating over. That is,

$$\int_B 1\, d\mu = \int_B d\mu = \mu(B).$$

(2) If you integrate a function with respect to a delta measure, then you get the value of the function on the support of the measure. That is,

$$\int f\,d(\delta_x) = f(x).$$

Now, some basic results about weak and weak* convergence. We will use some very basic topological space in an attempt to eliminate technical details.

Proposition A.1. Let $(\mu_n) \subset \text{Prob}(S)$ where S is a locally compact, second countable space. Suppose $\mu_n \to \nu$ weakly. Then $\nu \in \text{Prob}(S)$.

Proof. We simply need to show that $\nu(S) = 1$. But since $\int f\,d\mu_n \to \int f\,d\nu$ for every bounded continuous function, simply pick $f \ni f(x) = 1$ for all x. Then $\int\limits_S f\,\mu_n = \mu_n(S) = 1$ for all n.

$$\begin{aligned} &\Rightarrow \quad \mu_n(S) \to \int f\,d\nu = \nu(S) \\ &\Rightarrow \quad \nu(S) = 1 \qquad \square \end{aligned}$$

Example A.1. Notice Prop. 3.1 does not hold for vague convergence.

In fact, let $S = \mathbb{R}^+$. Then S is a locally compact semigroup. Let $\mu_n = \delta_n$, $n = 1,2,\ldots$. Then $\mu_n \xrightarrow{w^*} z$ where z is the zero measure. That is, $z(i) = 0$ for all i.

To see this, let f be any continuous function with compact support. Let $\text{supp}(f) = K$. Then K is closed and bounded and is contained in some interval of the form $[0,M]$. Then

$$\begin{aligned} \int f\,d\mu_n &= \int_K f\,d\mu_n \\ &= \int_M f\,d\mu_n \\ &= \begin{cases} f(n) & n \le M \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Thus for all continuous f with compact support,

$$\begin{aligned} &\int f\,d\mu_n \to 0 \\ &\Rightarrow \quad \mu_n \to z. \end{aligned}$$

Now, I can think of (μ_n) living on the positive integers $\mathbb{Z}^+$ (a perfectly good semigroup) and can visualize them as elements of $\mathbb{R}^\infty$. That is,

$$\mu \cong \overline{\mu} = (\mu(1), \mu(2), \mu_(3), \ldots).$$

Then

$$\mu_1 = \delta_1 \cong \overline{\delta}_1 = (1,0,0,\ldots)$$
$$\mu_2 \cong \overline{\mu}_2 = (0,1,0,\ldots)$$
$$\vdots$$
$$\mu_n \cong \overline{\mu}_n = (0,0,0,\ldots,1,\ldots).$$

Thus we have shown that the above sequence converges to $(0,0,0,\ldots)$ in the topology of vague or weak* convergence on measures. Also, notice that (μ_n) can not converge weakly to z by Proposition 3.1.

Midterm. Let S be a countable discrete space and $(\mu_n) \subset \text{Prob}(S)$. Suppose $\mu_n \to \nu$ weakly. Show that for every subset $B \subset S$

$$\mu_n(B) \to \nu(B).$$

Example A.2. Pointwise and uniform convergence of functions.

$$f_n(x) = x^n \text{ on } [0,1].$$

A.2 Some Initial Results of Convergence

In all that follows, S will be a compact semigroup and $(\mu_n) \subset \text{Prob}(S)$ such that $\mu_{k,n} \to \nu_k$ weakly for all k, and $\mathscr{F} = \{\lambda \mid \nu_{n_i} \to \lambda \text{ for some } (n_1)\}$ is the set of tail limits of (μ_n).

Examples of Compact Algebraic Structures.

(1) $[0,1]$ under multiplication (abelian semigroup)
(2) $[0,k)$ under addition mod k (abelian group)
(3) $\{z \in \mathbb{C} \mid \|z\| = 1\}$ (unit circle in the complex plane)
under multiplication (abelian group)
$z = e^{i\theta}$
(4) $n \times n$ stochastic matrices (semigroup)
(5) $n \times n$ doubly stochastic matrices (semigroup)
(6) $\{f : x \to x \mid x \text{ a normed vector space}\}$
$\|f(x) - f(y)\| \le c\|x - y\|, \quad c < 1$
contractions (semigroup)

Theorem A.1. Let $\overline{S} = \bigcup_{\nu \in \mathscr{F}} S(\nu)$ where $\mathscr{F}$ is the set of tail limits. Then the following are true:

(a) $\overline{S}$ is completely simple;
(b) $S(\nu) = X_\nu \times G \times Y$ for each $\nu \in \mathscr{F}$;
(c) $\overline{S} = \overline{X} \times G \times Y$ where $\overline{X} = \bigcup_{\nu \in \overline{S}} X_\nu$.

Proof. Let $a \in \overline{S}$ and consider $\overline{S}a\overline{S}$. WLOG assume $a \in S(\nu_\alpha)$, $\nu_\alpha \in \mathscr{F}$ then since $S(\nu_\alpha)S(\nu_\beta) = S(\nu_\alpha)$ for all β, $a\overline{S} \subset S(\nu_\alpha)$. But $\overline{S}S(\nu_\alpha) = \bigcup_{\nu \in \mathscr{F}} S(\nu)S(\nu_\alpha) = \bigcup_{\nu \in \mathscr{F}} S(\nu) = \overline{S}$. Thus $\overline{S}a\overline{S} = \overline{S}$ for all a and $\overline{S}$ is simple.

Now suppose e, f are idempotents with

$$ef = fe = e.$$

If $e, f \in S(\nu)$ for some fixed ν, then $ef = fe \Rightarrow e = f$, since $S(\nu)$ is completely simple. Thus suppose $e \in S(\nu_\alpha)$, $f \in S(\nu_2)$. But $ef \in S(\nu_\alpha)S(\nu_\beta) = S(\nu_\alpha)$ and $fe \in S(\nu_\beta)S(\nu_\alpha) = S(\nu_\beta)$

$$\begin{aligned} &\Rightarrow \quad e \in S(\nu_\beta) \\ &\Rightarrow \quad e = f \text{ since both are in the completely} \\ &\qquad\quad \text{simple semigroup } S(\nu_\beta) \\ &\Rightarrow \quad \text{all idempotents are primitive} \end{aligned}$$

(b) Choose $S(\nu)$ arbitrarily and pick $e \in S(\nu)$. Decompose both $S(\nu)$ and $\overline{S}$ with respect to e. That is, let

$$\begin{aligned} X_1 &= E[S(\nu)e], \quad H = eS(\nu)e, \quad Y_1 = E[eS(\nu)] \\ X &= E[\overline{S}e], \qquad G = e\overline{S}e, \qquad Y = E[e\overline{s}] \end{aligned}$$

Now for an arbitrary $S(\nu_\alpha)$

$$\begin{aligned} S(\nu_\alpha)S(\nu) &= S(\nu_\alpha) \\ S(\nu)S(\nu_\alpha) &= S(\nu) \end{aligned}$$

thus if $(x, g, y) \in S(\nu_\alpha)$

$$\begin{aligned} (X_1 \times H \times Y_1)(x, g, y) &\subset X_1 \times H \times Y_1 \\ \Rightarrow \quad y &\in Y_1. \end{aligned}$$

Now recall that $Y_1X_1 \subset H$. Thus

$$(x, g, y)(X_1 \times H \times Y_1) = \{x\} \times gH \times Y_1 \subset S(\nu_\alpha).$$

Consider g. Since $(e, e, e) \in X_1 \times H \times Y_1$ then

$$\begin{aligned} &(e, e, e)(x, g, y) \in X_1 \times H \times Y_1 \\ \Rightarrow \quad &(e, e(ex)g, y) \in X_1 \times H \times Y_1 \end{aligned}$$

but $ex = e$ since $e \in X$ and X is a left zero semigroup. But this implies that $g \in H$ and from this we get that $\{x\} \times H \times Y_1 \subset S(\nu_\alpha)$ from which $S(\nu_\alpha) = X_{\nu_\alpha} \times H \times Y_1$ follows easily.

(c) Finally, since

$$\begin{aligned}\overline{S} = \bigcup_{\nu\in\mathscr{F}} S(\nu) &= \bigcup_{\nu\in\mathscr{F}} X_\nu \times H \times Y_1 \\ &= X \times G \times Y.\end{aligned}$$

Then

$$H = G, \quad Y_1 = Y \text{ and } X = \bigcup_{\nu\in\mathscr{F}} X_\nu.$$

□

Now consider the case where S is a compact group. If ν_∞, ν_i' are tail limits of a convergent sequence then

$$\begin{aligned}\nu_\infty &= \nu_\infty \times \nu_\infty' \\ \nu_\infty' &= \nu_\infty' \times \nu_\infty.\end{aligned} \tag{A.1}$$

Since S is a group and ν_∞, ν_∞' are idempotents then $\nu_\infty = w_{H_1}$, $\nu_\infty' = w_{H_2}$, the Haar measure are compact subgroups H_1, H_2. Thus the above equations become

$$\begin{aligned}w_{H_1} &= w_{H_1} * w_{H_2} \\ w_{H_2} &= w_{H_2} * w_{H_1}\end{aligned} \tag{A.2}$$

which means that

$$\begin{aligned}H_1 = H_1 \cdot H_2 &\sim \bigcup_{h_1\in H_1} \cdot H_2 \\ H_2 = H_2 \cdot H_1.&\end{aligned}$$

Since $e \in H_1 \Rightarrow e \cdot H_2 \subset H_1 \Rightarrow H_2 \subset H_1$, likewise $H_1 \subset H_2 \Rightarrow H_1 = H_2 = H$, say. Thus w_H is the unique tail idempotent.

Theorem A.2. Suppose S is a compact group and $(\mu_n) \subset \mathrm{Prob}(S)$ with $\mu_{k,n} \to \nu_k$ for all k. Then for some subgroup $H \subset S$,

$$\nu_k \to w_H.$$

Proof. Recall that if every subsequence has a subsequence that converges to the same point, then the original sequence averages to that point.

Now let $(\nu_{n_i}) \subset (\nu_k)$ be any subsequence. Since $\mathrm{Prob}(S)$ is compact, then (ν_{n_i}) contains a limit point λ and $\nu_{n_i} \to \lambda$ for some $(\nu_{n_i}) \subset (\nu_{n_i})$.

Now $\lambda = \lambda^2 \Rightarrow \lambda = w_H$ for some $H \subset S$, but w_H is the unique tail idempotent. Thus all subsequences converge to $w_H \Rightarrow \nu_k \to w_H$.

Now suppose the convolution products do not converge.

For each fixed k, $(\mu_{k,n})$ is an infinite sequence of element in the compact set $\mathrm{Prob}(S)$. Let $k = 1$. Then there exists a subsequence (n_{0i}) such that $\mu_{0,n_{0i}}$ converges to some probability measure ν_0. Consider $(\mu_{1,n_{0,i}})$. This is an infinite set in a compact space and has a limit point, say ν_1, and there exists a

subsequence $(n_{1i}) \subset (n_{0i})$ such that $\mu_{1,n_{1i}} \to \nu_1$. But since (n_{1i}) is a subsequence of (n_{0i}), $\mu_{0,n_{1i}} \to \nu_0$ as well.

Continuing in this manner, we generate a nested collection of subsequences $(n_{0i}) \supset (n_{1i}) \supset (n_{2i}) \ldots$. Let $(n_i) = \bigcap_{j=0}^{\infty} (n_{ji})$. Then $\mu_{k,n_i} \to \nu_k$ for all K. □

Finally, consider (ν_{ni}). Then this is also an infinite set in a compact space and it has a limit point, say ν_∞. Choosing one final subsequence allows us to state the following theorem.

Theorem A.3 (Csiszar's Theorem). Let $(\mu_n) \subset \text{Prob}(S)$ and let (m_i) be a given sequence of integers. Then there exists $(n_i) \subset (m_i)$ such that

(1) $u_{k,n_i} \to \nu_k, \quad \nu_{n_i} \to \nu_\infty$;
(2) $\nu_k = \nu_k * \nu_\infty$;
(3) $\nu_\infty = \nu_\infty^2$.

Proof. Our previous discussion proved (1).

To prove (2) notice for $k < m < n_i$,

$$\mu_{k,n_i} = \mu_{k,m} * \mu_{m,n_i}$$

letting $n_i \to \infty$

$$\nu_k = \mu_{k,m} * \nu_m.$$

But this is true for all m, $m > k$. Thus letting $m = n_i$,

$$\nu_k = \mu_{k,n_i} * \nu_{n_i} \Rightarrow \nu_k = \nu_k * \nu_\infty.$$

Since k was arbitrary, letting $k = n_i$ we get $\nu_{n_i} = \nu_{n_i} * \nu_\infty$, which implies that $\nu_\infty = \nu_\infty^2$. □

(Mention tightness)

What about limit points that are not idempotent. Suppose (S_i) is an arbitrary subsequence and λ be a limit point of (ν_{s_i}), then for some $(r_i) \subset (s_i)$, $\nu_{r_i} \to \lambda$.

Now consider the subsequence (r_i) in the original products. Taking subsequences when appropriate (and renaming them (r_i) for convenience) we get,

$$\mu_{k,r_i} \to \nu'_k \quad \nu'_{r_i} \to \nu'_\infty$$
$$\nu'_k = \nu'_k * \nu'_\infty.$$

Finally, let λ' be such that $\nu'_{n_i} \to \lambda'$. Then we have the following theorem.

Theorem A.4.

$$\begin{array}{llll} \text{(1a)} & \nu_k = \nu'_k * \lambda & \text{(1b)} & \nu'_k = \nu_k * \lambda' \\ \text{(2a)} & \lambda = \lambda * \nu_\infty & \text{(2b)} & \lambda' = \lambda' * \nu'_\infty \\ \text{(3a)} & \lambda' = \nu_\infty * \lambda' & \text{(3b)} & \lambda = \nu'_\infty * \lambda \\ \text{(4a)} & \nu_\infty = \lambda' * \lambda & \text{(4b)} & \nu'_\infty = \lambda * \lambda'. \end{array}$$

Proof. Begin with the equations

$$(*)\quad \nu_k = \mu_{k,m} * \nu_m \text{ and } (**)\quad \nu'_k = \mu_{k,m} * \nu'_m.$$

Then letting $m = r_i$ in equation $(*)$ and letting $m = n_i$ in equation $(**)$ we get equations (1a) and (1b). □

Homework. Derive (2a) through (4b).

Consider the equation $\nu_k = \nu'_k * \lambda$. From the derivation it is clear that (ν_k) and (ν'_k) were completely arbitrary. Thus for any two sequences of limits points $(\nu_k), (\nu'_k)$ there exist tail limits λ, λ' such that λ transforms (ν'_k) into (ν_k) and λ' transforms (ν_k) into (ν'_k).

Look at the equations from an algebraic perspective.

Take, for example,

$$\nu_k = \nu'_k * \lambda.$$

Now ν'_∞ is a left identity for λ and ν_∞ is a right identity for λ. Thus the above equation can be written

$$\nu_k = \nu'_k * (\nu'_\infty * \lambda * \nu_\infty).$$

Now suppose conditions could be found such that $\nu'_\infty * \lambda * \nu_\infty = \nu'_\infty$. Then the following derivation would be possible

$$\begin{aligned} \nu_k &= \nu'_k * \lambda \\ &= \nu'_k * (\nu'_\infty * \lambda * \nu_\infty) \\ &= \nu'_k * \nu'_\infty \\ &= \nu'_k \end{aligned}$$

since ν'_∞ is a right identity for ν'_k.

Now this is true for all k, (under the above assumption) that $(\mu_{k,ni})$ and (μ_{k,r_i}) were arbitrary subsequences of $(\mu_{k,n})$ and they each converge to (ν_k) for all k. Thus $\mu_{k,n} \to \nu_k$ for all K.

Thus we have the following general theorem.

Theorem A.5. Let $(\mu_n) \subset \text{Prob}(S)$, where S is a locally compact topological semigroup. Let λ be an arbitrary tail limit of (μ_n). Then $\mu_{k,n}$ converges for all k weakly iff there exist "appropriate" tail idempotents ν'_∞, ν_∞ such that

$$\nu'_\infty = \nu'_\infty * \lambda * \nu_\infty.$$

Recall that when $\mu_{k,n}$ converges, the set of tail limits form a right zero semigroup. Then if $\nu'_\infty, \lambda, \nu_\infty$ are tail limits

$$\nu'_\infty = \nu'_\infty * \lambda * \nu_\infty.$$

Just a note of clarification. When $\nu_{k,n}$ does not converge then the set of tail limits, $\mathscr{F}$, must be defined in the following way

$$\mathscr{F} = \{\lambda \mid \lambda \text{ is a limit point of some } (\nu_k) \text{ where } \mu_{k,n_i} \to \nu_k \text{ for some } (n_i)\}$$

Suppose S is abelian. Then for any measurable set B,

$$Bx^{-1} = \{y \mid yx \in B\} = \{y \mid xy \in B\} = x^{-1}B.$$

Thus for all B

$$\begin{aligned}\mu_1 * \mu_2(B) &= \int \mu_1(Bx^{-1})\mu_2(dx) \\ &= \int \mu_1(x^{-1}B)\mu_2(dx) \\ &= \mu_2 * \mu_1(B).\end{aligned}$$

Thus Prob(S) is an abelian semigroup.

Notice, however, that ν_∞, ν'_∞ in the convolution equations were arbitrary with

$$\nu_\infty = \lambda' * \lambda \quad \nu'_\infty = \lambda * \lambda'.$$

But if S is abelian $\nu_\infty = \lambda' * \lambda = \lambda * \lambda' = \nu'_\infty$ and each sequence (μ_n) has a unique tail limit.

Theorem A.6. Let S be a locally compact abelian semigroup and $(\mu_n) \subset \text{Prob}(S)$. Let $\mathscr{F}$ be the set of tail limits of (μ_n) and $G = \overline{\langle \mathscr{F} \rangle}$ be the closed semigroup generated by $\mathscr{F}$. Then G is a group.

Proof. Midterm assignment. □

Theorem A.7. Suppose S is a locally compact abelian semigroup and $(\mu_n) \subset \text{Prob}(S)$. Suppose there exists an $x \in S$ such that for all $k \geq k_0$ and some $\delta > 0$

$$\liminf_{n\to\infty} \mu_{k,n}(N(x)) > \delta$$

Then $\mu_{k,n} \to \nu_k$ for all k as $n \to \infty$.

Proof. We will prove the theorem in the case when S is discrete. Let x be such that $\liminf_{n\to\infty} \mu_{k,n}(x) \geq \delta > 0$ for all $k \geq k_0$.

Step 1 Suppose (ν_k) is such that $\mu_{k,n_i} \to \nu_k$ for all k. Then $\mu_{k,n_i}(x) \to \nu_k(x)$. Suppose $\nu_k(x) < \delta$. Then there exists an $I \ni i \geq I$, $\mu_{k,n_i}(x) < \delta$. Thus the sequence of numbers $(\mu_{k,n_i}(x))$ has a limit point r where $r \leq \delta$.

$$\begin{aligned}&\Rightarrow \quad \liminf_n \mu_{k,n}(x) \leq \delta \\ &\Rightarrow \quad \nu_k(x) \geq \delta \text{ for all } k \geq k_0.\end{aligned}$$

Now let λ be any limit point of (ν_k). Since $\nu_k(x) \geq \delta$ for all $k \geq k_0$, $\lambda(x) > 0$. Thus we have shown that $x \leq S(\lambda)$ for every tail limit of λ of (μ_n).

Step 2 Now consider $S(\nu_\infty)$. Since ν_∞ is idempotent, $S(\nu_\infty)$ is a completely simple semigroup. Let $X \times G \times Y$ be a Rees product for $S(\nu_\infty)$.
Let $e_1 \cong (x_1, (y_1x_1)^{-1}, y_1)$, $e_2 \cong (x_2, (y_2x_2)^{-1}, y_2)$ be two idempotents of $S\nu_\infty \cong X \times G \times Y$. Then

$$e_1e_2 \cong (x_1, (y_1x_1)^{-1}, y_1)(x_2, (y_2x_2)^{-1}y_2)$$
$$e_2e_1 \cong (x_2, (y_2x_2)^{-1}, y_2)(x_1, (y_1x_1)^{-1}y_1)$$

Then if $e_1e_2 = e_2e_1$ it is clear that $x_1 = y_2$, $y_1 = y_2$. But then $e_1 = e_2 = e$ (say). Then $eS(\nu_\infty)e = S\nu_\infty$ and $S(\nu_\infty)$ is a group. Thus an abelian completely simple semigroup is a group. But if ν_∞ is an idempotent measure supported on a group, then it is the Haar measure on that group.

Step 3 Now recall the convolution equations.

$$\lambda * \lambda' = \nu_\infty$$

Now $x \in S(\lambda')$. Thus $S(\lambda * L\delta_x) = S_\lambda * x \subset S_{(\lambda)} \cdot S_{(\lambda)} = S(\nu_\infty)$. Therefore if we call $S(\nu_\infty) = H$, H a group, then the probability measure $\lambda * \delta_x \in \mathrm{Prob}(H)$. But $\nu_\infty = w_H$ from Step 2, and since w_H acts like a zero on $\mathrm{Prob}(H)$

$$(\lambda * \delta_x) * w_H = w_H.$$

Finally,

$$\begin{aligned}\nu_k &= \nu_k' * \lambda = \nu_k' * \lambda * w_H \\ &= \nu_k' * \lambda * (\delta_x * w_H) \\ &= \nu_k' * (\lambda * \delta_x) * w_H \\ &= \nu_k' * w_H = \nu_k'\end{aligned}$$

and we get convergence. □

A.3 Convolution Powers

Prelim Convolution powers, S_0, algebra of measures.

Theorem A.8. Let $\mu \in P(S)$ where S is a compact semigroup. (Suppose $\{\mu^n \mid n \geq 1\}$ is tight.) Let $K = \{\lambda \mid \lambda \text{ is a weak limit point of } (\mu^n)\}$ and

$$S_0 = U\{S(\lambda) \mid \lambda \in K\}$$
$$S_1 = \overline{S}_0 \text{ in general)}$$

Then the following are true:

(a) The sequence $\frac{1}{n}\sum_{k=1}^{n} \mu^k$ converges to $\nu \in P(s)$ with $\nu = \nu \times \nu = \mu \times \nu = \nu * \mu$.

(b) The set K is a group with respect to convolution.
(c) $S(\nu) = S_0$.
(d) If η is the identity of K, then $S(\eta) \subset S_0$ and $S_0 = X \times H \times Y$, $S(\eta) = X \times H_1 \times Y$, $H_1 \Delta H$.
(e) The sequence μ^n converges weakly iff there does not exist a closed normal subgroup H_0 of H such that

$$S(\mu) \cdot (X \times H_0 \times Y) \subset S \times gH_0 \times Y$$

for some $g \in H - H_0$.
(f) The sequence μ^n converges weakly iff there does not exist a closed normal subgroup H_0 of H such that $YX \subset H_0$ and $eS(\mu)e \subset gH_0$ for some $g \in H - H_0$ where e is the identity of H.

Proof. (a) For a finite measure ν

$$\|\nu\| = \sup_{\substack{\|f\| \le 1 \\ f \in C_k(S)}} \int f \, d\nu \text{ if } \nu \in \text{Prob}(S), \quad \|\nu\| = 1.$$

Now notice that

$$\begin{aligned} \mu * \mu_n &= \mu * \left[\frac{1}{n} \sum_{k=1}^{n} \mu^k \right] \\ &= \frac{1}{n} \sum_{k=1}^{n} \mu^{k+1} \\ &= \frac{1}{n} [\mu^2 + \mu^3 + \ldots + \mu^{n+1}] \end{aligned}$$

thus $\mu_n = \mu * \mu_n + \frac{1}{n}[\mu - \mu^{n+1}]$ and $\|\mu_n - \mu * \mu_n\| = \frac{1}{n}\|\mu - \mu^{n+1}\|$

$$\begin{aligned} \Rightarrow \quad \lim_{n \to \infty} \|\mu_n - \mu * \mu_n\| &= \lim_{n \to \infty} \frac{1}{n} \|\mu - \mu^{n+1}\| \\ &\le \lim_{n \to \infty} \frac{1}{n} \left[\|\mu\| + \|\mu^{n+1}\| \right] \\ &= 0. \end{aligned}$$

Thus let ν be any limit point of (μ_n)

$$\mu_{n_i} \to \nu \left[\mu_{n_i}(x) = \mu * \mu_{n_i}(x) + \frac{1}{n} [\mu(x) - \mu^{n+1}(x)] \right].$$

Then

$$\begin{aligned} & \nu = \mu * \nu \\ \Rightarrow \quad & \mu * \nu = \mu^2 * \nu = \nu \\ \text{and} \quad & \nu = \mu^k * \nu = \nu * \mu^k. \end{aligned}$$

Then

$$\begin{aligned}\mu_n * \nu &= \left(\frac{1}{n}\sum_{k=1}^{n}\mu^k\right) * \nu \\ &= \frac{1}{n}\sum_{k=1}^{n}\mu^k * \nu \\ &= \frac{1}{n}(n\nu) = \nu.\end{aligned}$$

Suppose (μ_n) had two limit points γ_1, ν_2, $\gamma_1 \neq \nu_2$; $\mu_{n_i} \to \nu_1$, $\mu_{m_i} \to \nu_2$.

$$\begin{aligned}\nu_1 = \mu_n * \nu_1 \Rightarrow \nu_1 &= \mu_{m_i} * \nu_1 \\ \nu_1 &= \nu_2 * \nu_1\end{aligned}$$

but

$$\begin{aligned}\nu_2 = \nu_2 * \mu_n \Rightarrow \nu_2 &= \nu_2 * \mu_{n_i} \\ \Rightarrow \nu_2 &= \nu_2 * \nu_1 \\ \Rightarrow \nu_2 &= \nu_1 \quad (= \nu \text{ say}).\end{aligned}$$

Also, it is clear that $\nu * \nu = \nu$. Thus the support of ν is a completely simple semigroup $X \times G \times Y$.

Notice that G must be finite, in the discrete case, since $\nu = \alpha * w_G * \beta$ where w_G is uniform on a discrete group G.

$$\begin{aligned}& w_G(Bx^{-1}) = w_{\mathbb{B}}(B) \\ \Rightarrow \quad & w_G(x) = w_G(y) \text{ for all } x, y \in G \\ \Rightarrow \quad & \text{if } w_G(x) = k\end{aligned}$$

then if $|\mathbb{G}| = \infty$, $w_G(G) = \infty$. $\otimes$

Now recall $S = \bigcup_{n=1}^{\infty} (S(\mu))^n$ then

$$\begin{aligned}S \cdot S(\nu) &= \bigcup_{n=1}^{\infty} (S(\mu))^n \cdot S(\nu) \\ &= S(\nu) = S(\nu) \cdot S.\end{aligned}$$

Thus $S(\nu)$ is an ideal of S and by simplicity is the minimal ideal of S.

Now assume S is discrete. Then for $B \subset S$

$$\mu_n(B) \to \nu(B).$$

Thus

$$\frac{1}{n}\sum_{k=1}^{n}\mu^k(S(\nu)) \to 1.$$

Thus, given $\varepsilon > 0$, there exists m such that $\mu^m(S(\nu)) > 1 - \varepsilon$.

Suppose not. Then for some $\varepsilon > 0$, $\mu^m(S(\nu)) \le 1-\varepsilon$ for all $m \ge 1$. Then

$$\begin{aligned}\frac{1}{n}\sum_{k=1}^{n}\mu^k(S(\nu)) &\le \frac{1}{n}n(1-\varepsilon)\\ &= 1-\varepsilon\end{aligned}$$

for each n

$$\Rightarrow \quad \nu(S(\nu)) \le 1-\varepsilon. \ \otimes$$

Now since $S(\nu)$ is an ideal of S

$$S(\nu) \supset S\cdot S(\nu)$$

and for all n

$$\begin{aligned}\mu^{n+m}(S(\nu)) &\ge \mu^{n+m}(S\cdot S(\nu))\\ &\ge \mu^n(S)\cdot\mu^m(S(\nu))\\ &> 1-\varepsilon\end{aligned}$$

$$\Rightarrow \quad \liminf_n \mu^n(S(\nu)) = 1$$

$$\Rightarrow \quad \lim \mu^n(S(\nu)) = 1. \qquad \square$$

Homework. Thus if λ is a weak limit point of (μ^n) then $S(\lambda) \subset S(\nu)$.

Thus $S_0 = \bigcup_{\lambda\in K} S(\lambda) \subset S(\nu)$.

Now the set of limit points of (μ^n) is an ideal of $(\overline{\mu^n})$. Thus S_0 is an ideal of S, since $\mu^k * \lambda$ is also a limit point of (μ^n). (Thus $S(\mu^k * \lambda) \subset S_0$.) Recall

$$S = \bigcup_{k=1}^{\infty} S(\mu^k).$$

But, since ν is idempotent, $S(\nu)$ is completely simple, which implies that $S_0 = S(\nu)$.

(d) Now let η be the identity of K where K is the set of limit points of (μ^n). Since $S(\eta) \subset S(\nu)$ and both $S(\eta)$ and $S(\nu)$ are completely simple, choose $e \in S(\eta)$ and construct the Rees product structure with respect to e of both sets

$$\begin{aligned}S(\eta) &= X_1 \times H_1 \times Y_1\\ S(\nu) &= X \times H \times Y.\end{aligned}$$

But for each $(x', g', y') \in S(\lambda^{-1})$

$$\begin{aligned}&(X \times U_\lambda \times Y)(X \times H_1 g' \times Y) \subset X \times H_1 \times Y\\ \Rightarrow \quad &(H_1 g)(H_1 g') \subset H_1 \text{ for each } H_1 g \subset U_\lambda\end{aligned}$$

in particular

$$H_1geg' \subset H_1$$
$$H_1gg' \subset H_1$$
$$H_1g \subset H_1g'$$

thus U_λ consists of a single coset of H_1. Thus $S(\lambda) = X \times H_1g \times Y$.

Homework. Show that $S(\lambda) = X \times gH_1 \times Y$. Thus whenever $(x,g,y) \in S(\lambda)$, $\lambda \in K$

$$gH_1 = H_1g.$$

But since $S(\nu) = \bigcup_{\lambda \in K} S(\lambda) = X \times H \times Y$. Then for all $g \in H$, $gH_1 = H_1g$ which shows that H_1 is normal in H.

Now $S(\nu) = \bigcup_{\lambda \in K} S(\lambda)$ and for each $\lambda \in K$ $\exists$ $\lambda^{-1} \in K$ such that

$$\lambda * \lambda^{-1} = \eta.$$

Now let $(x,g,y) \in S(\nu) \Rightarrow (x,g,y) \in S(\lambda)$ for some λ. Then $(x,g,y) \cdot (x',g',y') \in S(\eta)$ for each $(x',g',y') \in S(\lambda^{-1})$. Thus $x \in X$. Similarly since $(x',g',y') \cdot (x,g,y) \in S(\eta)$, $y \in Y_1$, thus $X_1 = X$, $Y_1 = Y$.

Notice that since $\eta * \lambda = \lambda = \lambda * \eta$, for $(x,g,y) \in S(\lambda)$.

$$\begin{aligned}
& (X \times H_1 \times Y)(x,g,y) \subset S(\lambda) \text{ but } YX \subset H_1 \\
\Rightarrow \quad & X \times H_1g \times \{y\} \subset S_\lambda \\
\text{and} \quad & (X \times H_1g \times \{y\}) \cdot \{x\} \times \{(yx)^{-1}\} \times Y \subset S_\lambda \\
\Rightarrow \quad & X \times H_1g \times Y \subset S_\lambda \\
\Rightarrow \quad & X \times U_\lambda \times Y \subset S_\lambda \\
\text{where} \quad & U_\lambda = \bigcup H_1g.
\end{aligned}$$

Then we have shown that the supports of all the limits measures of (μ^n) are on either $X \times H_1 \times Y$ or $X_1 \times H_1g \times Y$.

Now suppose $\lambda^2 \neq \lambda$. Can λ be supported on $X \times H_1 \times Y$. Suppose it was. Recall $\lambda * \eta = \lambda = \eta * \lambda$. Thus for $u \in S(\lambda)$, $v \in S(\eta)$

$$\lambda(Bu^{-1}v^{-1}) = \lambda(Bu^{-1}).$$

But we are supposing that $S(\eta) = S(\lambda)$. Thus,

$$\begin{aligned}
\lambda^2(Bu^{-1}) &= \int \lambda(Bu^{-1}v^{-1})\lambda(dv) \\
&= \lambda(Bu^{-1}) \int \lambda(dv) \\
&= \lambda(Bu^{-1}).
\end{aligned}$$

Thus

$$\begin{aligned}\lambda^2(B) &= \int \lambda^2(Bu^{-1})\eta(du)\\ &= \int \lambda(Bu^{-1})\eta(du)\\ &= \lambda * \eta(B) = \lambda(B).\end{aligned}$$

But this is true for all Borel sets B

$$\Rightarrow \quad \lambda^2 = \lambda \Rightarrow \lambda = \eta.$$

Now for part (e) and (f) of the theorem. If $\mu * \eta = \eta$, where η is the identity of K then $\mu^n * \eta = \eta$ for all $n \geq 1$. Then by the continuity of convolution, $\lambda * \eta = \eta$ for all $\lambda \in K$. But since K is a group, and η the identity, then $\lambda = \eta$. Thus $K = \{\eta\}$ and (μ^n) converges.

We showed earlier that K is an ideal of (μ^n). Thus if (μ^n) does not converge, then $\mu * \eta = \lambda \neq \eta$, $S(\mu) \cdot S(\eta) = X \times H_1 g \times Y$. That is, $S(\mu) \cdot (X \times H_1 \times Y) = X \times H_1 g \times Y$.

Now suppose there exists a closed normal subgroup H_1 of H such that $YX \subset H_1$ and $S(\mu) \cdot X \times H_1 \times Y \subset X \times H_1 g \times Y$, $g \in H - H_1$. Then

$$\begin{aligned}S(\mu^2) \cdot X \times H_1 \times Y &\subset (S(\mu))^2 \cdot X \times H_1 \times Y\\ &\subset S(\mu)[X \times H_1 g \times Y]\\ &= S(\mu)(X \times H_1 \times Y)(X \times H_1 g \times Y)\\ &\subset (X \times H_1 g \times Y)(X \times H_1 g \times Y)\\ &= X \times H_1 g^2 \times Y\end{aligned}$$

and in general,

$$S(\mu^n) \cdot (X \times H_1 \times Y) \subset X \times H_1 g^n \times Y$$

$\{H_1 g^n \mid n \geq 1\}$ has a finite number of elements. This is true, since we are assuming discreteness and $S(\nu) = X \times H \times Y$, $\nu = \alpha * w_H * \beta$ but it is a fact that the Haar measure is only supported on compact sets and (compact, discrete) $\equiv$ (finite). Thus $\{H_1 g^n\}$ forms a finite cyclic group and for infinitely many n, say (n_i)

$$H_1 g^{n_i} = H_1$$

and

$$H_1 g^{m_i} = H_1 g$$

for some (m_i). Thus if β is any probability measure on $X \times H_1 \times Y$,

$$\begin{aligned}S(\mu^{n_i} * \beta) &\subset X \times H_1 g^{n_i} \times Y\\ &= X \times H_1 \times Y\end{aligned}$$

and

$$\begin{aligned}S(\mu^{m_i} * \beta) &\subset X \times H_1 g^{m_i} \times Y\\ &= X \times H_1 g \times Y.\end{aligned}$$

But

$$(X \times H_1 \times Y) \cap (X \times H_1 g \times Y) = \emptyset$$
$$\Rightarrow \quad (\mu^n * \beta) \text{ does not converge}$$
$$\Rightarrow \quad (\mu^n) \text{ does not converge.}$$

Homework. Show part (f).

Corollary A.1. Let G be a finite group and $u \in \text{Prob}(G)$. Suppose $e \in S(\mu)$. Then (μ^n) converges.

Proof. Let H be any normal subgroup of G then

$$S(\mu) \cdot H = \bigcup_{g \in S(\mu)} gH \supset eH = H.$$

Thus (μ^n) converges. □

Application. Convergence of doubly stochastic matrices.

Appendix B
The Symmetric-Function Excursus

The ladder chain belongs in the main book. The extended symmetric-function material has a separate home in the SymmetricFunctions project, specifically LadderProcesses_SymmetricFunctions.tex. Monomial, power-sum, homogeneous, and Schur-function calculations should be included here only when used by the semigroup narrative, rather than importing the whole independent subject. The original source is retained and indexed; this appendix is a scope statement, not a completed transfer of that paper.

Appendix C
Exact Computation and the Example Bank

The checked examples and exact verification records are in the project's Examples folder. The four-state example supplies a two-partition kernel with local C_2; the six-state example supplies explicit reset words and a solvable Lie algebra; the nine-state example supplies local S_3, canonical choices of different ranks, and parity-coupled tensor groups. Each should be revisited at the theorem it illustrates, not introduced only in this appendix. A future computational edition should supply common input/output routines, test cases, and selected exercise solutions.

C.1 Running all recolorings of the four-state graph

Start with $R = [4312]$, $B = [3443]$ and right-action composition. Then

$$A = \frac{1}{2}\begin{pmatrix} 0&0&1&1\\ 0&0&1&1\\ 1&0&0&1\\ 0&1&1&0 \end{pmatrix}, \qquad \Delta = \frac{1}{2}\begin{pmatrix} 0&0&-1&1\\ 0&0&1&-1\\ 1&0&0&-1\\ 0&1&-1&0 \end{pmatrix}, \qquad \pi = \frac{1}{6}(1,1,2,2).$$

For a subset S of source vertices, swap the red and blue outgoing edges precisely at S. Writing $G_S = \mathrm{diag}(\varepsilon_1, \ldots, \varepsilon_4)$, with $\varepsilon_x = -1$ at swapped sources and $+1$ elsewhere, gives

$$A_S = A, \qquad \Delta_S = G_S\Delta, \qquad R_S = A + \Delta_S, \quad B_S = A - \Delta_S.$$

The stationary row is unchanged. In particular $\mathrm{rank}\, A = 3$ and $d = \mathrm{rank}\, \Delta_S = 3$ for every recoloring.

Fixed target-pair structure

Using Definition 1.2, the compact source-indexed list and the underlying undirected graph are

$$\boxed{\mathscr{T}(A) = (34, 34, 14, 23), \qquad H_A : \ 1-4-3-2.}$$

The two occurrences of 34 come from sources 1 and 2. They represent a repeated auxiliary edge, not parallel edges in the original directed graph. This fixed header is recorded once, not repeated in each recoloring row.

The target-pair graph has one connected component, and it is bipartite, with sides $\{1,3\}$ and $\{2,4\}$. Consequently

$$\ker \Delta_S = \operatorname{span}\{(1,1,1,1)^{\mathsf{T}}\}, \qquad \ker A = \operatorname{span}\{(1,-1,1,-1)^{\mathsf{T}}\}.$$

Thus $\operatorname{rank}\Delta_S = \operatorname{rank} A = 3$ for all sixteen recolorings. There are exactly two labeled CC choices, interchanged by a global color swap: direct every red edge into $\{1,3\}$ and every blue edge into $\{2,4\}$, or reverse these target types. They are rows $k = 14$ and $k = 4$ of the table, respectively.

The graph-intrinsic canonical component partition is the single block $\{1,2,3,4\}$, whereas its chosen bipartition has two sides. Neither is to be identified automatically with a kernel partition. Most importantly, this same auxiliary graph supports kernel ranks one, two and three. It determines the ranks of A and Δ, and CC existence, but not the kernel rank or whether π generates the full row space of Δ_S through positive powers. The latter is exactly what the ranked test in the recoloring table measures.

A one-at-a-time enumeration scheme

For $k = 0, \ldots, 15$, use the binary-reflected Gray code $g_k = k \operatorname{xor} \lfloor k/2 \rfloor$. Bit $x-1$ specifies whether source x is swapped relative to the original coloring. Consecutive masks differ in exactly one bit, so consecutive colorings differ by a single elementary recoloring. Every labeled coloring occurs once; global red/blue interchanges are retained, not identified.

For each coloring compute the cycle matrix

$$\Sigma_S = \begin{pmatrix} \pi\Delta_S \\ \pi\Delta_S^2 \\ \pi\Delta_S^3 \end{pmatrix}.$$

The coloring is called ranked here precisely when $\operatorname{rank}\Sigma_S = d$. The sequence starts with exponent one; π is not an additional row of this test. Independently enumerate the transformations generated by R_S, B_S; the smallest image

k	Swapped sources S	R_S	B_S	r_K	rank Σ_S	Reset word
0	$\varnothing$	[4312]	[3443]	2	1	–
1	{1}	[3312]	[4443]	1	3	*RB*
2	{1,2}	[3412]	[4343]	2	1	–
3	{2}	[4412]	[3343]	1	3	*RB*
4	{2,3}	[4442]	[3313]	1	3	*RB*
5	{1,2,3}	[3442]	[4313]	1	3	*RRB*
6	{1,3}	[3342]	[4413]	3	1	–
7	{3}	[4342]	[3413]	1	3	*BBR*
8	{3,4}	[4343]	[3412]	2	1	–
9	{1,3,4}	[3343]	[4412]	1	3	*BR*
10	{1,2,3,4}	[3443]	[4312]	2	1	–
11	{2,3,4}	[4443]	[3312]	1	3	*BR*
12	{2,4}	[4413]	[3342]	3	1	–
13	{1,2,4}	[3413]	[4342]	1	3	*RRB*
14	{1,4}	[3313]	[4442]	1	3	*RB*
15	{4}	[4313]	[3442]	1	3	*BBR*

Table C.1 All sixteen labeled recolorings in Gray-code order. Swapped sources are measured relative to the initial coloring. Words act from left to right. A dash means that no reset word exists. Every displayed reset word has minimum length.

size is the kernel rank r_K. A breadth-first enumeration also gives shortest reset words.

There are ten rank-one kernels, four rank-two kernels, and two rank-three kernels. In this example, exactly the ten rank-one colorings are ranked. Thus the sufficient ranked criterion detects every synchronizing coloring in this table; this computation does not establish a general converse. The two CC colorings occur at $k = 4$ and $k = 14$; their global color interchange exchanges them. Complementary swap sets always give a global color interchange, hence the same kernel and cycle-matrix ranks. The complete exact records, including kernel cardinalities, are supplied in the accompanying JSON and CSV files.

Appendix D
Alternative Proofs and Historical Arguments

These inherited blocks are preserved for mathematical review, not as assertions of current research status or novelty. In particular, a proposed technique for the generalized problem is not a completed proof. Alternative proofs of shared results should be selected or cross-referenced after the core exposition is stabilized.

D.1 Introduction

Suppose one is given a strongly connected digraph (that is, there is a directed path between any two vertices). One can think of the vertices of the graph as buildings, for example, and the directed arcs as unnamed, one-way roads connecting the buildings. Suppose, further, that in each building there is a person. Under what conditions can one color the roads, so that the same set of instructions allows each person to get to the same building.

This problem, whose precise formulation is given below, was first posed by Adler, Goodwyn, and Weiss [1] in a paper more than twenty years ago. Due to the informal interpretation given above, it has become known as the road coloring problem, and the set of directions which gets everyone to the same building is known as the synchronizing instruction.

Various partial results exist (as our list of references indicates). The techniques employed by these papers vary considerably, but almost all have a heavy combinatorial flavor. The main contribution of this paper is to translate the problem into semigroup theory and to use the fundamental theorem of finite semigroups to both generalize the problem and to simplify the proofs of many of the past results. This unification of technique permits the ability to see many diverse results from a common perspective. In particular, the structure theory of the minimal ideal (kernel) of a finite semigroup plays an essential role. We feel that the kernel will eventually play an important role

in the complete solution of the problem. All of the semigroup theory used is elementary and can be found, for example, in Clifford and Preston [7].

The structure of this paper will be as follows. After this introduction, we will provide the necessary semigroup theory to state the problem precisely. Next we will include a section that shows how the structure of the kernel allows us to translate certain graph theoretic properties into the language of semigroups. In addition, we will generalize the problem in this semigroup context. Finally, we will present new semigroup proofs of several of the most well known results. Examples to illustrate various key ideas will be used throughout the paper.

D.2 Semigroup Proofs of Selected Results

In this final section we present new proofs of several well-known results. Part one of the next theorem was originally proven by Friedman in [9]. To the best of our knowledge, part two of this theorem is a new result.

Theorem D.1. Let $G = (V, \mathscr{E})$ be a strongly connected, d-out digraph with adjacency matrix A. Suppose $w = (w_1, w_2, \ldots, w_n)$ is such that $wA = dw$ and let $W = \sum_{i=1}^{n} w_i$ be the Friedman weight of G. We can and do assume that each w_i is a positive integer.

Let $S = \langle R_1, R_2, \ldots, R_d \rangle$ be a coloring semigroup of G with kernel $\mathscr{K}$ having rank k. Then

(1) $k \mid W$
(2) $k \mid |\mathscr{H}|$ where $\mathscr{H}$ is any maximal group in $\mathscr{K}$.

Proof. If $S = \langle R_1, R_2, \ldots, R_d \rangle$ then

$$\frac{1}{d}A = \frac{1}{d}\sum_{j=1}^{d} R_j.$$

Notice $\frac{1}{d}A$ is stochastic, and since strongly connected

$$\lim_{m\to\infty} \frac{1}{m}\sum_{k=1}^{m}\left(\frac{1}{d}A\right)^k = A_0,$$

where A_0 is a rank one strictly positive stochastic matrix with each row equal to $w_0 = \left(\frac{1}{W}\right)w$.

Now consider the element in the semigroup algebra of binary stochastic matrices

$$\frac{1}{d}R_1 + \frac{1}{d}R_2 + \cdots + \frac{1}{d}R_d.$$

Since the coloring semigroup is finite, there is an equivalence between powers of this element and convolution powers of the probability measure μ where

$$\mu(R_i) = \frac{1}{d}.$$

It is known that

$$\lim_{m\to\infty}\left(\frac{1}{m}\sum_{k=1}^{m}\mu^k\right) = \lambda$$

where the support of the measure λ is precisely the kernel $\mathscr{K}$ of the coloring semigroup S. (See Högnäs and Mukherjea [12], pages 79 and 92, for more details.) Thus

$$A_0 = \sum_{K\in\mathscr{K}} \lambda(K)K.$$

Now λ is an idempotent probability measure which is well-known to be locally uniform on each maximal group $\mathscr{H}$.

Precisely, let $E \in E(\mathscr{K})$, then since A_0 is a rank one stochastic matrix,

$$EA_0E = A_0E.$$

Also, $\delta_E * \lambda * \delta_E$, which is an idempotent probability measure on its support $E\mathscr{K}E$ (the reason being $\lambda * \delta_E * \lambda = \lambda$, see [12]), is uniform on $E\mathscr{K}E$; thus

$$A_0E = \sum_{H\in E\mathscr{K}E} \frac{1}{|\mathscr{H}|}H. \qquad (*)$$

Since $\mathrm{rank}(E) = k$, renumber the range elements of E to be $\{1,\ldots,k\}$. Then

$$E = \left[\begin{array}{c|c} I_{k\times k} \\ \hline T \end{array}\;\middle|\; 0\right]$$

where T is $(b-k)\times k$. Also, each $H \in \mathscr{H} = E\mathscr{K}E$ has the same zero columns as E, being elements in the same group.

Write $\mathscr{H} = \{H_1, H_2, \ldots, H_m\}$ and let $\mathscr{H}_1 = \{H_1, H_2, \ldots, H_t\}$ be all the elements such that

$$(H_j)_{1,1} = 1 \qquad 1 \le j \le t;$$

then by $(*)$ above

$$(A_0E)_{1,1} = \frac{t}{|\mathscr{H}|}.$$

But A_0E has identical rows, thus

$$(A_0E)_{2,1} = \frac{t}{|\mathscr{H}|}.$$

Now $\mathscr{H}_1$ is a subgroup of $\mathscr{H}$ that fixes the element 1. Let H_{t+1} be such that

$$(H_{t+1})_{1,2} = 1$$

then $\mathscr{H}_1 H_{t+1} = \{H_{t+1}, \ldots, H_{2t}\}$ is a coset of $\mathscr{H}_1$.

Notice for j such that $t+1 \leq j \leq 2t$

$$\left(H_j^{-1}\right)_{2,1} = 1;$$

thus

$$(A_0 E)_{1,2} = \frac{t}{|\mathscr{H}|}$$

thus,

$$A_0 E = \left[\begin{array}{ccc|c} \frac{t}{|\mathscr{H}|} & \frac{t}{|\mathscr{H}|} \cdots & \frac{t}{|\mathscr{H}|} & \\ \vdots & & \vdots & \\ \vdots & & \vdots & 0 \\ \frac{t}{|\mathscr{H}|} & \cdots \cdots & \frac{t}{|\mathscr{H}|} & \end{array}\right]$$

Since $A_0 E$ is a stochastic matrix, then

$$\frac{tk}{|\mathscr{H}|} = 1$$

which implies

$$k \Big| |\mathscr{H}|.$$

Also, since

$$A_0 = \begin{pmatrix} w_0 \\ w_0 \\ \vdots \\ w_0 \end{pmatrix}$$

where

$$w_0 = \left(\frac{w_1}{W}, \frac{w_2}{W}, \cdots, \frac{w_n}{W}\right) \qquad w_i \in Z^+$$

then

$$A_0 E = \left[\begin{array}{cccc|c} \sum\limits_{E_{i,1}=1} \frac{w_i}{W}, & \sum\limits_{E_{i,2}=1} \frac{w_i}{W}, & \cdots, & \sum\limits_{E_{i,k}=1} \frac{w_i}{W} & \\ \vdots & \vdots & \vdots & \vdots & \\ \vdots & \vdots & \vdots & \vdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \end{array}\right]$$

Let

$$\sum_{E_{i,1}=1} w_i = W_0.$$

Then

$$\frac{W_0}{W} = \frac{t}{|\mathscr{H}|};$$

but since $|\mathscr{H}| = tk$, then

$$kW_0 = W$$

and thus,

$$k \mid W. \qquad \square$$

In the remainder of the paper, we will make the simplifying assumption that the out degree of each vertex is two. We are leading up to O'Brien's result, in which he assumes that a cycle of prime length exists within the graph. Under this assumption, O'Brien also showed that the general problem can be reduced to one of out degree two (see [16] for details). The cycle C will, for the purposes of notation, always in the rest of this paper consist of vertices $1 \to 2 \to 3 \cdots \to p \to 1$. If p is assumed to be prime, it will be explicitly stated.

Definition D.1. A binary stochastic matrix R will be called a C-matrix for G iff

(1) $A = R + B$
(2) $(R)_{i,i+1} = (R)_{p,1} = 1$ for $i = 1, \ldots, p-1$.
(3) For each $w \in V$, there is an $i \in C$ such that $(R^t)_{w,i} = 1$, for some t sufficiently large.

It is clear from this definition that for t_0 sufficiently large

$$R^{t_0} = \left[\begin{array}{c|c} \frac{I_{p\times p}}{M} & 0 \end{array} \right]$$

where $I_{p\times p}$ is the $p \times p$ identity matrix and M is a binary stochastic $(b-p) \times p$ matrix. In fact, the powers of R generate an abelian semigroup whose kernel is a cyclic group of order p. R^{t_0} is simply the identity element for this cyclic group.

Definition D.2. A vertex $w \in V$ is called periodic with respect to a C-matrix R iff whenever $A_{w,u} = A_{w,v} = 1$, then there exists $k \geq 1$ such that for some $i \in C$,

$$\left(R^k\right)_{u,i} = \left(R^k\right)_{v,i} = 1.$$

If w is not periodic with respect to R, then it is called aperiodic.

Theorem D.2. If G is aperiodic and R is any C-matrix of G, then there exists a vertex $w \in V$ which is aperiodic with respect to R.

Proof. Suppose each $w \in V$ is periodic with respect to a given C-matrix R. Then if u, v are such that

$$A_{w,u} = A_{w,v} = 1.$$

Then for some k $(= k(w))$,

$$\left(R^k\right)_{u,i} = \left(R^k\right)_{v,i} = 1.$$

Thus

$$\left(BR^k\right)_{w,i} = \left(R^{k+1}\right)_{w,i} = 1.$$

But once the w-th row of BR^k and R^{k+1} are equal, they remain equal under additional powers of R. Since this is true for each vertex, there exists a $\bar{k} = \max_w(k(w))$, such that for all $k \geq \bar{k}$

$$BR^k = R^{k+1}.$$

This implies,

$$\begin{aligned}\left(\frac{1}{2}A\right)R^k &= \left(\frac{1}{2}R + \frac{1}{2}B\right)R^k \\ &= R^{k+1}.\end{aligned}$$

Recall that A is irreducible and aperiodic or, equivalently,

$$\left(\frac{1}{2}A\right)^t > 0 \quad \text{for some } t.$$

However, since

$$\left(\frac{1}{2}A\right)^t R^k = R^{k+t}, \quad t \geq 1,$$

the $p \times p$ submatrix of R^{k+t} (corresponding to the cycle) is strictly positive. Since R^{k+t} is a binary stochastic matrix, we have our contradiction. □

Theorem D.3. Let R be a C-matrix for G and let $\mathscr{K}$ be the kernel of $S = \langle R, B\rangle$. Then the rank of $\mathscr{K}$ divides p. Here p need not be a prime.

Proof. Let

$$M = R^t = \left[\begin{array}{ccccc|c} 0 & 1 & 0 & \cdots & 0 & \\ 0 & 0 & 1 & \cdots & \vdots & \\ \vdots & & & & \vdots & \\ 0 & \cdots & \cdots & \cdots & 1 & 0 \\ 1 & 0 & \cdots & \cdots & 0 & \\ -- & -- & -- & -- & -- & \\ \multicolumn{5}{c|}{D} & \end{array}\right].$$

Let $z \in \mathscr{K}$ and let $\mathscr{G} = zM\mathscr{K}zM$. Suppose $E = E^2$ is the local identity of $\mathscr{G}$. Notice that each zero column of M is a zero column of E.

Suppose that $\mathsf{rank}(E) = k$ and that k does not divide p. Then

$$p = kr + s, \quad 1 \leq s < k.$$

If we can show that, as a result of this assumption, there exists a positive integer m such that

$$\mathsf{rank}\,(EM^mE) < k,$$

then we are done. □

Let the non-zero columns of E be the columns $i_1 < i_2 \cdots < i_k \leq p$ and define

$$P_j := \left\{ \ell : E_{\ell,i_j} = 1, 1 \leq \ell \leq p \right\}.$$

Since the P_j's are disjoint and

$$\mathsf{card}\left(\bigcup_{j=1}^{k} P_j \right) = p < (r+1)k$$

there must exist at least one j_0 such that $\mathsf{card}\left(P_{j_0}\right) \leq r$. With no loss of generality, assume that $j_0 = 1$.

Claim. For some m, $1 \leq m \leq p$,

$$(M^mE)_{i_j,i_1} = 0 \quad 1 \leq j \leq k.$$

Notice that, because of the action of the cycle, if $(M^mE)_{i_j,i_1} = 1$, then $i_j + m$ must be in P_1. Therefore, m has at most r choices for each j. Thus, there are at most rk choices and if m is different from each of these choices so that for this m, $i_j + m$ is not in P_1 for any j, $1 \leq j \leq k$

$$(M^mE)_{i_j,i_1} = 0.$$

It follows that the i_1th column of EM^mE is zero and this implies that

$$\mathsf{rank}\,(EM^mE) < k.$$

Theorem D.4. Suppose $G = (V, \mathscr{E})$ is aperiodic, has no parallel edges and a simple cycle, C, of prime length p. Then there exists a C-matrix R whose coloring semigroup $S = \langle R, B \rangle$ contains a synchronizing instruction.

Proof. Let $A = R + B$. Construct the C-matrix R in the following way. If for i not in C, there are j in C and k not in C such that

$$A_{i,j} = A_{i,k} = 1,$$

then we choose so that $R_{i,j} = 1 = B_{i,k}$. Now if there are any vertices outside of C not connected to C by the previous rule, simply connect them to C by the shortest path to define R.

Case 1. Every point in $V - C$ is periodic with respect to R.

Notice that if every point in $V - C$ is periodic with respect to R, then by the assumption of no parallel edges, and by construction the $(V-C) \times C$ block of the matrix B is all zero. (If a vertex in $V - C$ is connected to two distinct vertices in C, it must be aperiodic.) We know that since every point in $V - C$ is periodic, $BR^n = R^{n+1}$, for sufficiently large n, when we restrict the matrices to $V - C$. Notice that because of our convention mentioned earlier, for every positive integer p the $(V-C) \times V$ block of $B^{p+1}R^n$ is equal to the product of the $(V-C) \times (V-C)$ block of B and the $(V-C) \times V$ block of B^pR^n. Thus, by induction, it is easy to see that for any word H in B and R of length r,

$$\begin{aligned} &HR^n \text{ restricted to the } (V-C) \times V \text{ block} \\ &= R^{n+r} \text{ restricted to the } (V-C) \times V \text{ block.} \end{aligned}$$

This means that $[(1/2)A]^p (R^n) = R^{n+p}$, when we restrict both matrices to the $(V-C) \times V$ block, for some sufficiently large n, but for all positive integers p. But this is a contradiction as the left side converges as p goes to infinity, but the right side does not unless $p = 1$. The case $p = 1$ is trivial to synchronize.

Case 2. There is at least one aperiodic point outside C with respect to R.

First observation: Let w_1 be aperiodic, w_1 not in C. Consider all those vertices v_k outside C such that

$$v_k R^k = w_1.$$

If there is no such positive k, then we work with w_1; otherwise, let k_0 be the largest such k. If v_{k_0} is aperiodic, then we will replace w_1 by v_{k_0} and we'll still call it w_1. If v_{k_0} is periodic, we will not change anything and still work with our old w_1. Thus, there is no loss of generality in assuming that w_1 is aperiodic, w_1 not in C, and if there is a v_k such that $v_k R^k = w_1$ for positive integer k, then such a v_k must be periodic.

Second observation: Let us consider the shortest path from C to w_1. We consider a word H (in R and B) such that $(C)H = \{w_1, w_2, \ldots, w_p\}$, and for $i > 1$, the H-path from C to w_i does not contain w_1. In other words, if we interchange R and B only at the vertex w_1, then $(C)H$ remains unaffected.

Final observation: Decompose $(C)H = \{w_1\} \cup Y_k \cup Z_k$, where $Y_k = \{w_2, \ldots, w_k\}$ and $Z_k = \{w_{k+1}, \ldots, w_p\}$, such that for $1 < j < k+1$, $w_j R^{s_j - 1} = w_1$ and w_j is

periodic, where $s_2, s_3, \ldots, s_k$ are distinct positive integers, and for each $j > k$, there is no positive integer p such that $w_j R^p = w_1$. [Note: There cannot be two different w_i in Y_k leading to w_1 in the same number of R-steps, as otherwise we can easily reduce the rank of the kernel.] Thus, every vertex in Y_k is periodic; and when we apply the map R^{n+1} on $(C)H$, then the vertices in Z_k remain unaffected if we interchange R and B only at w_1.

Suppose Z_k is empty (in this case, recoloring is not needed). Choose n large so that $(V)R^n$ is inside C and $w_1 R^{n+1} = 1$. Then, Y_k is mapped onto $\{2, 3, \ldots, p\}$ by both R^{n+1} and BR^n. However, $w_1 BR^n > 1$.

Suppose Z_k is not empty. Choose n large as before. Let $w_j R^{n+1} = q_j$. Then, $w_1 BR^{n+1} = q_{j_0}$, for some $j_0 > k$. Now we recolor at the vertex w_1 and then again consider $(C)HR^{n+1}$, with the new R. Notice that the image of Z_k remains the same before and after coloring under the map R^{n+1}; however, q_{j_0}, the present image of w_1 after recoloring, creates a contradiction. □

Finally, we state a result where the elementary proof is left to the reader. It is included since it dispenses with a large number of examples quite easily. Notice no assumption concerning parallel edges is required.

Theorem D.5. Let $G = (V, E)$ have a simple cycle C of prime length p and adjacency matrix A. Suppose one of the following conditions holds:

(1) there exists $j \notin C$ such that

$$\sum_{i=1}^{p} A_{i,j} \geq 2; \text{ or}$$

(2) there exists $j \in C$ such that

$$\sum_{i=1}^{p} A_{i,j} \geq 3.$$

Then G has a coloring semigroup that contains a synchronizing instruction.

Acknowledgements

The authors are grateful to the members of the Discrete Mathematics Seminar at the University of South Florida (USF) for their kind attention to a series of lectures given by the authors. In particular, conversations with W.E. Clark, N. Jonaska, and S. Suen were always interesting. The authors would also like to thank Göran Högnäs for helpful comments.

This work was done while the first author was visiting USF while on sabbatical from Southern Illinois University at Carbondale (SIUC). Both the generosity of SIUC and the hospitality of USF were very much appreciated.

D.3 A technique to complete the solution to the generalized road coloring conjecture

Using the result from the previous section, a technique suggests itself to solve the general problem. Suppose the graph is periodic of order p. Look at the collection of all minimal coloring semigroups. Suppose the kernels are of rank t. There are two possibilities. If one of the minimal kernels is a right group, from the previous result, we immediately deduce that $p = t$. If no minimal coloring semigroup has a kernel which is a right group, one must show that some recoloring of one of the minimal semigroups produces a kernel of rank strictly less than t.

Thus, if one can prove the following conjecture, the proof of the generalized road coloring problem would be complete.

Conjecture D.1. Let S be a coloring semigroup. Suppose the kernel of S is not a right group. Then S is not a minimal coloring semigroup.

Consider the following example of recoloring in the instance that a kernel is not a right group.

Example D.1. Let G be given by the following adjacency matrix which is colored as indicated

$$A = \begin{bmatrix} 0&1&0&1 \\ 0&0&1&1 \\ 1&1&0&0 \\ 1&0&1&0 \end{bmatrix} = \begin{bmatrix} 0&0&0&1 \\ 0&0&0&1 \\ 1&0&0&0 \\ 1&0&0&0 \end{bmatrix} + \begin{bmatrix} 0&1&0&0 \\ 0&0&1&0 \\ 0&1&0&0 \\ 0&0&1&0 \end{bmatrix} = r_1 + b_1$$

Let $S_1 = \langle r_1, b_1 \rangle$. Then S_1 is, in fact, a completely simple semigroup of matrices of rank two. S_1 consists of the following eight matrices.

$$S_1 = \left\{ \begin{bmatrix} 1&0&0&0 \\ 1&0&0&0 \\ 0&0&0&1 \\ 0&0&0&1 \end{bmatrix}, \begin{bmatrix} 0&0&0&1 \\ 0&0&0&1 \\ 1&0&0&0 \\ 1&0&0&0 \end{bmatrix}, \begin{bmatrix} 0&1&0&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 0&0&1&0 \end{bmatrix}, \begin{bmatrix} 0&0&1&0 \\ 0&0&1&0 \\ 0&1&0&0 \\ 0&1&0&0 \end{bmatrix} \right.$$
$$\left. \begin{bmatrix} 1&0&0&0 \\ 0&0&0&1 \\ 1&0&0&0 \\ 0&0&0&1 \end{bmatrix}, \begin{bmatrix} 0&0&0&1 \\ 1&0&0&0 \\ 0&0&0&1 \\ 1&0&0&0 \end{bmatrix}, \begin{bmatrix} 0&0&1&0 \\ 0&1&0&0 \\ 0&0&1&0 \\ 0&1&0&0 \end{bmatrix}, \begin{bmatrix} 0&1&0&0 \\ 0&0&1&0 \\ 0&1&0&0 \\ 0&0&1&0 \end{bmatrix} \right\}$$

Since S_1 (which is its own kernel), is not a right group, the functions within it do not all have the same partition. There are two partitions in this example.

$$\pi_1 = \{\{1,2\},\{3,4\}\}, \quad \pi_2 = \{\{1,3\},\{2,4\}\}$$

In general there will be many such partitions. There will always be at least one vertex v, which is paired with distinct vertices in two of these partitions.

(In this example, any vertex would suffice. That is, 1 is paired with 2 in π_1, but with 3 in π_2.) Perhaps recoloring at such a vertex v will allow one to reduce the rank. Computer experiments offer some hope this is true. Certainly, in this case, recoloring at 1 produces the following.

$$\begin{bmatrix} 0&1&0&1 \\ 0&0&1&1 \\ 1&1&0&0 \\ 1&0&1&0 \end{bmatrix} = \begin{bmatrix} 0&1&0&0 \\ 0&0&0&1 \\ 1&0&0&0 \\ 1&0&0&0 \end{bmatrix} + \begin{bmatrix} 0&0&0&1 \\ 0&0&1&0 \\ 0&1&0&0 \\ 0&0&1&0 \end{bmatrix} = r_2 + b_2$$

A simple computation will show that

$$r_2 b_2 r_2 = \begin{bmatrix} 1&0&0&0 \\ 1&0&0&0 \\ 1&0&0&0 \\ 1&0&0&0 \end{bmatrix}.$$

Thus $S_2 = \langle r_2, b_2 \rangle$ contains a rank one matrix, or equivalently, a synchronizing instruction.

D.4 Conclusion

In this paper, the author has shown how the structure theory of the kernel of a coloring semigroup continues to shed light on a longstanding problem in theoretical computer science. In addition, directed graphs, in the form of state transition diagrams, are used pervasively in both deterministic symbolic dynamics and Markov chain theory. The continuing success of semigroup theory in studying the road coloring problem offers hope that these methods could apply to open problems in these other areas, as well.

References

1. R.L. Adler, L.W. Goodwyn, and B. Weiss, Equivalence of topological Markov shifts, Israel Journal of Mathematics, 27, 49–63, (1977)
2. G. Budzban, Semigroups and the generalized road coloring problem, Semigroup Forum 69, 201–208, (2004).
3. G. Budzban and P. Feinsilver, Completely Simple Semigroups, Lie Algebras, and the Road Coloring Problem, accepted for publication, Semigroup Forum.
4. G. Budzban and A. Mukherjea, A semigroup approach to the road coloring problem. In: Contemporary Mathematics, vol. 261, Amer. Math. Soc., pp. 195–207. Providence, R.I. (2000).
5. A. Carbone, Cycles of relatively prime length and the road coloring problem, Israel J. Math. 123 (2001), 303–316.
6. W.E. Clark, Remarks on the kernel of a matrix semigroup, Czechoslovak Math. Journal, 15, 305–310, (1965).
7. A.H. Clifford and G.B. Preston, The Algebraic Theory of Semigroups. AMS, 1, Providence (1961).
8. K. Culic, J. Kari, and J. Karhumaki, A note on synchronized automata and road coloring problem, Internat. J. Found. Comp. Sci. 3, 459–471, (2002).
9. J. Friedman, On the road coloring problem, Proceedings of the AMS, 110, 1133–1135, (1990).
10. R. Hegde and K. Jain, A min-max theorem about the road coloring conjecture, submitted to Discrete Math.
11. G. Högnäs, Random semigroup acts on a finite set, J. Australian Math Society, 23, 481–498, (1977).
12. G. Hognas and A. Mukherjea, Probability Measures on Semigroups: Convolution Products, Random Walks, and Random Matrices. Plenum Press, New York (1995).
13. N. Jacobson, Lie algebras, Dover, 1968.
14. N. Jonoska and S. Suen, Monocyclic decomposition of graphs and the road coloring problem, Congressum Numerantium, 110, 201–209, (1995).
15. J. Kari, Synchronizing finite automata on Eulerian digraphs, Proc. Math. Found. of Comp. Sci., LNCS 2136, 432–438, (2001).
16. G.L. O'Brien, The road coloring problem, Israel Journal of Mathematics, 39, 145–154, (1981).
17. D. Perrin and M.P. Schtzenberger, Synchronizing prefix codes and automata and the road coloring problem, symbolic dynamics and applications. In: Contemporary Mathematics, vol. 135, Amer. Math. Soc., pp. 295–318. Providence, R.I. (1992).
18. M. Petrich, Introduction to Semigroups. Charles E. Merrill Publishing, Columbus (1973).

19. H. Radjavi & P. Rosenthal, Simultaneous triangularization, Universitext, Springer, Berlin, 2000.
20. O.S. Rothaus, Semigroups of finite matrices, Semigroup Forum, 49, 59–65, (1994).
21. A. N. Trahtman, The road coloring problem, Israel Journal of Mathematics (2009). DOI: 10.1007/s11856-009-0062-5.

Subject Index

Abel limit, 90
absorbing state, 91

canonical coloring, 57
collision
  filtration, 79
  free quotient, 79
  probability, 47
coloring
  canonical, 57
  definition, 3
  recoloring, 139
  semigroup, 18
  synchronizing, see kernel, minimum rank
common-letter dynamics, 79
commutator, 109
compatible field
  normalized inclusion, 91
completely simple semigroup, 9
convolution powers
  convergence, 29
cycle matrix, 69

divisibility
  kernel rank and cycle length, 71

equipartition
  kernel blocks, 52
  stationary measure, 52
  tensor-square proof, 42

fixed-point equations
  kernel expectations, 50
Friedman weight, 72
fundamental inverse, 72

graph
  double construction, 99
  edge-state lift, 99
  target-pair, 4

idempotent
  average of distinguished idempotents, 49
inclusion matrix, 91
independent-particle dynamics, 79

kernel
  idempotents, 47
  limiting measure, 29
  minimum rank, 20
  partition marginals, 47
  partition projections, 48
  partition structure, 20
  range marginals, 47
  range projections, 48
  range structure, 20
  rank from representation levels, 91

letter probabilities
  interpolation, 73
Lie constant, 109
limiting projection
  higher representations, 90

Markov chain
  permanental, 93

operators
  centered range expectation, 50
  collision expectation, 47
  pair survival, 50
  potential, 72
  weighted range expectation, 47

opposite-letter dynamics, 95

partial circulant
  generators, 113
  kernel structure, 114
  minimum rank, 114
  residue blocks, 113
permanent, 93
potential operator, 72
probability measure
  convolution, 23
projection
  partition-fixed, 48
  range-fixed, 48

random walk
  on a semigroup, 23
rank
  coloring criterion, 63
  cycle matrix, 69
  kernel, 20
  matrix, 69
  ranked condition, 69
ranked transformation, 69
recoloring
  four-state example, 139
Rees representation, 11
Rees–Sushkewitch theorem, 11
right action, 3

semigroup
  coloring, 18
  completely simple, 9
  definition, 7
  examples, 7
semigroup algebra, 27
stationary distribution
  perturbation, 73
symmetric power, 79
synchronization, see coloring, synchronizing

target-pair graph, 4
tensor coupling, 95
tensor power, 79
trace identities, 82
transformation
  matrix representation, 3
transversal, 47

weak convergence, 121
weighted block average, 48

zeon algebra
  collision-free interpretation, 79
  definition, 81
  pair representation, 81
  zero-diagonal lift, 82